# MATHEMATICS

# AREA UNDER CURVE

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# AREA UNDER CURVE

## JEE EXERCISES

## Exercise J-I

### AREA BOUNDED BETWEEN TWO CURVES

1. The area between the curve $xy = a^2$, x-axis, $x = a$ and $x = 2a$ is :

(A) $a \log 2$ (B) $a^2 \log 2$ (C) $2a \log 2$ (D) None of these

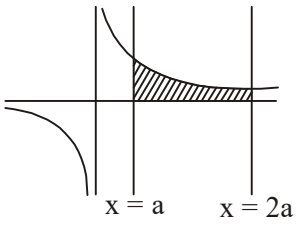
वक्र $xy = a^2$, x-अक्ष, $x = a$ तथा $x = 2a$ के मध्य क्षेत्रफल है :

(A) $a \log 2$ (B) $a^2 \log 2$ (C) $2a \log 2$ (D) कोई नहीं

**Ans.** B

**Sol.** $xy = a^2$ x-axis $x = a$ and $x = 2a$

$$A = \int_a^{2a} y\,dx$$




$$= \int_a^{2a} \frac{a^2}{x}dx = a^2 \log x\Big|_a^{2a}$$

$$= a^2 \log \frac{2a}{a}$$

$$= a^2 \log 2$$

2. Area under the curve $y = \sin 2x + \cos 2x$ between $x = 0$ and $x = \frac{\pi}{4}$, is :

(A) 2 sq. units (B) 1 sq. units (C) 3 sq. units (D) 4 sq. units

वक्र $y = \sin 2x + \cos 2x$ तथा $x = 0$ तथा $x = \frac{\pi}{4}$, के मध्य क्षेत्रफल है :

(A) 2 वर्ग इकाई (B) 1वर्ग इकाई (C) 3 वर्ग इकाई (D) 4 वर्ग इकाई

**Ans.** B

**Sol.** $y = \sin 2x + \cos 2x$ $x = 0$ & $\frac{\pi}{4}$

$$A = \int_0^{\frac{\pi}{4}} (\sin 2x + \cos 2x)dx$$

$$= -\frac{\cos 2x}{2} + \sin \frac{2x}{2}\Big|_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

3. The area bounded between the curve $|y| = 1 - x^2$ is :

(A) 2/3 (B) 4/3 (C) 8/3 (D) None of these

वक्र $|y| = 1 - x^2$ के द्वारा परिबद्ध क्षेत्रफल है :

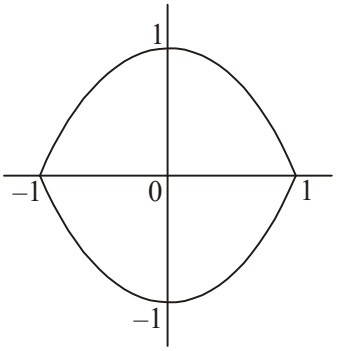
(A) 2/3 (B) 4/3 (C) 8/3 (D) इनमें से कोई नहीं

**Ans.** C

**Sol.** $|y| = 1 - x^2$

$y = 1 - x^2 \Rightarrow x^2 = -(y - 1)$

$|y| = 1 - x^2$




$$A = 2\int_{-1}^{1} y\,dx$$

$$= 2\int_{-1}^{1} \left(1 - x^2\right) dx$$

$$= 2\left(x - \frac{x^3}{3}\right)_{-1}^{1} = \left(2 - \frac{1}{3} - \left(-1 + \frac{1}{3}\right)\right) = \frac{8}{3}$$

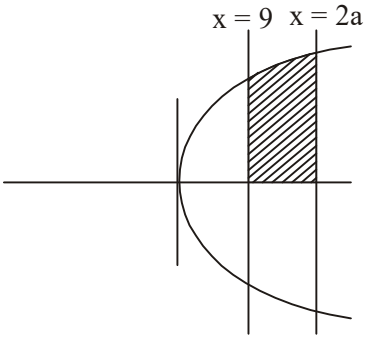
4. The area bounded by the parabola $y^2 = 4ax$, x-axis and $x = 2a$ and latus rectum is :

परवलय $y^2 = 4ax$, x-अक्ष तथा $x = 2a$ तथा नाभि लम्ब के मध्य क्षेत्रफल है :

(A) $2a^2\left(\sqrt{2} - 1\right)$ (B) $4a^2\left(2\sqrt{2} - 1\right)$ (C) $\frac{4}{3}a^2\left(2\sqrt{2} - 1\right)$ (D) $\frac{8a^2}{3}\left(2\sqrt{2} - 1\right)$

**Ans.** C

**Sol.** $y^2 = 4ax$ x-axis $x = 2a$ & latus rectum




$$y = \int_{a}^{2a} \sqrt{4ax}\,dx$$

$$= 2\sqrt{a}\left[\frac{2}{3}x^{\frac{3}{2}}\right]_{a}^{2a} 4$$

$$= 2\sqrt{a}\left[\frac{2}{3}\cdot 2\sqrt{2}a^{\frac{3}{2}} - \frac{2}{3}a^{\frac{3}{2}}\right]$$

$$= \frac{4}{3}\left[2\sqrt{2}a^2 - a^2\right]$$

$$= \frac{4}{3}a^2\left[2\sqrt{2}-1\right]$$

5. The area bounded between parabola $x^2 = 4y$ and $y = |x|$ is :

परवलय $x^2 = 4y$ तथा $y = |x|$ के मध्य क्षेत्रफल है :

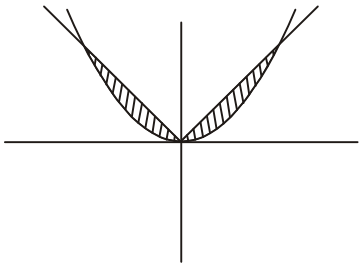
(A) 2/3 (B) 4/3 (C) 8/3 (D) 16/3

**Ans.** D

**Sol.** $x^2 = 4y$

$$\text{Area} = 2\int_0^4 (y_1 - y_2)dx$$

$$= 2\int_0^4 \left(x - \frac{x^2}{4}\right)dx$$



$$= 2\left(\frac{x^2}{2} - \frac{x^3}{12}\right)\Bigg|_0^4$$

$$= 2\left[\left(8 - \frac{16}{3}\right) - 0\right]$$

$$= 2 \times \frac{8}{3}$$

$$= \frac{16}{3}$$

6. The area of the figure bounded by the parabola $y = x^2 + 1$ and the straight line $x + y = 3$ is :

$y = x^2 + 1$ तथा सरल रेखा $x + y = 3$ के मध्य क्षेत्रफल है-

(A) 3/2 (B) 5/2 (C) 7/2 (D) 9/2

**Ans.** D

**Sol.** $y = x^2 + 1$ $x + y = 3$

Point of Intersection $3 - x = x^2 + 1$

$x^2 + x - 2 = 0$

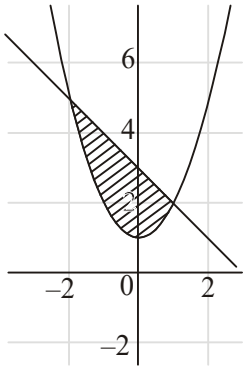
$(x + 2)(x - 1) = 0$

$x = -2, 1$

$$A = \int_{-2}^{1} (y_1 - y_2)\,dx$$

$$= \int_{-2}^{1} \Big((3-x)-\left(x^2+1\right)\Big)dx$$

$$= \int_{-2}^{1} \left(2-x-x^2\right)dx$$




$$= 2x - \frac{x^2}{2} - \frac{x^3}{3}\Big|_{-2}^{1}$$

$$A = \left(2 - \frac{1}{2} - \frac{1}{3}\right) - \left(-4 - \frac{4}{2} + \frac{8}{3}\right)$$

$$= \frac{12-3-2}{6} - \frac{-24-12+16}{6}$$

$$= \frac{7}{6} + \frac{20}{6}$$

$$= \frac{27}{6}$$

$$= \frac{9}{2}$$

7. Common area between the parabolas $y = 2x^2$ and $y = x^2 + 4$ is :

(A) 16/3 (B) 8/3 (C) 32/3 (D) None of these

परवलयों $y = 2x^2$ तथा $y = x^2 + 4$ का उभयनिष्ठ क्षेत्रफल है :

(A) 16/3 (B) 8/3 (C) 32/3 (D) इनमें से कोई नहीं

**Ans.** C

**Sol.** $y = 2x^2$ $y = x^2 + 4$

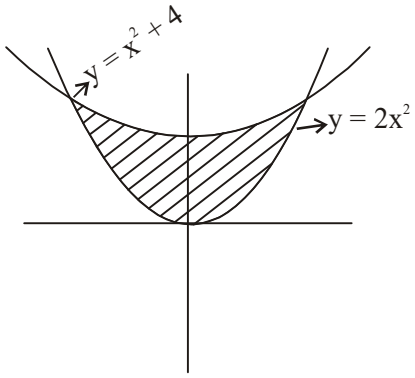
Point of intersection

$2x^2 = x^2 + 4$

$x^2 = 4$

$x = \pm 2$

$$\text{Area} = 2\int_0^2 (y_2 - y_1)\,dx$$




$$A = 2\int_0^2 \left(\left(x^2 + 4\right) - \left(2x^2\right)\right)dx$$

$$= 2\int_0^2 \left(4 - x^2\right)dx$$

$$= 2\left[4x - \frac{x^3}{3}\right]_0^2$$

$$= 2\left(8 - \frac{8}{3}\right)$$

$$= \frac{32}{3}$$

8. Area bounded by the curves $y = 2^x$, $y = 2x - x^2$, $x = 0$ and $x = 2$ is :

वक्रों $y = 2^x$, $y = 2x - x^2$, $x = 0$ तथा $x = 2$ के मध्य क्षेत्रफल है :

(A) $\frac{3}{\log 2} - \frac{4}{3}$ (B) $\frac{3}{\log 2} + \frac{4}{3}$ (C) $3\log 2 + \frac{4}{3}$ (D) $3\log 2 - \frac{4}{3}$

**Ans.** A

**Sol.** $y = 2x$ $y = 2x - x^2$ $x = 0$ $x = 2$

$y = 2^x$

$y = 2x - x^2$

$-y = x^2 - 2x$

$-(y - 1) = (x - 1)^2$

$$\text{Area} = \int_0^2 (y_2 - y_1)\,dx$$

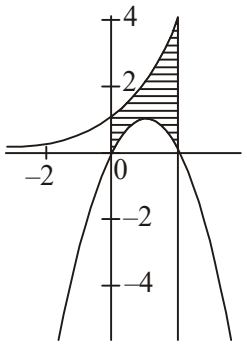
$$= \int_0^2 \left(2^x - \left(2x - x^2\right)\right) dx$$

$$= \int_0^2 \left(2^x - 2x + x^2\right) dx$$

$$= \frac{2^x}{\ln 2} - x^2 + \frac{x^3}{3}\Big|_0^2$$

$$= \frac{4}{\ln 2} - 4 + \frac{8}{3} - \frac{1}{\ln 2}$$

$$= \frac{3}{\ln 2} - \frac{4}{3}$$



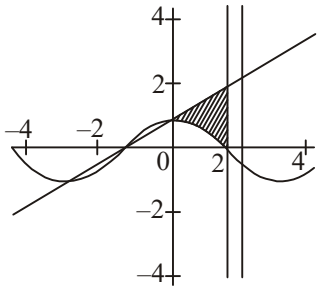

9. The area of the closed figure bounded by $y = \cos x$, $y = 1 + (2/\pi)\, x$ and $x = \pi/2$ is :

$y = \cos x$, $y = 1 + (2/\pi)\, x$ तथा $x = \pi/2$ के मध्य क्षेत्रफल है :

(A) $\frac{3\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi - 4}{4}$ (D) $\frac{\pi - 4}{4}$

**Ans.** C

**Sol.** $y = \cos x$ $\quad y = 1 + \frac{2}{\pi}x$ $\quad x = \frac{\pi}{2}$




$$A = \int_0^{\frac{\pi}{2}} \left(y_2 - y_1\right) dx$$

$$= \int_0^{\frac{\pi}{2}} \left(\left(1 + \frac{2}{\pi}x\right) - \cos x\right) dx$$

$$= x + \frac{x^2}{\pi} - \sin x\Big|_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} + \frac{\pi}{4} - 1$$

$= \frac{3\pi}{4} - 1$

10. The area enclosed between the curves $y = \log_e (x + e)$, $x = \log_e \left(\frac{1}{y}\right)$ and the x-axis is :

वक्रों $y = \log_e (x + e)$, $x = \log_e \left(\frac{1}{y}\right)$ तथा x-अक्ष द्वारा बने क्षेत्र का क्षेत्रफल होगा :

(A) 1 (B) 2 (C) 3 (D) 4

**Ans.** B

**Sol.** $y = \log_e (x + e)$ $x = \log_e \left(\frac{1}{y}\right)$ x-axis

$x = \log_e \left(\frac{1}{y}\right)$

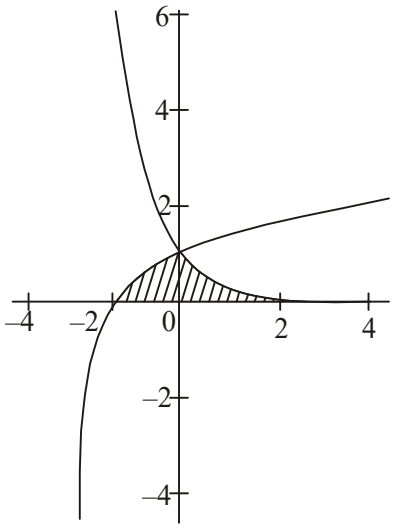
$A = \int_{1-e}^{0} \log_e (x + e)\,dx - \int_{0}^{\infty} e^{-x}\,dx$

$= (x + e)\log_e \left(\frac{x+e}{e}\right)\Big|_{1-e}^{0} - -e^{-x}\Big|_{0}^{\infty}$

$= \left(-\log_e \frac{1}{e}\right) - (-1)$

$= 1 + 1$

$= 2$




11. Area of the circle $(x - 2)^2 + (y - 3)^2 = 32$ which lies below the line $y = x + 1$ is :

(A) $\int_{-2}^{6} \left[(x+1) + \sqrt{32 - (x-2)^2} + 3\right] dx$ (B) $\int_{-2}^{6} \left[\left\{\sqrt{32 - (x-2)^2} + 3\right\} - (x+1)\right] dx$

(C) $16\pi$ (D) None of these

वृत्त $(x - 2)^2 + (y - 3)^2 = 32$ जो कि रेखा $y = x + 1$ के नीचे स्थित है, का क्षेत्रफल है :

(A) $\int_{-2}^{6} \left[(x+1) + \sqrt{32 - (x-2)^2} + 3\right] dx$ (B) $\int_{-2}^{6} \left[\left\{\sqrt{32 - (x-2)^2} + 3\right\} - (x+1)\right] dx$

(C) $16\pi$ (D) इनमें से कोई नहीं

**Ans.** C

**Sol.** $(x-2)^2+(y-3)^2=32$ $\quad y=x+1$

Point of intersection

$(x-2)^2+(x-2)^2=32$

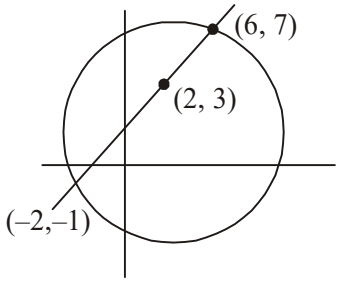
$(x-2)^2=16$

$x-2=\pm4$

$x=6,-2$

$x=6 \quad y=7 \quad (6,7)$

$x=-2 \quad y=-1 \quad (-2,-1)$




center is on the line, so line is diameter

$$A=\frac{\pi r^2}{2}=\frac{\pi\times32}{2}=16\pi$$

12. If $y=2\sin x+\sin 2x$ for $0\le x\le 2\pi$, then the area enclosed by the curve and the x-axis is :
(A) 6 (B) 8 (C) 12 (D) none

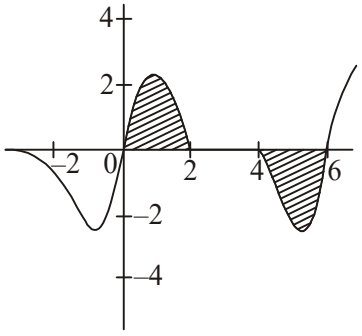
यदि $y=2\sin x+\sin 2x$, $0\le x\le 2\pi$ के लिये, तो वक्र तथा x-अक्ष द्वारा परिबद्ध क्षेत्रफल होगा :
(A) 6 (B) 8 (C) 12 (D) कोई नहीं

**Ans.** B

**Sol.** $y=2\sin x+\sin 2x$ $\quad$ x-axis $\quad x\in[0,2\pi]$

$y=2\sin x(1+\cos x)$

$$=2\sin x\cdot 2\cos^2\frac{x}{2}$$




$$A=\int_0^{\pi}(2\sin x+\sin 2x)\,dx+(-)\int_{\pi}^{2\pi}2\sin x+\sin 2x$$

$$=-2\cos x-\frac{\cos 2x}{2}\Big|_0^{\pi}-\left[-2\cos x-\frac{\cos 2x}{2}\right)_{\pi}^{2\pi}$$

$$=+2-\frac{1}{2}+2+\frac{1}{2}-\left[-2-\frac{1}{2}-\left(2-\frac{1}{2}\right)\right)$$

$=8$

13. The area of the closed figure bounded by $x = -1$, $x = 2$ and $y = \begin{cases} -x^2 + 2 \ , & x \le 1 \\ 2x - 1 \ , & x > 1 \end{cases}$ and the abscissa axis, is :

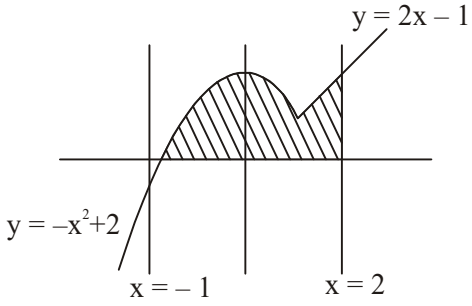
वक्र $x = -1$, $x = 2$, $y = \begin{cases} -x^2 + 2 \ , & x \le 1 \\ 2x - 1 \ , & x > 1 \end{cases}$ तथा x-अक्ष द्वारा परिबद्ध क्षेत्रफल होगा :

(A) $\frac{7}{3}$ (B) $\frac{10}{3}$ (C) $\frac{13}{3}$ (D) $\frac{16}{3}$

**Ans.** D

**Sol.** $y = \begin{cases} -x^2 + 2 & x \le 1 \\ 2x - 1 & x > 1 \end{cases}$, $x = -1, x = 2$

$$A = \int_{-1}^{1} \left(-x^2 + 2\right) dx + \int_{1}^{2} (2x - 1)\, dx$$




$$= 2\int_{0}^{-1} \left(-x^2 + 2\right) dx + x^2 - x\Big|_{1}^{2}$$

$$= 2\left( \frac{-x^3}{3} + 2x \right)_{0}^{1} + \left((4 - 2) - (1 - 1)\right)$$

$$= 2\left( -\frac{1}{3} + 2 \right) + 2$$

$$= \frac{10}{3} + 2$$

$$= \frac{16}{3}$$

14. Area between curve $y = \frac{x}{1 + x^2}$ and x axis from $x = -1$ to $x = 1$ is :

वक्र $y = \frac{x}{1 + x^2}$ तथा x- अक्ष, $x = -1$ से $x = 1$ तक परिबद्ध क्षेत्रफल होगा :

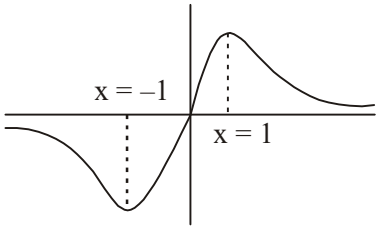
(A) $\ln 2$ (B) $\ln 4$ (C) $\pi/2$ (D) $\pi$

**Ans.** A

**Sol.** $y = \frac{x}{1+x^2}$ x-axis $x = -1, 1$

$$\frac{dy}{dx} = \frac{1+x^2-2x^2}{\left(1+x^2\right)^2}$$

$$= \frac{1-x^2}{\left(1+x^2\right)^2}$$




$$A = 2\int_0^1 ydx$$

$$= 2\int_0^1 \frac{x}{1+x^2}dx$$

$$= \log\left(1+x^2\right)\Big|_0^1$$

$$= \log 2$$

15. The area of the figure bounded by the curves $y = e^x$, $y = e^{-x}$ and the straight line $x = 1$ is :

(A) $e + \frac{1}{e}$ (B) $e - \frac{1}{e}$ (C) $e + \frac{1}{e} - 2$ (D) none

वक्र $y = e^x$, $y = e^{-x}$ तथा सरल रेखा $x = 1$ द्वारा परिबद्ध क्षेत्रफल होगा :

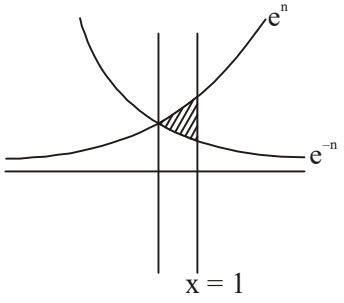
(A) $e + \frac{1}{e}$ (B) $e - \frac{1}{e}$ (C) $e + \frac{1}{e} - 2$ (D) कोई नहीं

**Ans.** C

**Sol.** $y = e^x$ $y = e^{-x}$ $x = 1$

$$A = \int_0^1 e^x dx - \int_0^1 e^{-x}dx$$

$$= \left(e^x\right)_0^1 - \left(-e^{-x}\right)_0^1$$



$$= (e-1) - \left(-\frac{1}{e}+1\right)$$

$$= e + \frac{1}{e} - 2$$

16. The area bounded by $y = x^2$ and $y = 2 - x^2$ is :

वक्र $y = x^2$ तथा $y = 2 - x^2$ द्वारा परिबद्ध क्षेत्रफल होगा :

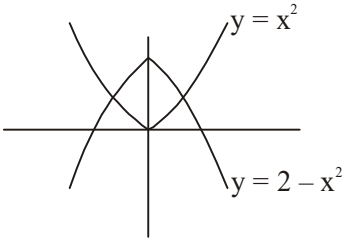
(A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{32}{3}$ (D) $\frac{64}{3}$

**Ans.** A

**Sol.** $y = x^2$ $y = 2 - x^2$

Point of intersection $x = \pm 1$

$$A = 2\int_0^1 (y_2 - y_1)\,dx$$




$$= 2\int_0^1 \left(2 - x^2 - x^2\right) dx$$

$$= 2\left(2x - \frac{2x^3}{3}\right)_0^1$$

$$= 2\left(2 - \frac{2}{3}\right)$$

$$= \frac{8}{3}$$

17. The area bounded by the curves $y^2 = 3x$ & $y^2 = 36 - 9x$ is :

(A) 16 (B) 18 (C) 21.5 (D) none

वक्र $y^2 = 3x$ तथा $y^2 = 36 - 9x$ द्वारा परिबद्ध क्षेत्रफल होगा :

(A) 16 (B) 18 (C) 21.5 (D) कोई नहीं

**Ans.** A

**Sol.** $y^2 = 3x$ $y^2 = 36 - 9x$

$= -9(x-4)$

Point of intersection

$3x = 36 - 9x$

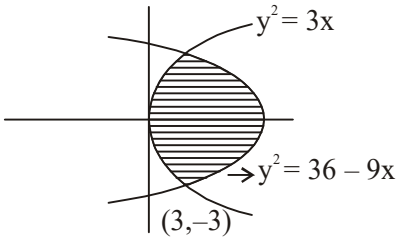
$12x = 36$

$x = 3$

$y^2 = 9$

$y = \pm 3$

$(3, 3) \qquad (3, -3)$




$$A = 2\int_0^3 (x_2 - x_1)\,dy$$

$$= 2\int_0^3 \left(\frac{36-y^2}{9}\right) - \left(\frac{y^2}{3}\right) dy$$

$$= 2\left[\left.\frac{36y - \frac{y^3}{3}}{9}\right|_0^3 - \left.\frac{y^3}{9}\right|_0^3\right]$$

$$= 2\left[\frac{108-9}{9} - 3\right]$$

$$= 2\left[\frac{99}{9} - 3\right]$$

$= 2\,[11 - 3]$

$= 16$

18. The ratio in which the area enclosed by the curve $y = \cos x \left(0 \le x \le \frac{\pi}{2}\right)$ in the first quadrant is divided by the curve $y = \sin x$, is :

(A) $\left(\sqrt{2} - 1\right) : 1$ (B) $\left(\sqrt{2} + 1\right) : 1$ (C) $\sqrt{2} : 1$ (D) none

प्रथम चतुर्थांश में वक्र $y = \cos x\left(0 \le x \le \frac{\pi}{2}\right)$ द्वारा परिबद्ध क्षेत्रफल को वक्र $y = \sin x$ जिस अनुपात में विभाजित करता है, वह

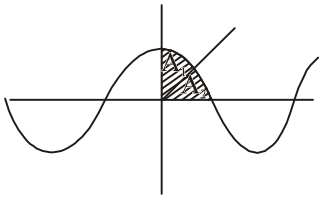
vuqikr gksxk:

(A) $\left(\sqrt{2}-1\right):1$ (B) $\left(\sqrt{2}+1\right):1$ (C) $\sqrt{2}:1$ (D) कोई नही

**Ans.** C

**Sol.** $y = \cos x$ $y = \sin x$

$$\frac{A_1}{A_2} = ?$$



$$A_1 = \int_0^{\frac{\pi}{4}} (\cos x - \sin x)$$

$$= \sin x + \cos x\Big|_0^{\frac{\pi}{4}}$$

$$= \sqrt{2} - 1$$

$$A_2 = \int_0^{\frac{\pi}{4}} \sin x dx + \int_0^{\frac{\pi}{2}} \cos x dx$$

$$= -\cos x\Big|_0^{\frac{\pi}{4}} + \sin x\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= -\frac{1}{\sqrt{2}} + 1 + 1 - \frac{1}{\sqrt{2}}$$

$$= 2 - \sqrt{2}$$

$$= \sqrt{2}\left(\sqrt{2}-1\right)$$

$$\frac{A_2}{A_1} = \frac{\sqrt{2}\left(\sqrt{2}-1\right)}{\sqrt{2}-1} = \frac{\sqrt{2}}{1}$$

19. The area bounded by the curve $y^2 + x^4 = x^2$ is :

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) none

वक्र $y^2 + x^4 = x^2$ से घिरा हुआ क्षेत्रफल होगा :

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) कोई नहीं

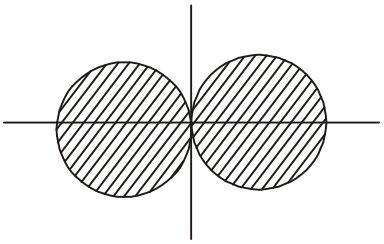
**Ans.** B

**Sol.** $y^2 + x^4 = x^2$

$y^2 = x^2 - x^4$

$y^2 = x^2 (1 - x^2)$

$y = \pm x\sqrt{1 - x^2}$



Symmetric about x-axis and y-axis both

$$A = 4\int_0^1 y\,dx$$

$$= 4\int_0^1 x\sqrt{1 - x^2}\,dx$$

$$= 2\cdot\left( -\frac{2}{3}\left(1 - x^2\right)^{\frac{3}{2}} \right)_0^1$$

$$= 2\left(\frac{2}{3}\right) = \frac{4}{3}$$

20. Area bounded by the curves $y = |x| - 2$ and $y = 1 - |x-1|$ is equal to :

(A) 4 sq. units (B) 6 sq. units (C) 2 sq. units (D) 8 sq. units

वक्रों $y = |x| - 2$ तथा $y = 1 - |x-1|$ के मध्य घिरा हुआ क्षेत्रफल होगा :

(A) 4 वर्ग ईकाई (B) 6 वर्ग ईकाई (C) 2 वर्ग ईकाई (D) 8 वर्ग ईकाई

**Ans.** A

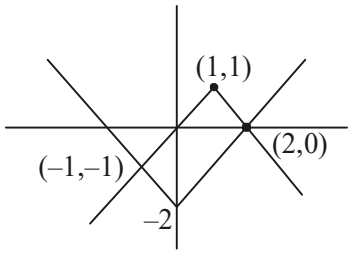
**Sol.** $y = |x| - 2$ $y = 1 - |x - 1|$

I $y = |x| - 2$

$$y = \begin{cases} x-2 & x \geq 0 \\ -x-2 & x < 0 \end{cases}$$

II $y = 1 - |x-1|$

$$y = \begin{cases} 2-x & x \geq 1 \\ x & x < 1 \end{cases}$$




$A = l \times b$

$= 2\sqrt{2} \times \sqrt{2} = 4$

21. Area bounded by the parabola $y = x^2 - 2x + 3$ and tangents drawn to it from the point P(1, 0) is equal to :

(A) $4\sqrt{2}$ sq. units (B) $\frac{4\sqrt{2}}{3}$ sq. units (C) $\frac{8\sqrt{2}}{3}$ sq. units (D) $\frac{16}{3}\sqrt{2}$ sq. units

परवलय $y = x^2 - 2x + 3$ तथा बिन्दु P(1, 0) से इस पर खींची गई स्पर्श रेखा के मध्य का क्षेत्रफल होगा :

(A) $4\sqrt{2}$ वर्ग ईकाई (B) $\frac{4\sqrt{2}}{3}$ वर्ग ईकाई (C) $\frac{8\sqrt{2}}{3}$ वर्ग ईकाई (D) $\frac{16}{3}\sqrt{2}$ वर्ग ईकाई

**Ans.** B

**Sol.** $y = x^2 - 2x + 3$ tangent from P (1, 0)

$y - 2 = (x-1)^2$

Tangent

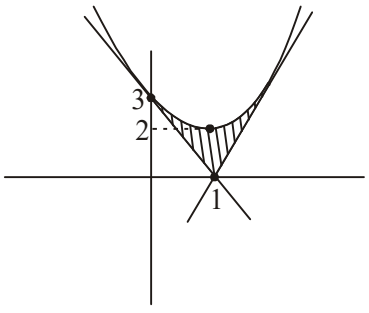
$$x = \frac{y-2}{m} + \frac{m}{4}$$

(1, 0)

$$= \frac{-2}{m} + \frac{m}{4}$$

$0 = -8 + m^2$

$m = \pm 2\sqrt{2}$




Tangents

$y = 2\sqrt{2}(x-1)$ & $y = -2\sqrt{2}(x-1)$

Point of intersection of $y - 2 = (x-1)^2$ & $y = \pm 2\sqrt{2}(x-1)$

$$y-2=\frac{y^2}{8}\Rightarrow y^2-8m+16=0$$

$$y=4$$

$$A=2\int_1^{1+\sqrt{2}}(y_1-y_2)\,dx$$

$$=2\int_1^{1+\sqrt{2}}\left[2+(x-1)^2-2\sqrt{2(x-1)}\right]dx$$

$$x-1=t$$

$$=2\int_0^{\sqrt{2}}\left(2+t^2-2\sqrt{2}t\right)dt$$

$$=2\left(2t+\frac{t^3}{3}-\sqrt{2}t^2\right)_0^{\sqrt{2}}$$

$$=2\left(2\sqrt{2}+\frac{2\sqrt{2}}{3}-2\sqrt{2}\right)$$

$$=\frac{4\sqrt{2}}{3}$$

22. Area bounded by the curves $y=\sin x$, tangent drawn to it at $x=0$ and the line $x=\frac{\pi}{2}$, is equal to :

(A) $\frac{\pi^2-4}{2}$ sq. units (B) $\frac{\pi^2-8}{8}$ sq. units (C) $\frac{\pi^2-2}{4}$ sq. units (D) $\frac{\pi^2-2}{2}$ sq. units

वक्रों $y=\sin x$, तथा इस पर $x=0$ पर स्पर्श रेखा तथा रेखा $x=\frac{\pi}{2}$ के मध्य का क्षेत्रफल होगा :

(A) $\frac{\pi^2-4}{2}$ वर्ग ईकाई (B) $\frac{\pi^2-8}{8}$ वर्ग ईकाई (C) $\frac{\pi^2-2}{4}$ वर्ग ईकाई (D) $\frac{\pi^2-2}{2}$ वर्ग ईकाई

**Ans.** B

**Sol.** $y=\sin x$ Tangent at $x=0$ & line $x=\frac{\pi}{2}$

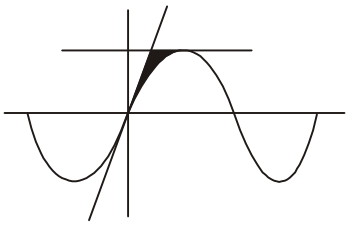
$$\frac{dy}{dx} = \cos x$$

$$m_T\left(x = 0\right) = 1$$

at $x = 0$ $\quad y = 0$

Tangent $y = mx$

$y = x$



$$A = \int_0^{\frac{\pi}{2}} \left(-\sin x + x\right) dx$$

$$= \left( +\cos x + \frac{x^2}{2} \right)\Bigg|_0^{\frac{\pi}{2}}$$

$$= \left( +\cos\frac{\pi}{2} + \frac{\pi^2}{8} \right) - (1)$$

$$= +\frac{\pi^2}{8} - 1 = \frac{\pi^2 - 8}{8}$$

23. Value of the parameter a such that the area bounded by $y = a^2x^2 + ax + 1$, coordinate axes and the line $x = 1$, attains it's least value, is equal to :

प्राचल 'a' के किस मान के लिए वक्र $y = a^2x^2 + ax + 1$, निर्देशांक अक्षों को $x = 1$ से घिरे हुये क्षेत्रफल का न्यूनतम मान होगा :

(A) $-\frac{1}{4}$ (B) $-\frac{1}{2}$ (C) $-\frac{3}{4}$ (D) $-1$

**Ans.** C

**Sol.** $y = a^2x^2 + ax + 1$ x-axis & y-axis line $x = 1$

$D = a^2 - 4(a^2)$

$= -3a^2 < 0$

$$A = \int_0^1 y\,dx$$

$$A = \int_0^1 \left(a^2x^2 + ax + 1\right)dx$$

$$A = \frac{a^2x^3}{3} + \frac{ax^2}{2} + x\Big|_0^1$$

$$A = \frac{a^2}{3} + \frac{a}{2} + 1$$

$$\frac{dA}{da} = \frac{2a}{3} + \frac{1}{2} = 0$$

$$a = -\frac{3}{4}$$

$$A = \frac{9}{16\times 3} - \frac{3}{8} + 1$$

$$= \frac{3}{16} - \frac{6}{16} + \frac{16}{16}$$

$$= \frac{13}{16}$$

24. Area enclosed by the curve $y = f(x)$ defined parametrically as $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$ is equal to :

(A) $\pi$ sq. units (B) $\frac{\pi}{2}$ sq. units (C) $\frac{3\pi}{4}$ sq. units (D) $\frac{3\pi}{2}$ sq. units

यदि वक्र $y = f(x)$ प्राचलिक रूप में $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$ प्रकार से परिभाषित है, तो इस से घिरा हुआ क्षेत्रफल होगा :

(A) $\pi$ वर्ग इकाई (B) $\frac{\pi}{2}$ वर्ग ईकाई (C) $\frac{3\pi}{4}$ वर्ग ईकाई (D) $\frac{3\pi}{2}$ वर्ग ईकाई

**Ans.** A

**Sol.** $y = f(x)$

$$x = \frac{1-t^2}{1+t^2} \quad \& \quad y = \frac{2t}{1+t^2}$$

Let $t = \tan\theta$

$$x = \frac{1-\tan^2\theta}{1+\tan^2\theta}$$

$$x = \cos 2\theta$$

and $$y = \frac{2\tan\theta}{1+\tan^2\theta}$$

$$y = \sin 2\theta$$

$x^2 + y^2 = 1$

Area = $\pi r^2$

$= \pi \times 1$

$= \pi$

25. The area of the region bounded by the curves $y = |x - 1|$ and $y = 3 - |x|$ is:

(A) 6 sq. units (B) 2 sq. units (C) 3 sq. units (D) 4 sq. units

वक्र $y = |x - 1|$ तथा $y = 3 - |x|$ के द्वारा परिबद्ध क्षेत्रफल है :

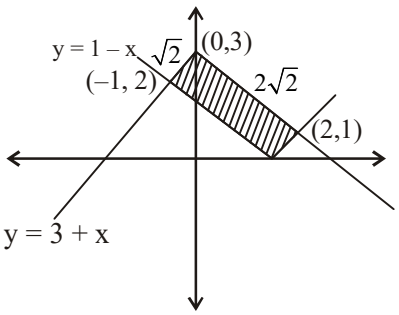
(A) 6 वर्ग इकाई (B) 2 वर्ग इकाई (C) 3 वर्ग इकाई (D) 4 वर्ग इकाई

**Ans.** D

**Sol.** $y = |x - 1|$ $y = 3 - |x|$

$$y = \begin{cases} x-1 & x \geq 1 \\ 1-x & x < 1 \end{cases}$$

$$y = \begin{cases} 3-x & x > 0 \\ 3+x & x \leq 0 \end{cases}$$




Area = $\sqrt{2} \times 2\sqrt{2}$

$= 4$

26. The area enclosed between the curve $y = \log_e(x + e)$ and the coordinate axes is :

उस क्षेत्रफल, जो कि वक्र $y = \log_e(x + e)$ और अक्षों के बीच में बना है, उसका मान होगा :

(A) 1 (B) 2 (C) 3 (D) 4

**Ans.** A

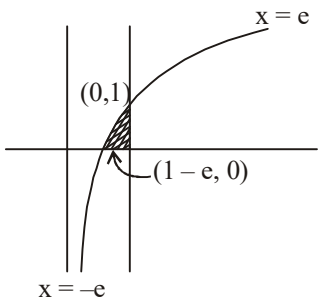
**Sol.** $y = \log_e (x + e)$

$$A = \int_{1-e}^{0} \log_e (x+e)\,dx$$

$$= (x+e)\log_e \left(\frac{x+e}{e}\right)_{1-e}^{0}$$

$= (1 - e + e)\,(1)$

$= 1$




27. The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes. If $S_1$, $S_2$, $S_3$ are respectively the areas of these parts numbered from top to bottom; then $S_1 : S_2 : S_3$ is :

परवलय $y^2 = 4x$ तथा $x^2 = 4y$, रेखाओं $x = 4$, $y = 4$ तथा निर्देशांक अक्षों से घिरे वर्गाकार क्षेत्र को बांटते हैं। यदि ऊपर से नीचे तक r hu [ k Mkad sØe' k/kksa ds Øe esa $S_1$, $S_2$ तथा $S_3$ हैं, तो $S_1 : S_2 : S_3$ का मान है :

(A) 1 : 2 : 1 (B) 1 : 2 : 3 (C) 2 : 1 : 2 (D) 1 : 1 : 1

**Ans.** D

**Sol.** $y^2 = 4x$ $x^2 = 4y$

$$S_1 = \int_0^4 x\,dy$$

$$= \int_0^4 \frac{y^2}{4}\,dy$$

$$= \frac{y^3}{3x^4}\Big|_0^4$$

$$= \frac{64}{3\times 4} = \frac{16}{3}$$

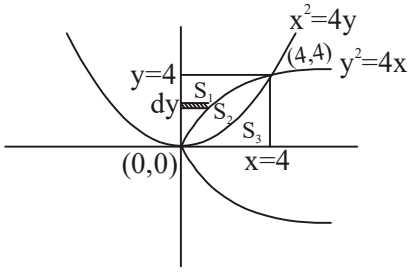
from symmetry

$$S_2 = \frac{16}{3}$$

$S_2 = 16 - S_1 - S_3$

$$= 16 - \frac{16}{3} - \frac{16}{3}$$

$$= 16 - \frac{32}{3}$$



$= \frac{16}{3}$

$S_1 : S_2 : S_3 = 1 : 1 : 1$

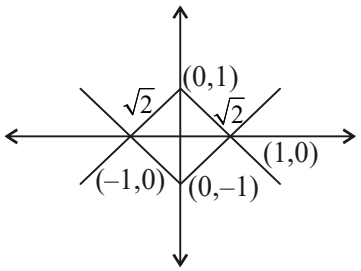
28. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is :

वक्रों $y = |x| - 1$ और $y = -|x| + 1$ से परिबद्ध क्षेत्रफल है :

(A) 1 (B) 2 (C) $2\sqrt{2}$ (D) 4

**Ans.** B

**Sol.** $y = |x| - 1$ $y = -|x| + 1$




$$y = \begin{cases} x - 1 & x > 0 \\ x - 1 & x \le 0 \end{cases}$$

$A = \sqrt{2} \times \sqrt{2} = 2$

29. If the area bounded by $y = ax^2$ and $x = ay^2$ , $a > 0$, is 1, then a =

यदि $y = ax^2$ और $x = ay^2$ , $a > 0$ से परिबद्ध क्षेत्रफल 1 है, तब a =

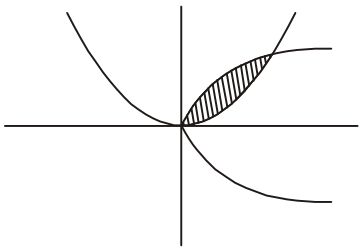
(A) 1 (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{\sqrt{3}}$

**Ans.** B

**Sol.** $y = ax^2$ $x = ay^2$

$x^2 = \frac{y}{a}$ $y^2 = \frac{x}{a}$

Enclosed area $= \frac{16}{3}\left(\frac{1}{4a} \times \frac{1}{4a}\right)$



$A = \frac{1}{3a^2}$

$1 = \frac{1}{3a^2}$

$a = \pm\frac{1}{\sqrt{3}}$

$\because a > 0$ $a = \frac{1}{\sqrt{3}}$

30. The area bounded by the parabolas $y = (x + 1)^2$ and $y = (x - 1)^2$ and the line $y = 1/4$ is :

(A) 4 sq. units (B) $\frac{1}{6}$ sq. units (C) $\frac{4}{3}$ sq. units (D) $\frac{1}{3}$ sq. units

परवलयों $y = (x + 1)^2$ और $y = (x - 1)^2$ और रेखा $y = 1/4$ से परिबद्ध क्षेत्रफल है :

(A) 4 वर्ग इकाई (B) $\frac{1}{6}$ वर्ग इकाई (C) $\frac{4}{3}$ वर्ग इकाई (D) $\frac{1}{3}$ वर्ग इकाई

**Ans.** D

**Sol.** $y = (x + 1)^2$

$y = (x - 1)^2$

$$y = \frac{1}{4}$$

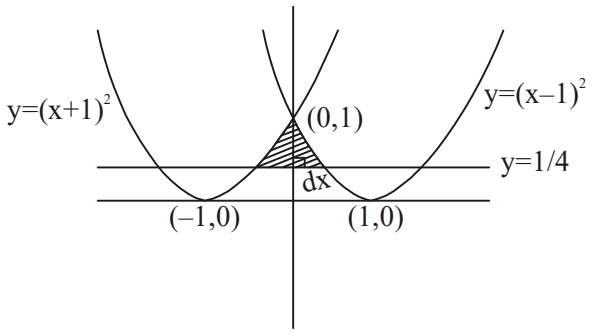
$$A = \int_{\frac{1}{4}}^{1} \left(-\sqrt{y} + 1\right) dy$$

$$\because y = (x-1)^2$$

$$x = \begin{cases} -\sqrt{y} + 1 & \text{for} \quad x < 1 \\ \sqrt{y} + 1 & \text{for} \quad x \geq 1 \end{cases}$$

$$= 2\int_{\frac{1}{4}}^{1} \left(-\sqrt{y} + 1\right) dy$$

$$= 2\left( -\frac{2}{3} y^{\frac{3}{2}+y} \right)_{\frac{1}{4}}^{1}$$




$$= 2\left[ \left( \frac{-2}{3} + 1 \right) - \left( -\frac{2}{3} \times \frac{1}{8} + \frac{1}{4} \right) \right]$$

$$= 2\left[ \frac{1}{3} - \left( -\frac{1}{12} + \frac{1}{4} \right) \right]$$

$$= 2\left[ \frac{1}{3} - \left( \frac{1}{6} \right) \right]$$

$$= 2\left( \frac{1}{6} \right) = \frac{1}{3}$$

31. The area of the region between the curves $y = \sqrt{\frac{1+\sin x}{\cos x}}$ and $y = \sqrt{\frac{1-\sin x}{\cos x}}$ bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$ is :

वक्रों $y = \sqrt{\frac{1+\sin x}{\cos x}}$ और $y = \sqrt{\frac{1-\sin x}{\cos x}}$ तथा रेखाओं $x = 0$ और $x = \frac{\pi}{4}$ से परिबद्ध क्षेत्र का क्षेत्रफल है :

(A) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}}dt$ (B) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

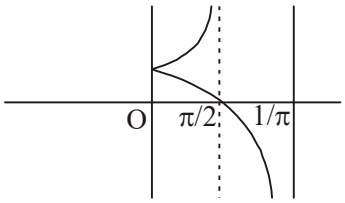
(C) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$ (D) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}}dt$

**Ans.** B

**Sol.** Area $= \int_0^{\pi/4} \left( \sqrt{\frac{1+\sin x}{\cos x}} - \sqrt{\frac{1-\sin x}{\cos x}} \right) dx$

$$= \int_0^{\pi/4} \frac{\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)-\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)}{\sqrt{\cos^2\frac{x}{2}-\sin^2\frac{x}{2}}}dx$$

$$= \int_0^{\pi/4} \frac{2\sin\frac{x}{2}}{\sqrt{\cos^2\frac{x}{2}-\sin^2\frac{x}{2}}}dx = \int_0^{\pi/4} \frac{2\tan\frac{x}{2}}{\sqrt{1-\tan^2\frac{x}{2}}}dx$$




Let $\tan\frac{x}{2} = t$

$$\sec^2\frac{x}{2}dx = 2dt \Rightarrow dx = \frac{2dt}{(1+t^2)}$$

$\therefore$ Area $= \int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

32. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is :

वक्र $y^2 = x$ तथा $y = |x|$ द्वारा परिबद्ध क्षेत्रफल है :

(A) $\frac{2}{3}$ (B) 1 (C) $\frac{1}{6}$ (D) $\frac{1}{3}$

**Ans.** C

**Sol.** $y^2 = x$ $y = |x|$

$$A = \int_0^1 \left(y - y^2\right) dy$$

$$= \frac{y^2}{2} - \frac{y^3}{3}\Big|_0^1$$

$$= \frac{1}{2} - \frac{1}{3}$$

$$= \frac{1}{6}$$

33. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to :

वक्रों $x + 2y^2 = 0$ तथा $x + 3y^2 = 1$ के बीच घिरे तलीय क्षेत्र का क्षेत्रफल है :

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$

**Ans.** C

**Sol.** $y^2 = -\frac{x}{2}$ $y^2 = -\frac{1}{3}(x-1)$

$x = -2y^2$ $x = 1 - 3y^2$

POI

$$y^2 = -\frac{x}{2} = -\frac{1}{3}(x-1)$$

$3x = 2x - 2$

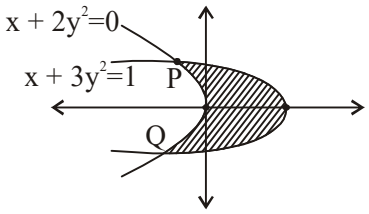
$x = -2$

$y^2 = 1 \Rightarrow y = \pm 1$

P (–2,1)

Q (–2,–1)

$$I = \int_{y_1}^{y_2} (x_2 - x_1)\,dy$$




$$= 2\int_0^1 \left((1-3y^2)-(-2y^2)\right)dy$$

$$= 2\int_0^1 (1-y^2)\,dy$$

$$= 2\left(y - \frac{y^3}{3}\right)_0^1$$

$$= \frac{4}{3}$$

34. The area of the region bounded by the parabola $(y-2)^2 = x-1$, the tangent to the parabola at the point (2, 3) and the x – axis is : **[AIEEE2009]**

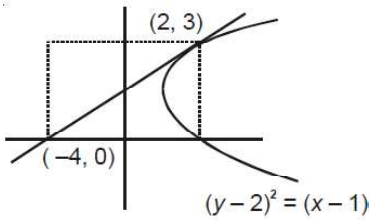
परवलय $(y-2)^2 = x-1$, परवलय के बिन्दु (2, 3) पर खींची गई स्पर्श रेखा तथा x – अक्ष द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है : **[AIEEE2009]**

(A) 3 (B) 6 (C) 9 (D) 12

**Ans.** C

**Sol.** The equation of tangent at (2, 3) to the given parabola is $x = 2y - 4$

Required area $= \int_0^3 \left\{(y-2)^2 + 1 - 2y + 4\right\}dy$




$$= \left[\frac{(y-2)^3}{3} - y^2 + 5y\right]_0^3$$

$$= \frac{1}{3} - 9 + 15 + \frac{8}{3}$$

$= 9$ sq. units

35. The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$ x-axis and lying in the first quadrant is: **[JEE(Main)2013]**

वक्रों $y = \sqrt{x}$, $2y - x + 3 = 0$ तथा x- अक्ष से घिरे उस क्षेत्र, जो प्रथम चतुर्थांश में स्थित है, का (वर्ग इकाई में) क्षेत्रफल है : **[JEE(Main)2013]**

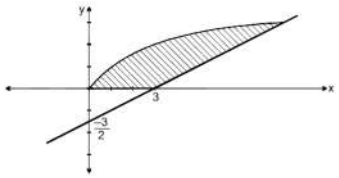
(A) 9 (B) 36 (C) 18 (D) 27/4

**Ans.** A

**Sol.** $y = \sqrt{x}$ ...(1)

and $2y - x + 3 = 0$ ...(2)

On solving both $y = -1, 3$



$$\text{Required area} = \int_0^3 \left\{(2y+3) - y^2\right\} dy$$

$$= y^2 + 3y - \frac{y^3}{3}\Big|_0^3$$

$= 9 + 9 - 9$

$= 9$

36. Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, and $\alpha, \beta$ $(\alpha < \beta)$ be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. unit) bounded by the curve $y = (gof)(x)$ and the lines $x = \alpha$, $x = \beta$ and $y = 0$ is :

**[JEE(Main)2018]**

माना $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, तथा $\alpha, \beta$ $(\alpha < \beta)$ द्विघाती समीकरण $18x^2 - 9\pi x + \pi^2 = 0$ के मूल हैं। तो वक्र $y=(gof)(x)$ तथा j§ kv lsa $x = \alpha$, $x = \beta$ तथा $y = 0$ द्वारा घिरे क्षेत्र का क्षेत्रफल (वर्ग इकाइयों में) है : **[JEE(Main)2018]**

(A) $\frac{1}{2}\left(\sqrt{3} - \sqrt{2}\right)$ (B) $\frac{1}{2}\left(\sqrt{2} - 1\right)$ (C) $\frac{1}{2}\left(\sqrt{3} - 1\right)$ (D) $\frac{1}{2}\left(\sqrt{3} + 1\right)$

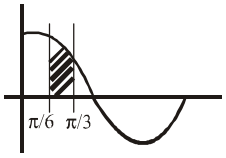
**Ans.** C

**Sol.** $gof(x) = \cos x$

$18x^2 - \pi x + \pi^2 = 0$

$(6x - \pi)(3x - \pi) = 0$

$x = \frac{\pi}{6}, x = \frac{\pi}{3}$




$$A = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos x \, dx = \frac{\sqrt{3} - 1}{2}$$

37. Let the straight line $x = b$ divide the area enclosed by $y = (1 - x^2)$, $y = 0$ and $x = 0$ into two parts $R_1 (0 \le x \le b)$ and $R_2 (b \le x \le 1)$ such that $R_1 - R_2 = \frac{1}{4}$. Then b equals : **[JEE-2011]**

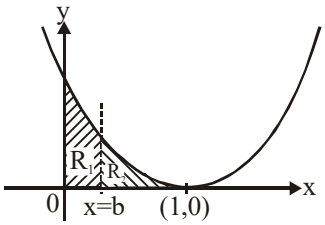
यदि रेखा x = b वक्रों $y = (1 - x^2)$, y = 0 तथा x = 0 को दो भागों में $R_1(0 \le x \le b)$ तथा $R_2(b \le x \le 1)$ में इस प्रकार विभाजित

d j r kgSd $R_1 - R_2 = \frac{1}{4}$ तो b का मान होगा : **[JEE-2011]**

(A) 3/4 (B) 1/2 (C) 1/3 (D) 1/4

**Ans.** B

**Sol.** $\because \quad R_1 - R_2 = \frac{1}{4}$




$$\Rightarrow \int_0^b (1-x)^2 dx - \int_b^1 (1-x)^2 dx = \frac{1}{4}$$

$$\Rightarrow -\left(\frac{(1-x)^3}{3}\right)_0^b + \left(\frac{(1-x)^3}{3}\right)_b^1 = \frac{1}{4}$$

$$\Rightarrow -\left\{\frac{(1-b)^3}{3} - \frac{1}{3}\right\} - \frac{(1-b)^3}{3} = \frac{1}{4}$$

$$\Rightarrow \frac{1}{3} - \frac{2}{3}(1-b)^3 = \frac{1}{4} \Rightarrow \frac{2}{3}(1-b)^3 = \frac{1}{12}$$

$$\Rightarrow (1-b)^3 = \frac{1}{8} \Rightarrow 1 - b = \frac{1}{2}$$

$$\Rightarrow b = \frac{1}{2}$$

**SHIFTING OF ORIGIN BASED QUESTIONS**

38. The line y = mx bisects the area enclosed by the curve $y = 1 + 4x - x^2$ and the lines x = 0, x = 2 and y = 0. Then the value of m is :

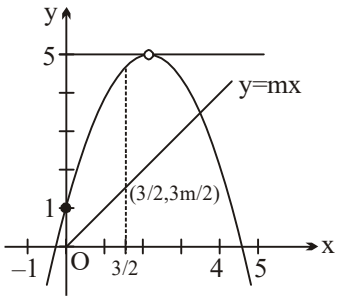
(A) $\frac{13}{6}$ (B) $\frac{6}{13}$ (C) $\frac{11}{6}$ (D) none

यदि रेखा y = mx, वक्र $y = 1 + 4x - x^2$ तथा रेखाऐं x = 0, x = 2 तथा y = 0 द्वारा परिबद्ध क्षेत्रफल को समद्विभाजित करती है, तो m का मान होगा :

(A) $\frac{13}{6}$ (B) $\frac{6}{13}$ (C) $\frac{11}{6}$ (D) कोई नही

**Ans.** A

**Sol.** $2 \cdot \frac{1}{2}\left(\frac{3}{2}\cdot\frac{3m}{2}\right) = \int_0^{3/2}(1+4x-x^2)\,dx$




$$\frac{9m}{4} = x+2x^2-\frac{x^3}{3}\Big]_0^{3/2} = \frac{3}{2}+\frac{9}{2}-\frac{27}{8}\cdot\frac{1}{3} = 6-\frac{9}{8} = \frac{39}{8}$$

$$m = \frac{13}{6}$$

39. Area enclosed by the curve $|x-2|+|y+1|=1$ is equal to :

(A) 4 sq. units (B) 6 sq. units (C) 2 sq. units (D) 8 sq. units

वक्र $|x-2|+|y+1|=1$ से घिरा हुआ क्षेत्रफल होगा :

(A) 4 वर्ग ईकाई (B) 6 वर्ग ईकाई (C) 2 वर्ग ईकाई (D) 8 वर्ग ईकाई

**Ans.** C

**Sol.** $|x-2|+|y-1|=1$

$(0,0) \to (2,1)$

$|x|+|y|=1$

Area = $\sqrt{2}\times\sqrt{2}$

$= 2$

**AREA REPRESENTED BY INEQUALITIES**

40. The area of the region of the plane bounded by $\max(|x|,|y|) \le 1$ and $xy \le \frac{1}{2}$ is :

$\max(|x|,|y|) \le 1$ तथा $xy \le \frac{1}{2}$ द्वारा निरूपित क्षेत्र का क्षेत्रफल होगा :

(A) $\frac{1}{2}+\ln 2$ (B) $\frac{31}{4}$ (C) $1+2\ln 2$ (D) $3+\ln 2$

**Ans.** D

**Sol.** $\max(|x|,|y|) \le 1$ $\qquad xy \le \frac{1}{2}$

$$\text{Area} = 2\times 2 - 2\int_{\frac{1}{2}}^{1}\left(1-\frac{1}{2}x\right)dx$$

$$= 4 - 2\left(x-\frac{1}{2}\ell nx\right)_{\frac{1}{2}}^{1}$$

$$= 4 - 2 + 2\left(\frac{1}{2}-\frac{1}{2}\ell n\frac{1}{2}\right)$$

$$= 2 + 1 - \ell n\frac{1}{2}$$

$$= 3 + \ell n\, 2$$

41. Area of the region which consists of all the points satisfying the conditions $|x-y|+|x+y|\le 8$ and $xy \ge 2$, is equal to:

(A) $4(7 - l n8)$ sq. units (B) $4\,(9 - l n8)$ sq. units

(C) $2(7 - l n8)$ sq. units (D) $2\,(9 - l n\,8)$ sq. units

प्रतिबंधों $|x - y| + |x + y| \le 8$ तथा $xy \ge 2$ को संतुष्ट करने वाले सभी बिन्दु से बना हुआ कुल क्षेत्रफल होगा :

(A) 4;7 $l n8)$ वर्ग इकाई (B) 4 ;9 $l n8)$ वर्ग इकाई

(C) 2;7 $l n8)$ वर्ग इकाई (D) 2 ;9 $l n\,8)$ वर्ग इकाई

**Ans.** A

**Sol.** $|x-y|+|x+y| \le 8$ $xy \ge 2$

$$\text{Required Area} = 2\times\int_{\frac{1}{2}}^{4}\left(4-\frac{2}{x}\right)dx$$

$$= 2\left(4x - 2\ell nx\right)_{\frac{1}{2}}^{4}$$

$$= 4\left(8 - \ell n4 - 1 + \ell n\left(\frac{1}{2}\right)\right)$$

$$= 4(7 - \ell n8)$$

42. A point P moves in xy plane in such a way that $[|x|] + [|y|] = 1$, where [.] denotes the greatest integer function. Area of the region representing all possible positions of the point P is equal to :

(A) 4 sq. units (B) 16 sq. units (C) $2\sqrt{2}$ sq. units (D) 8 sq. units

एक बिन्दु P xy तल में इस प्रकार गति करता है कि [|x|] + [|y|] = 1, जहाँ [.] महत्म पूर्णांक फलन है, तो P के सभी सम्भव मानों से घिरा gq k{ksk Qy gkxk:

(A) 4 वर्ग ईकाई (B) 16 वर्ग ईकाई (C) $2\sqrt{2}$ वर्ग ईकाई (D) 8 वर्ग ईकाई

**Ans.** D

**Sol.** $\left[|x|\right]+\left[|y|\right]=1$

1st Quadrant

[x] + [y] = 1

2nd Quadrant

[–x] + [y] = 1

3rd Quadrant

[–x] + [–y] = 1

4th Quadrant

[x] + [–y] = 1

Area = 2 × 2 = 4

43. The area defined by $|y| \le e^{-|x|} - \frac{1}{2}$ in cartesian co-ordinate system, is :

कार्तिय–निर्देशांक निकाय में $|y| \le e^{-|x|} - \frac{1}{2}$ से परिभाषित क्षेत्रफल होगा :

(A) (4 – 2*l*n 2) (B) (4 – *l*n 2) (C) (2 – *l*n 2) (D) (2 – 2 *l*n 2)

**Ans.** D

**Sol.** For $y \ge 0$,

$$\therefore\ y = e^{-|x|} - \frac{1}{2} = \begin{cases} e^{-x} - \frac{1}{2} \text{ if } x \ge 0 \\ e^{x} - \frac{1}{2} \text{ if } x < 0 \end{cases} \quad \text{.....(1)}$$

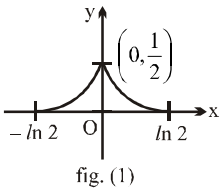
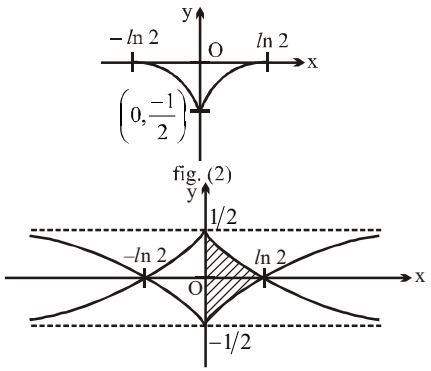


fig. (1)

and for $y < 0$,

$$-y = \frac{1}{2} - e^{-|x|} = \begin{cases} \frac{1}{2} - e^{-x} & \text{if } x \geq 0 \\ \frac{1}{2} - e^{x} & \text{if } x < 0 \end{cases} \quad \text{.....(2)}$$




Combining (1) and (2) we have the required area given as

$$A = 4\int_0^{\ln 2}\left(e^{-x} - \frac{1}{2}\right)dx = 4\left(\frac{1}{2} - \frac{1}{2}\ln 2\right) = (2 - 2\ln 2)$$

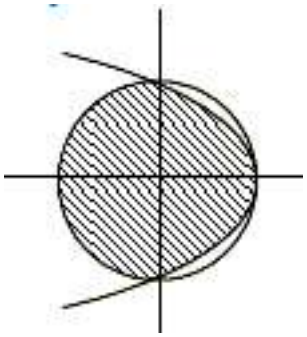
44. The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$ is : **[JEE(Main)2014]**

$A = \{(x, y) : x^2 + y^2 \leq 1$ तथा $y^2 \leq 1 - x\}$ के द्वारा प्रदत्त क्षेत्र का क्षेत्रफल है : **[JEE(Main)2014]**

(A) $\frac{\pi}{2} - \frac{2}{3}$ (B) $\frac{\pi}{2} + \frac{2}{3}$ (C) $\frac{\pi}{2} + \frac{4}{3}$ (D) $\frac{\pi}{2} - \frac{4}{3}$

**Ans.** C

**Sol.**



Required area $= \pi - 2\int_0^1\left(\sqrt{1-x^2} - \sqrt{1-x}\right)dx$

$$= \pi - 2\left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2}\sin^{-1}x + \frac{2}{3}(1-x)^{3/2}\right]_0^1 = \pi - 2\left[\frac{\pi}{4} - \frac{2}{3}\right] = \frac{\pi}{2} + \frac{4}{3}$$

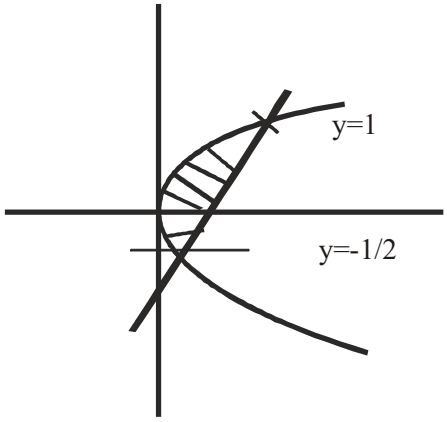
45. The area (in sq. units) of the region described by $[(x, y) : y^2 \leq 2x \text{ and } y \geq 4x - 1]$ is : **[JEE(Main)2015]**

$[(x, y) : y^2 \leq 2x$ तथा $y \geq 4x–1]$ द्वारा परिभाषित क्षेत्र का क्षेत्रफल (वर्ग इकाईयों) में हैं : **[JEE(Main)2015]**

(A) $\frac{1}{32}$ (B) $\frac{5}{64}$ (C) $\frac{15}{64}$ (D) $\frac{9}{32}$

**Ans.** D

**Sol.** $y^2 \leq 2x$ $y \geq 4x - 1$




$$\text{Req Area} = \int_{-\frac{1}{2}}^{1}\left(\frac{y+1}{4}-\frac{y^2}{2}\right)dy$$

$$=\frac{9}{32}$$

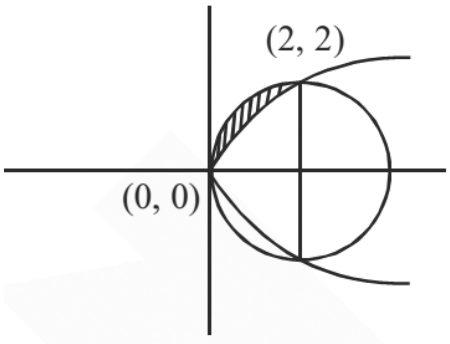
46. The area (in sq. units) of the region $\{(x, y\} : y^2 \geq 2x$ and $x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is: **[JEE(Main)2016]**

क्षेत्र $\{(x, y\} : y^2 \geq 2x$ तथा $x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ का क्षेत्रफल (वर्ग इकाईयों में) है : **[JEE(Main)2016]**

(A) $\pi - \frac{4\sqrt{2}}{3}$ (B) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$ (C) $\pi - \frac{4}{3}$ (D) $\pi - \frac{8}{3}$

**Ans.** D

**Sol.**




$$\text{Area} = \frac{1}{4}(\text{Area of circle}) - \int_0^2 \sqrt{2x}\,dx$$

$$= \pi - \frac{8}{3}$$

47. The area (in sq. units) of the region $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y$ and $y \leq 1 + \sqrt{x}\}$ is:

**[JEE(Main)2017]**

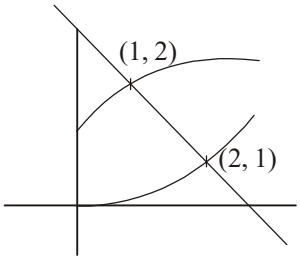
क्षेत्र $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y$ तथा $y \leq 1 + \sqrt{x}\}$ का क्षेत्रफल होगा : **[JEE(Main)2017]**

(A) $\frac{7}{3}$ (B) $\frac{5}{2}$ (C) $\frac{59}{12}$ (D) $\frac{3}{2}$

**Ans.** B

**Sol.** $\text{Area} = \int_0^1 \left\{\left(1+\sqrt{x}\right) - \frac{x^2}{4}\right\} dx + \int_1^2 \left\{\left(3-x\right) - \frac{x^2}{4}\right\} dx$

$= \frac{5}{2}$




**DETERMINATION OF UNKNOWN PARAMETERS**

48. If A is the area between the curve y = sin x and x-axis in the interval [0, $\pi/2$], then the area between y = sin 2x and x- axis in this interval will be :

(A) A (B) 2A (C) A/2 (D) None of these

यदि वक्र y = sin x तथा x-अक्ष के मध्य अन्तराल [0, $\pi/2$] में क्षेत्रफल A हो, तो y = sin 2x तथा x-अक्ष के मध्य इसी अन्तराल में {ksk Qy gksxk:

(A) A (B) 2A (C) A/2 (D) इनमें से कोई नहीं

**Ans.** A

**Sol.** $y = \sin x$

$$A = \int_0^{\frac{\pi}{2}} \sin x dx$$

$$= -\cos x\Big|_0^{\frac{\pi}{2}}$$

$= 1$

$$A' = \int_0^{\frac{\pi}{2}} \sin 2x dx$$

$$= -\frac{\cos 2x}{2}\Big|_0^{\frac{\pi}{2}}$$

$$= -\frac{1}{2}[-1-1] = 1 = A$$

49. If (a, 0); a > 0 is the point where the curve $y = \sin 2x - \sqrt{3}\sin x$ cuts the x-axis first, A is the area bounded by this part of the curve , the origin and the positive x-axis, then :

यदि (a, 0); a > 0 वह बिन्दु है जहाँ वक्र $y = \sin 2x - \sqrt{3}\sin x$, x-अक्ष को प्रथम बार काटता है तथा वक्र के इस भाग, मूलबिन्दु तथा /kukRed x-अक्ष द्वारा घिरा क्षेत्रफल A है, तब :

(A) 4A + 8 cosa = 7 (B) 4A + 8 sina = 7 (C) 4A – 8 sina = 7 (D) 4A – 8 cosa = 7

**Ans.** A

**Sol.** (a, 0) lies on the given curve

$\therefore \quad 0 = \sin 2a - \sqrt{3}\sin a \quad \Rightarrow \quad \sin a = 0 \text{ or } \cos a = \sqrt{3}/2$

$\Rightarrow \quad a = \frac{\pi}{6}$ (as a > 0 and the first point of intersection with positive X-axis)

and $A = \int_0^{\pi/6} (\sin 2x - \sqrt{3}\sin x)\,dx = \left(-\frac{\cos 2x}{2} + \sqrt{3}\cos x\right)_0^{\pi/6}$

$$= \left(-\frac{1}{4} + \frac{3}{2}\right) - \left(-\frac{1}{2} + \sqrt{3}\right) = \frac{7}{4} - \sqrt{3} = \frac{7}{4} - 2\cos a \quad \Rightarrow \quad 4A + 8\cos a = 7$$

**AREA BOUNDED BY INVERSE OF A FUNCTION**

50. Area of the region enclosed between the curves $x = y^2 - 1$ and $x = |y|\sqrt{1-y^2}$ is :

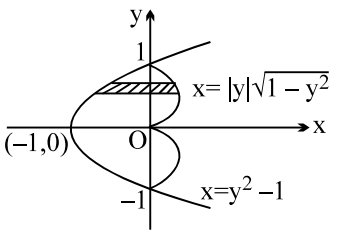
वक्रों $x = y^2 - 1$ तथा $x = |y|\sqrt{1-y^2}$ द्वारा परिबद्ध क्षेत्रफल होगा :

(A) 1 (B) 4/3 (C) 2/3 (D) 2

**Ans.** D

**Sol.** $A = 2\int_0^1 \left[y\sqrt{1-y^2} - (y^2-1)\right] dy$

$= 2$



**DETERMINATION OF FUNCTION**

51. The area bounded by the curve $y = f(x)$, the x-axis & the ordinates $x = 1$ & $x = b$ is $(b-1)\sin(3b+4)$. Then $f(x)$ is :

(A) $(x-1)\cos(3x+4)$ (B) $\sin(3x+4)$

(C) $\sin(3x+4)+3(x-1).\cos(3x+4)$ (D) none

यदि वक्र $y = f(x)$, x-अक्ष तथा कोटि $x = 1$ तथा $x = b$ द्वारा परिबद्ध क्षेत्रफल $(b-1)\sin(3b+4)$ है, तो $f(x)$ होगा :

(A) $(x-1)\cos(3x+4)$ (B) $\sin(3x+4)$

(C) $\sin(3x+4)+3(x-1).\cos(3x+4)$ (D) कोई नहीं

**Ans.** C

**Sol.** $\int_1^b f(x)\,dx = (b-1)\sin(3b+4)$

Apply Leibnitz

$f(b) = (b-1)\cos(3b+4)3 + \sin(3b+4)$

52. If the area bounded by the x-axis, curve $y = f(x)$ and the lines $x = 1$, $x = b$ is equal to $\sqrt{b^2+1} - \sqrt{2}$ for all $b > 1$, then $f(x)$ is :

यदि x-अक्ष, वक्र $y = f(x)$ तथा रेखाओं $x = 1$, $x = b$ के द्वारा परिबद्ध क्षेत्रफल $\sqrt{b^2+1} - \sqrt{2}$ ; सभी $b > 1$ के लिऐ हो, तो $f(x)$ होगा :

(A) $\sqrt{(x-1)}$ (B) $\sqrt{(x+1)}$ (C) $\sqrt{(x^2+1)}$ (D) $\frac{x}{\sqrt{1+x^2}}$

**Ans.** D

**Sol.** $\int_1^b f(x)\,dx = \sqrt{b^2+1} - \sqrt{2}$

$$f(b) = \frac{2b}{2\sqrt{b^2+1}} = \frac{b}{\sqrt{b^2+1}}$$

53. Let $f(x)$ be a non-negative continuous function such that the area bounded by the curve $y = f(x)$, x-axis and the ordinates $x = \frac{\pi}{4}$ and $x = \beta > \frac{\pi}{4}$ is $\left(\beta\sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta\right)$. Then $f\left(\frac{\pi}{2}\right)$ is :

माना f (x) एक इस प्रकार का अऋणात्मक सतत् फलन है, कि वक्र y = f (x), x-अक्ष तथा रेखाओं $x = \frac{\pi}{4}$ तथा $x = \beta > \frac{\pi}{4}$ से आबद्ध क्षेत्र का क्षेत्रफल $\left(\beta\sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta\right)$ है, तो $f\left(\frac{\pi}{2}\right)$ का मान है :

(A) $\left(\frac{\pi}{4} + \sqrt{2} - 1\right)$ (B) $\left(\frac{\pi}{4} - \sqrt{2} + 1\right)$ (C) $\left(1 - \frac{\pi}{4} - \sqrt{2}\right)$ (D) $\left(1 - \frac{\pi}{4} + \sqrt{2}\right)$

**Ans.** D

**Sol.** $\int_{\frac{\pi}{4}}^{\beta} f(x)\,dx = \beta\sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta$

$f(\beta) = \beta\cos\beta + \sin\beta - \frac{\pi}{4}\sin\beta + \sqrt{2}$

$f\left(\frac{\pi}{2}\right) = 0 + 1 - \frac{\pi}{4} + \sqrt{2}$

# Exercise J-II

## SUBJECTIVE / INTEGER TYPE QUESTIONS

### AREA BOUNDED BETWEEN TWO CURVES

1. Find the area bounded on the right by the line $x+y=2$, on the left by the parabola $y=x^2$ and below by the x-axis in first quadrant.

   क्षेत्रफल ज्ञात कीजिए, जो दांयी ओर से सरल रेखा $x+y=2$ से, बांयी ओर से परवलय $y=x^2$ से, नीचे की तरफ से x-अक्ष से परिबद्ध हो (प्रथम चतुर्थांश में)।

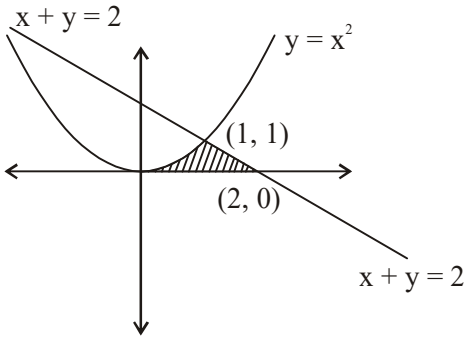
**Ans.** 5/6 sq. units

**Sol.** Line $x + y = 2$ and $y = x^2$

POI $x + y = 2$

$y = x^2$

(0, 0) and (1, 1)




$$\text{Area} = \int_0^1 y_1 dx + \int_1^0 y_2 dx$$

$$A = \int_0^1 x^2 dx + \int_1^2 (2-x) dx$$

$$= \frac{x^3}{3}\Big|_0^1 + 2x - \frac{x^2}{2}\Big|_1^2$$

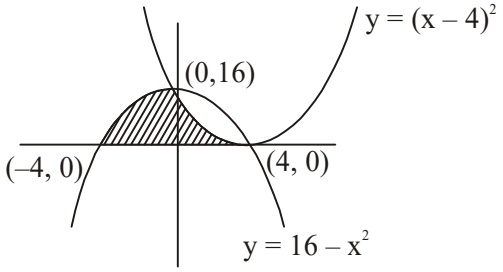
$$= \frac{1}{3} + 4 - 2 - 2 + \frac{1}{2} = \frac{5}{6}$$

2. Find the area of the region bounded by curves $f(x) = (x-4)^2$, $g(x) = 16 - x^2$ and the x - axis.

   वक्रों $f(x) = (x-4)^2$, $g(x) = 16 - x^2$ और x-अक्ष से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

**Ans.** 64

**Sol.** $f(x) = (x-4)^2$ $g(x) = 16 - x^2$ x–axis




$$A = \int_{-4}^{0} \left(16 - x^2\right) dx + \int_{0}^{4} \left(x - 4\right)^2 dx$$

$$= 16x - \frac{x^3}{3}\Bigg|_{-4}^{0} + \frac{\left(x-4\right)^3}{3}\Bigg|_{0}^{4}$$

$$= 64 + \frac{16}{3} - \frac{16}{3} = 64$$

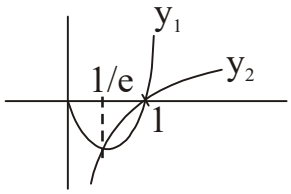
3. Compute the area of the region bounded by the curves y = e. x. $ln\,x$ & y = $ln\,x$/(e.$x$) where $ln$ e = 1.

वक्रों y = e. x. $ln\,x$ और y = $ln\,x$/(e.$x$) जहाँ $ln$ e = 1 से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

**Ans.** $(e^2 - 5)/4$ e sq. units

**Sol.** $y_1 = e \cdot x \cdot \ell nx$

$$y_2 = \frac{\ell nx}{ex}$$




$$A = \int_{\frac{1}{e}}^{1} \left( \frac{\ell nx}{ex} - ex\ell nx \right) dx$$

$$= \frac{1}{e} \frac{\left(\ell nx\right)^2}{2}\Bigg|_{\frac{1}{e}}^{1} - e\left[ \frac{x^2}{2}\ell nx - \frac{x^2}{4} \right]_{\frac{1}{e}}^{1}$$

$$= \frac{1}{e}\left(-\frac{1}{2}\right) - e\left[-\frac{1}{4} + \frac{1}{2e^2} + \frac{1}{4e^2}\right]$$

$$= -\frac{1}{2e} + \frac{e}{4} - \frac{3}{4e}$$

$$= \frac{e}{4} - \frac{5}{4e}$$

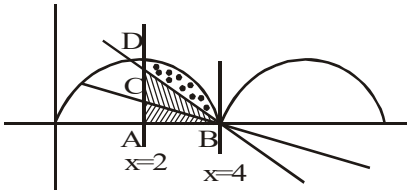
$$= \frac{e^2 - 5}{4e}$$

4. A figure is bounded by the curves $y = \left|\sqrt{2}\sin\frac{\pi x}{4}\right|$, $y = 0$, $x = 2$ & $x = 4$. At what angles to the positive x–axis straight lines must be drawn through $(4, 0)$ so that these lines partition the figure into three parts of the same size.

एक क्षेत्र वक्रों $y = \left|\sqrt{2}\sin\frac{\pi x}{4}\right|$, $y = 0$, $x = 2$ और $x = 4$ से परिबद्ध है। तब x–अक्ष की धनात्मक दिशा किस कोण से सरल रेखा जो कि $(4, 0)$ से जाती है खिचीं जाए की वह क्षेत्र को समान आकार के तीन भागों में विभाजित कर दे।

**Ans.** $\pi - \tan^{-1}\frac{2\sqrt{2}}{3\pi}$; $\pi - \tan^{-1}\frac{4\sqrt{2}}{3\pi}$

**Sol.** Let equation of line is $y = mx - 4m$




$$A = \int_2^4 \sqrt{2}\sin\frac{\pi}{4}x\,dx = \left[-\sqrt{2}\frac{4}{\pi}\cos\frac{\pi x}{4}\right]_2^4 = \frac{4\sqrt{2}}{\pi} \quad \text{... (i)}$$

$$\text{Also area of } \Delta ABC = \frac{1}{2}.2.(-2m_1) = -2m_1 \quad \text{... (ii)}$$

from (i) and (ii)

$$-2m_1 = \frac{4\sqrt{2}}{3\pi} \Rightarrow m_1 = \frac{-2\sqrt{2}}{3\pi}$$

$$\Rightarrow \tan(\pi - \theta_1) = \frac{-2\sqrt{2}}{3\pi} \Rightarrow \pi - \theta_1 = \tan^{-1}\frac{2\sqrt{2}}{3\pi}$$

$\Rightarrow q_1 = \pi - \tan^{-1}\dfrac{2\sqrt{2}}{3\pi}$ or $\dfrac{1}{2}.(2)(-2m_2) = \dfrac{8\sqrt{2}}{3\pi}$

$\Rightarrow m_2 = \dfrac{-4\sqrt{2}}{3\pi} \Rightarrow \tan(\pi - \theta_2) = \dfrac{-4\sqrt{2}}{3\pi}$

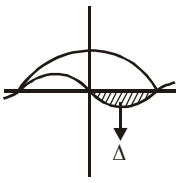
$\Rightarrow q_2 = \pi - \tan^{-1}\dfrac{4\sqrt{2}}{3\pi}$

5. Find the area bounded by the curves $y = \sqrt{1-x^2}$ and $y = x^3 - x$. Also find the ratio in which the y-axis divided this area.

वक्रों $y = \sqrt{1-x^2}$ और $y = x^3 - x$ से परिबद्ध क्षेत्रफल ज्ञात कीजिए। साथ ही वह अनुपात भी बताईये जबकि y-अक्ष क्षेत्रफल को विभाजित करता है।

**Ans.** $\dfrac{\pi}{2};\ \dfrac{\pi-1}{\pi+1}$

**Sol.** $\Delta = \left|\int_0^1 (x^3 - x)dx\right| = \dfrac{1}{4}$




$\text{Area} = \dfrac{\pi}{4} + \dfrac{1}{4} + \dfrac{\pi}{4} - \dfrac{1}{4} = \dfrac{\pi}{2}$

$\text{required ratio} = \dfrac{\frac{\pi}{4}+\frac{1}{4}}{\frac{\pi}{4}-\frac{1}{4}}$

6. The line $3x + 2y = 13$ divides the area enclosed by the curve, $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.

यदि सरल रेखा $3x + 2y = 13$, वक्र $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ के बन्द क्षेत्र को दो भागों में विभाजित करती है तब बड़े आकार के क्षेत्रफल तथा छोटे क्षेत्र के क्षेत्रफलों का अनुपात ज्ञात कीजिए।

**Ans.** $\dfrac{3\pi + 2}{\pi - 2}$

**Sol.** $3x + 2y = 13$

$9x^2 + 4y^2 - 18x - 16y - 11 = 0$

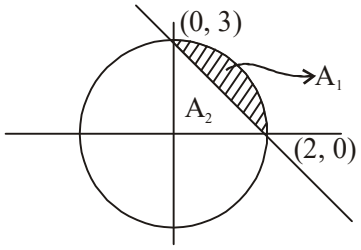
$9(x-1)^2 + 4(y-2)^2 - 36 = 0$

$$\frac{(x-1)^2}{4}+\frac{(y-2)^2}{9}=1$$

$x-1=x \qquad y-2=y$

$\frac{x^2}{4}+\frac{y^2}{9}=1$ and $3(x+1)+2(y+2)=13$

$$3x+2y=6$$




$$A_2=\frac{1}{2}\times 2\times 3=3$$

Area of ellipse = $\pi ab = \pi \times 2 \times 3$

$$= 6\pi$$

$$A_1=\frac{6\pi}{4}-3$$

$$=\frac{3\pi-6}{2}$$

Non-shaded ellipse area

$$=\frac{3}{4}\times 6\pi+3=\frac{9\pi+6}{2}$$

$$\text{Ratio}=\frac{\frac{9\pi+6}{2}}{\frac{3\pi-6}{2}}=\frac{3\pi+2}{\pi-2}$$ Ans.

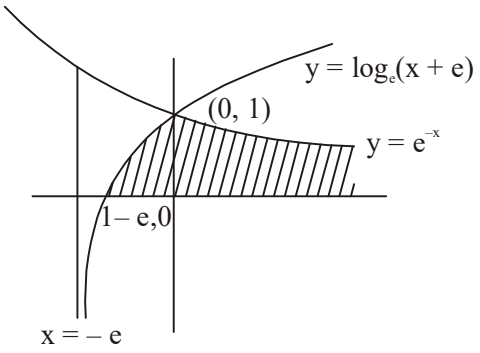
7. Find the area enclosed between the curves : $y=\log_e(x+e)$, $x=\log_e(1/y)$ & the x–axis.

वक्रों $y=\log_e(x+e)$, $x=\log_e(1/y)$ और x–अक्ष से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

**Ans.** 2 sq. units

**Sol.** $y = \log_e (x + e)$ and x-axis

$x = \log_e (y_y) = y = e^{-x}$




$$\int_{1-e}^{0} \log_e (x+e)\,dx + \int_{0}^{\infty} e^{-x}dx$$

$$= (x+e)\log_e\left(\frac{x+e}{e}\right)\Big|_{1-e}^{0} + -e^{-x}\Big|_{0}^{\infty}$$

A = O – 1(–1) + 1 = 2 Ans.

8. Compute the area of the curvilinear triangle bounded by the y-axis & the curve, y = tan *x* & y=(2/3)cos x.

वक्रों y = tan *x* और y=(2/3)cos x और y-अक्ष से परिबद्ध वक्रीय त्रिभुज का क्षेत्रफल ज्ञात कीजिए।

**Ans.** $\frac{1}{3} + \ell n\left(\frac{\sqrt{3}}{2}\right)$ sq. units

**Sol.** $y = \tan x \qquad y = \frac{2}{3}\cos x \qquad$ y-axis

POI

$$\tan x = \frac{2}{3}\cos x$$

$$3\tan x = 2\cos x$$

$$3\sin x = 2\cos^2 x$$

$3\sin x = 2 - 2\sin^2 x$

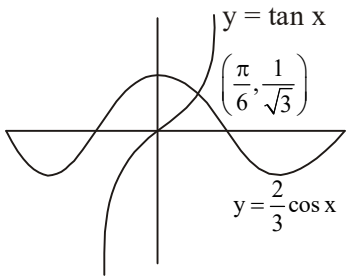
$2\sin^2 x + 3\sin x - 2 = 0$

$2\sin^2 x + 4\sin x - \sin x - 2 = 0$

$(2\sin x - 1)(\sin x + 2) = 0$

$\sin x = -1$

$x = \frac{\pi}{6}$




$$A = \int_0^{\frac{\pi}{6}} \left(\frac{2}{3}\cos x - \tan x\right) dx$$

$$= \frac{2}{3}\sin x - |n|\sec x\Big|_0^{\frac{\pi}{6}}$$

$$= \frac{2}{3}\times\frac{1}{2} - |n|\frac{2}{\sqrt{3}}\Big| = \frac{1}{3} + |n|\frac{\sqrt{3}}{2}\Big|$$

9. Find the area bounded by the curve $y = x e^{-x}$ ; $xy = 0$ and $x = c$ where c is the x-coordinate of the curve's inflection point.

वक्रों $y = x e^{-x}$ ; $xy = 0$ और $x = c$ से परिबद्ध क्षेत्रफल ज्ञात कीजिए। जहाँ c वक्र के नति परिवर्तन बिन्दु का x-निर्देशांक है।

**Ans.** $1 - 3e^{-2}$

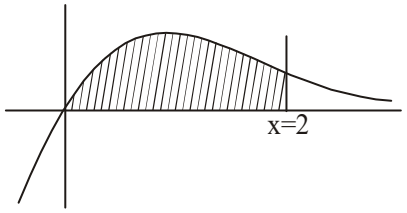
**Sol.** $y = xe^{-x}$

$y' = e^{-x} - x e^{-x} = (1 - x)e^{-x} \uparrow$ for $x < 1$

$y'' = -e^{-x} - [e^{-x} - xe^{-x}] = e^{-x}[-1 - 1 + x]$

$= (x - 2)e^{-x}$

for point of inflection $y'' = 0 \quad \Rightarrow \quad x = 2$




$$A = \int_0^2 xe^{-x}\,dx = -xe^{-x}\Big|_0^2 + \int_0^2 e^{-x}$$

$= (-2e^{-2}) - \left(e^{-x}\right)_0^2$

$= -2e^{-2} - (e^{-2} - 1) = 1 - e^{-2} - 2e^{-2} = 1 - 3e^{-2}$

10. Find the area bounded by the curve $y = x e^{-x^2}$, the x-axis, and the line x = c where y(c) is maximum.

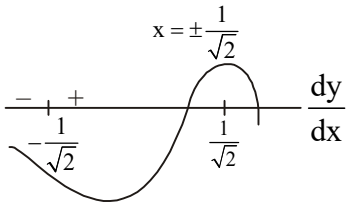
वक्र $y = x e^{-x^2}$, x-अक्ष तथा रेखा x = c परिबद्ध क्षेत्रफल ज्ञात कीजिए। (जहाँ y(c) अधिकतम है।)

**Ans.** $\frac{1}{2}(1 - e^{-1/2})$

**Sol.** $y = xe^{-x^2}$, x-axis $\quad x = c$

$$\frac{dy}{dx} = -2x^2e^{-x^2} + e^{-x^2} = 0$$

$$x = \pm\frac{1}{\sqrt{2}}$$




$$\left.\frac{dy}{dx}\right|_{max} \text{ at } x = \frac{1}{\sqrt{2}} \qquad c = \frac{1}{\sqrt{2}}$$

$$A = \int_0^{\frac{1}{\sqrt{2}}} xe^{-x^2}dx$$

$x^2 = t \quad \Rightarrow 2xdx = dt$

$$= \int_0^{\frac{1}{2}} e^{-t}\frac{dt}{2}$$

$$= \frac{1}{2}\left[-e^{-t}\right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2}\left(1 - e^{-\frac{1}{2}}\right) \text{ Ans.}$$

11. Consider a circle $x^2 + (y-1)^2 = 1$ and the parabola $y = -\frac{x^2}{4}$. The common tangents to the two curves constitute a triangle ABC, the point A and B being on the x-axis and C on the y-axis. CA produced touches the parabola at P and CB produced touches the parabola at Q.

(a) Find the equation of the common tangent BC.

(b) Find the area of the portion between the upper arc of the circle and the common tangents QC and PC.

(c) Find the area enclosed by the parabola $y = -\frac{x^2}{4}$, the x-axis and the lines AP and BQ.

माना वृत्त $x^2 + (y-1)^2 = 1$ और परवलय $y = -\frac{x^2}{4}$ दिया गया है। यदि वक्रों की उभयनिष्ठ स्पर्शरेखाऐं एक त्रिभुज ABC का निमार्ण करता है, जहाँ बिन्दु A और B, x-अक्ष पर है और C, y-अक्ष पर, यदि CA का बढ़ाया भाग परवलय का P पर स्पर्श करता है और CB का बढ़ाया भाग परवलय को Q पर स्पर्श करता है।

(a) उभयनिष्ठ स्पर्शरेखा BC ज्ञात कीजिए।

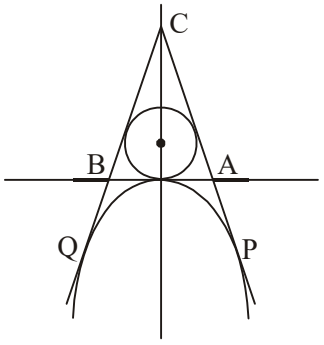
(b) वृत्त के ऊपरी चाप और उभयनिष्ठ स्पर्श रेखाओं QC और PC से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

(c) परवलय $y = -\frac{x^2}{4}$, x-अक्ष और रेखाओं AP और BQ से परिबद्ध क्षेत्रफल ज्ञात कीजिए।

**Ans.** (a) $\sqrt{3}x - y + 3 = 0$ (b) $\left(\sqrt{3} - \frac{\pi}{3}\right)$ (c) $\sqrt{3}$

**Sol.** $x^2 + (y-1)^2 = 1$ and $x^2 = -4y$

Common tangent at parabola




$x = \frac{y}{m} - m$

$y = mx + m^2$

COT from (0, 1)

$\left|\frac{1-m^2}{\sqrt{1+m^2}} = 1\right|$

$m^4 - 2m^2 + 1 = 1 + m^2$

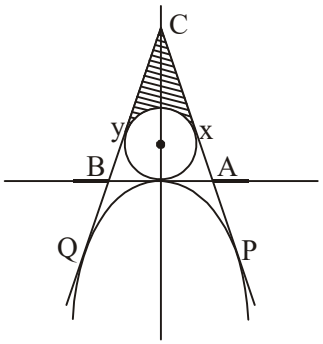
$m^4 = m^2$

$m = O, m = \sqrt{3}, m = -\sqrt{3}$

AB : $y = 0$ BC : $y = \sqrt{3}x + 3$ AC : $y = -\sqrt{3}x + 3$

$A\left(-\sqrt{3},0\right)$ $B\left(\sqrt{3},0\right)$ C(0, 3)

Point of contact on circle




12. If f (x) is monotonic in (a, b) then prove that the area bounded by the ordinates at x = a ; x = b ; y = f (x) and y = f(c), c ∈ (a, b) is minimum when $c = \frac{a+b}{2}$. Hence if the area bounded by the graph of $f(x)=\frac{x^3}{3}-x^2+a$, the straight lines x = 0, x = 2 and the x-axis is minimum then find the value of 'a'.

यदि f (x); (a, b) में एकदिष्ट फलन है, तो सिद्ध कीजिये कि x = a ; x = b ; y = f (x) और y = f (c), c ∈ (a, b) से परिबद्ध क्षेत्रफल न्यूनतम होगा यदि $c = \frac{a+b}{2}$ । अब यदि वक्र $f(x)=\frac{x^3}{3}-x^2+a$, सरल रेखा x = 0, x = 2 और x-अक्ष से परिबद्ध क्षेत्रफल न्यूनतम है, तो 'a' का मान ज्ञात कीजिये।

**Ans.** $a = \frac{2}{3}$

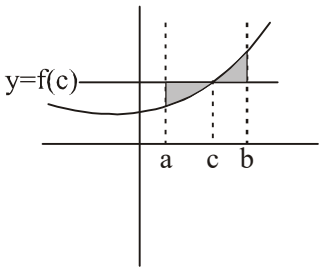
**Sol.** f(x)↑ (a, b)

Shaded area

$$A=\int_a^c\left(f(c)-(x)\right)dx+\int_c^b\left(f(x)-f(c)\right)dx$$

$$A=f(c)(c-a)-f(b-c)+\int_c^a f(n)+\int_c f(x)$$

$$A=f(c)(2c-a-b)+\underbrace{\int_c^a f(n)+\int_c^b f(x)}_{+ive}$$




f(c) (2c – a – b) = 0

$$c = \frac{a+b}{2}$$

$$f(x) = \frac{x^3}{3} - x^2 + a$$

$$f'(x) = x^2 - 2x$$

$-x = 0, x = 2$

Area will be minimum when graph will be symmetric about root $\alpha$.

$$\alpha = \frac{0+2}{2} = 1$$

$f(1) = 0$

$$-\frac{2}{3} + a = 0$$

$$a = \frac{2}{3}$$

13. Find the area of the region bounded by the curves $y = x^2$ , $y = |2 - x^2|$ and $y = 2$ , which lies to the right of the line $x = 1$.

वक्रों $y = x^2$ , $y = |2 - x^2|$ और रेखा $y = 2$ से परिबद्ध उस क्षेत्र का क्षेत्रफल ज्ञात कीजिये जो कि रेखा $x = 1$ के दायीं ओर है।

**Ans.** $\left(\frac{20}{3} - 4\sqrt{2}\right)$ sq. units

**Sol.** The given curves are $y = x2$

which is an upward parabola with vertex at (0, 0)

$y = |2 - x2|$

$$\text{or } y = \begin{cases} 2 - x^2 & \text{if} \quad -\sqrt{2} < x < \sqrt{2} \\ x^2 - 2 & \text{if} \quad x < -\sqrt{2} \text{ or } x > \sqrt{2} \end{cases}$$

or $x2 = -(y - 2)$; $\quad -\sqrt{2} < x < \sqrt{2}$ ... (2)

a downward parabola with vertex at (0, 2)

$x2 = y + 2$ ; $\quad x < -\sqrt{2}$ , $x > \sqrt{2}$ ... (3)

On upward parabola with vertex at (0, –2)

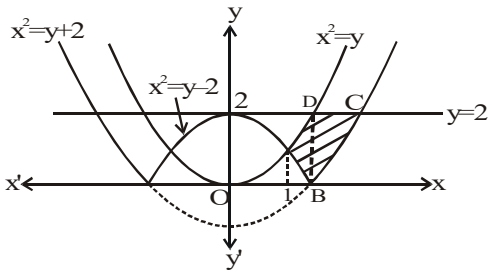
$y = 2$ ... (4)

Straight line parallel to x-axis

$x = 1$ ... (5)

Straight line parallel to y-axis

The graph of these curves is as follows.




$\therefore$ Required area = BCDEB

$$= \int_1^{\sqrt{2}} [x^2 - (2 - x^2)]dx + \int_{\sqrt{2}}^{2} [2 - (x^2 - 2)]dx$$

$$= \int_1^{\sqrt{2}} (2x^2 - 2)dx + \int_{\sqrt{2}}^{2} (4 - x^2)dx = \left(\frac{20}{3} - 4\sqrt{2}\right) \text{sq. units}$$

14. Find the area bounded by the curves $x^2 = y$, $x^2 = -y$ and $y^2 = 4x - 3$.

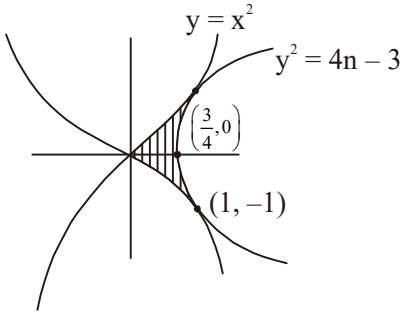
वक्रो $x^2 = y$, $x^2 = -y$ और $y^2 = 4x - 3$ से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिये।

**Ans.** $\frac{1}{3}$ sq. units

**Sol.** $x^2 = y$

$x^2 = -y$

$y^2 = x\left(x - \frac{3}{4}\right)$




POI

$y^2 = 4x - 3$ and $x^2 = y$

$x^4 = 4x - 3$

$x^4 - 4x + 3 = 0$

$(x - 1)(x^3 + x^2 + x - 3) = 0$

$$(x-1)(x-1)\left(\underbrace{x^2 + 2x + 3}_{D<0}\right) = 0$$

$n = 1$

$$A = 2\left[\int_0^1 x^2 dx - \int_{\frac{3}{4}}^1 \sqrt{4x-3}dx\right]$$

$$= 2\left[\frac{x^3}{3}\Big|_0^1 - \frac{2}{3}\frac{(4x-3)^{\frac{3}{2}}}{4}\Big|_{\frac{3}{4}}^1\right]$$

$$= 2\left(\frac{1}{3}-\frac{1}{6}\right)$$

$$= 2\times\frac{1}{6} = \frac{1}{3}$$

15. Let f(x) be a quadratic polynomial and a, b, c be distinct real numbers such that

$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}.$$

Let V be the point of maximum of the curve $y = f(x)$. If A and B are the points on this curve such that the curve meets the positive x-axis at A and the chord AB subtends a right angle at V, then find the area enclosed by the curve and the chord AB.

माना f(x) एक वर्ग बहुपद है और a, b, c तीन अलग–अलग वास्तविक संख्याएँ है, इस प्रकार है कि

$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}.$$

माना $y = f(x)$ का उच्चिष्ठ बिन्दु V है। यदि A और B वक्र पर दो बिन्दु इस प्रकार है कि वक्र x-अक्ष को बिन्दु A पर मिलता है तथा जीवा AB, V पर समकोण बनाती है, तब वक्र और जीवा AB से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

**Ans.** $\frac{125}{3}$ sq. units

**Sol.** We have, $\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}$

$\Rightarrow 4a2f(-1) + 4af(1) + f(2) = 3a2 + 3a$

$4b2f(-1) + 4bf(1) + f(2) = 3b2 + 3b$

$4c2f(-1) + 4cf(1) + f(2) = 3c2 + 3c$

Consider the equation

$4x2f(-1) + 4xf(1) + f(2) = 3x2 + 3x$

or $[4f(-1) -3]x2 + [4f(1) - 3]\, x + f(2) = 0$

Then clearly this equation is satisfied by

$x = a, b, c$

A quadratic equation satisfied by more than two values of x means it is an identity and hence

$4f(-1) - 3 = 0 \quad \Rightarrow \quad f(-1) = 3/4$

$4f(1) - 3 = 0 \quad \Rightarrow \quad f(1) = 3/4$

$f(2) = 0 \quad \Rightarrow \quad f(2) = 0$

Let $f(x) = px2 + qx + r$ [$f(x)$ being a quad. equation]

$$f(-1) = \frac{3}{4} \quad \Rightarrow \quad p - q + r = \frac{3}{4}$$

$$f(1) = \frac{3}{4} \quad \Rightarrow \quad p + q + r = \frac{3}{4}$$

$$f(2) = 0 \quad \Rightarrow \quad 4p + 2q + r = 0$$

Solving the above we get $q = 0$, $p = \frac{-1}{4}$, $r = 1$

$$\therefore\ f(x) = -\frac{1}{4}\, x2 + 1$$

It's maximum value occur at $f'(x) = 0$

i.e., $x = 0$ then $f(x) = 1 \qquad \therefore\ V(0, 1)$

A $(-2, 0)$ is the pt. where curve meet x-axis

Let B be the pt. $\left(h, \frac{4-h^2}{4}\right)$

As $\quad \angle AVB = 90°$

$mAV \times mBV = -1$

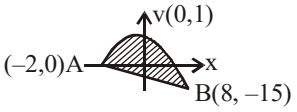
$$\Rightarrow \quad \frac{1}{2} \times \left(\frac{-h}{4}\right) = -1$$

$\Rightarrow \quad h = 8 \qquad \therefore\ B(8, -15)$

Equation of chord AB is

$$y + 15 = \frac{0-(-15)}{-2-8}(x - 8)$$

$\Rightarrow 3x + 2y + 6 = 0$




Required area is

the area of shadded
region given by

$$= \int_{-2}^{8} \left[ \left( -\frac{x^2}{4} + 1 \right) - \left( \frac{-6-3x}{2} \right) \right] dx$$

$$= \frac{125}{3} \text{ sq. units.}$$

**DETERMINATION OF UNKNOWN PARAMETERS**

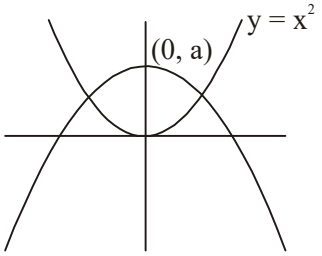
16. If the area enclosed by the parabolas $y = a - x^2$ and $y = x^2$ is $18\sqrt{2}$ sq. units. Find the value of 'a'.

यदि वक्रों $y = a - x^2$ और $y = x^2$ से परिबद्ध क्षेत्रफल $18\sqrt{2}$ वर्ग ईकाई है। तब a का मान ज्ञात कीजिए।

**Ans.** $a = 9$

**Sol.** $y = a - x^2$

$y = x^2$




$a \cdot x^2 = x^2$

$$x^2 = \frac{a}{2}$$

$$A = 2\int_{0}^{\sqrt{\frac{a}{2}}} \left( a \cdot x^2 - x^2 \right) dx$$

$$18\sqrt{2} = 2 \cdot \left[ ax - \frac{2x^3}{3} \right]_0^{\sqrt{\frac{a}{2}}}$$

$$9\sqrt{2} = \frac{a^{\frac{3}{2}}}{\sqrt{2}} - \frac{2 \cdot a^{\frac{3}{2}}}{3 \cdot 2\sqrt{2}}$$

$$18 = \frac{3a^{\frac{3}{2}} - a^{\frac{3}{2}}}{3}$$

$$a^{\frac{3}{2}} = 27$$

$a = 9$

17. Find the values of m (m > 0) for which the area bounded by the line $y = mx + 2$ and $x = 2y - y^2$ is : (i) 9/2 square units and (ii) minimum. Also find the minimum area.

m (m > 0) के मानों को ज्ञात कीजिए जबकि सरल रेखा $y = mx + 2$ और वक्र $x = 2y - y^2$ से परिबद्ध क्षेत्रफल : ; पद्ध9ध्2 वर्ग इकाई है (ii) न्यूनतम है। साथ ही न्यूनतम क्षेत्रफल ज्ञात कीजिए।

**Ans.** (i) $m = 1$ (ii) $m = \infty$ ; $A_{min} = 4/3$

**Sol.** $y = mn + 2$; $m > 0$ and $x = 2y - y^2$

POI

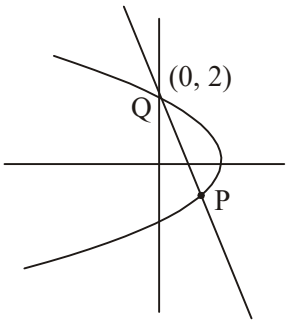
$y = mn + 2$

$$x = \frac{y-2}{m} = 2y - y^2$$

$$y - 2 = m(y-2)(-y)$$

$$y = -\frac{1}{m}, 2$$

$$x = -\frac{1-2m}{m^2}, 0$$




$$P\left(\frac{2m-1}{m^2}, -\frac{1}{3}\right) \text{ and } Q(0, 2)$$

$$\text{Area} = \int_{-\frac{1}{m}}^{2}\left[\left(2y - y^2\right) - \left(\frac{y-2}{m}\right)\right]dy$$

$$A = y^2 - \frac{y^2}{3} - \frac{y^2}{2m} + \frac{2y}{m}\Bigg|_{-\frac{1}{m}}^{2}$$

$$= \left(4 - \frac{8}{3} - \frac{2}{m} + \frac{4}{m}\right) - \left(\frac{1}{m^2} + \frac{1}{3m^3} - \frac{1}{2m^3} - \frac{2}{m^2}\right)$$

$$= \frac{4}{3} + \frac{2}{m} + \frac{1}{m^2} + \frac{1}{6m^3}$$

(i) $A = \frac{9}{2}$

$$\frac{9}{2} = \frac{4}{3} + \frac{2}{m} + \frac{1}{m^2} + \frac{1}{6m^3}$$

$27m^3 = 8m^3 + 12m^2 + 6m + 1$

$19m^3 - 12m^2 - 6m - 1 = 0$

$(m-1)(19m^2 + 7m + 1) = 0$

$D < 0$

$m = 1$

(ii) minimum area

$\frac{dA}{dm} = \frac{-2}{m^2} - \frac{2}{m^3} - \frac{1}{2m^4} = 0$

$\frac{4m^2 + 4m + 1 = 0}{2m^4}$

$(2m+1)^2 = 0$ or $m = \frac{1}{0}$

$m = -\frac{1}{2}$ or $m = \frac{1}{0}$

$m \neq -\frac{1}{2}$

So $m = \frac{1}{0}$

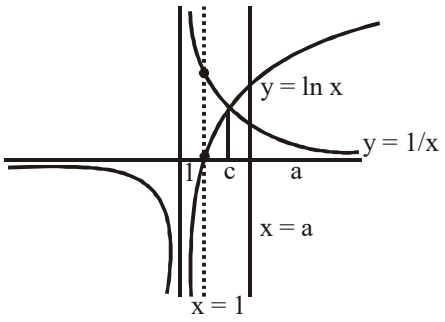
$Amin = \frac{4}{3}$

18. Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = ln\, x$ on the xy plane. Let $D_1$ denotes the region surrounded by $C_1$, $C_2$ and the line $x = 1$ and $D_2$ denotes the region surrounded by $C_1$, $C_2$ and the line $x = a$. If $D_1 = D_2$. Find the value of 'a'.

ढ $_1 : y = \frac{1}{x}$ और $C_2 : y = ln\, x$, xy समतल में दो वक्र है। माना $D_1$ वह क्षेत्र है जो कि $C_1$, $C_2$ और रेखा $x = 1$ से परिबद्ध है और $D_2$, $C_1$, $C_2$ और रेखा $x = a$ से परिबद्ध क्षेत्र है। यदि $D_1 = D_2$ हो तब a का मान ज्ञात कीजिए।

**Ans.** e

**Sol.**




$D_1 = D_2$

$$\int_1^C \left(\frac{1}{n} - \ln x\right) dx = \int_c^a \left(\ln x - \frac{1}{x}\right) dx = 0$$

$$\int_1^C \left(\frac{1}{x} - \ln x\right) dx + \int_c^a \left(\frac{1}{x} - \ln x\right) dx = 0$$

$$\int \left(\frac{1}{x} - \ln x\right) dx = 0$$

$$\ln x - x\left|x\left(\frac{x}{e}\right)\right|_1^a = 0$$

$$\ln a - a\ln\left(\frac{a}{e}\right) + \ln\left(\frac{1}{e}\right) = 0$$

$\ln a - a \ln a + a - 1 = 0$

$(\ln a - 1)(a - 1) = 0$

$\ln a = 1$ & $a = 1$

$a = e$

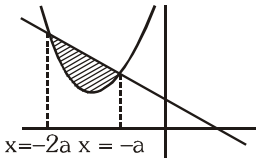
19. Find the value(s) of the parameter 'a' ($a > 0$) for each of which the area of the figure bounded by the straight line, $y = \frac{a^2 - ax}{1 + a^4}$ & the parabola $y = \frac{x^2 + 2ax + 3a^2}{1 + a^4}$ is the greatest.

; झ ०द्धके मानों का ज्ञात कीजिए जबकि सरल रेखा $y = \frac{a^2 - ax}{1 + a^4}$ और परवलय $y = \frac{x^2 + 2ax + 3a^2}{1 + a^4}$ से परिबद्ध क्षेत्रफल अधिकतम है।

**Ans.** $a = 3^{1/4}$

**Sol.** $A = \int_{-a}^{-2a} \frac{a^2 - ax - (x^2 + 2ax + 3a^2)}{1 + a^4} dx$

$= \frac{3}{2} \frac{a^3}{1 + a^4}$




Now $f(a) = \frac{3}{2} \frac{a^3}{1 + a^4}$

$\Rightarrow f'(a) = 0$

$\Rightarrow (1 + a4)\ 3a2 - a3\ 4a3 = 0$

$\Rightarrow$ amin = 0, amax = 31/4

20. For what value of 'a' is the area bounded by the curve $y = a^2x^2 + ax + 1$ and the straight line $y = 0$, $x = 0$ and $x = 1$ the least?

के किस मान के लिए वक्र $y = a^2x^2 + ax + 1$ और सरल रेखा $y = 0$, $x = 0$ और $x = 1$ से परिबद्ध क्षेत्रफल न्यूनतम होगा।

**Ans.** $a = -3/4$

**Sol.** Required area $= \int_0^1 y dx$

$$A = \int_0^1 \left(a^2x^2 + ax + 1\right) dx$$

$$A = \frac{a^2x^3}{3} + \frac{ax^2}{2} + x \Big|_0^1$$

$$A = \frac{a^2}{3} + \frac{a}{2} + 1$$

For min

$$\frac{dA}{da} = 0$$

$$\frac{2a}{3} + \frac{1}{2} = 0$$

$$\Rightarrow a = -\frac{3}{4}$$

21. Find the positive value of 'a' for which the parabola $y = x^2 + 1$ bisects the area of the rectangle with vertices (0, 0), (a, 0), (0, $a^2 + 1$) and (a, $a^2 + 1$).

का वह धनात्मक मान ज्ञात कीजिए जिसके लिए परवलय $y = x^2 + 1$ चतुर्भुज जिनके शीर्ष (0, 0), (a, 0), (0, $a^2 + 1$) और (a, $a^2 + 1$) से परिबद्ध क्षेत्रफल को समद्विभाजित करता है।

**Ans.** $\sqrt{3}$

**Sol.** Area of rectangle = $a(a^2 + 1)$

$= a^3 + a$

$$A = \int_0^a y\,dx$$

$$\frac{a^3 + a}{2} = \frac{a^3}{3}$$

$$a = 0, a = \pm\sqrt{3} \Rightarrow a = \sqrt{3}$$

22. Find the value of 'c' for which the area of the figure bounded by the curve, $y = 8x^2 - x^5$, the straight lines $x = 1$ & $x = c$ & the abscissa axis is equal to 16/3.

ब का मान ज्ञात कीजिए जबकि वक्र $y = 8x^2 - x^5$, सरल रेखाओं $x = 1$ और $x = c$ तथा भुज अक्ष से परिबद्ध क्षेत्रफल 16/3 वर्ग ईकाई है।

**Ans.** $C = -1$ or $\left(8 - \sqrt{17}\right)^{1/3}$

**Sol.** C-I $C < 1$

$$\int_c^1 \left(8x^2 + x^5\right) dx = \frac{16}{3}$$

$$\frac{8x^3}{3} - \frac{x^6}{6}\Bigg|_c^1 = \frac{16}{3}$$

$$\frac{8}{3} - \frac{1}{6} - \frac{8c^3}{3} + \frac{c^6}{6} = \frac{16}{3}$$

$$\frac{-8}{3}c^3 + \frac{c^6}{6} = \frac{17}{6} = \frac{17}{6}$$

$c^6 - 16c^3 - 17 = 0$

$c^3 = -1, 17$

$C = -1, 17^{1/3}$

$C < 1$

$C = -1$

C - II

$C \geq 1$

$$\int_1^C \left(8x^2 - x^5\right) dx = \frac{16}{3}$$

$$\frac{8x^3}{3} - \frac{x^6}{6}\Bigg|_1^C = \frac{16}{3}$$

$$\frac{8C^3}{3} - \frac{C^6}{6} = \frac{47}{6}$$

$C^6 - 16C^3 + 47 = 0$

$C^3 = t$

$t^2 - 16t + 47 = 0$

$$t = \frac{16 \pm \sqrt{256 - 188}}{2}$$

$$t = 8 \pm \sqrt{64 - 47}$$

$$t = 8 \pm \sqrt{17}$$

$$C^3 = 8 \pm \sqrt{17}$$

$$C = \left(8 + \sqrt{17}\right)^{1/3}, \left(8 - \sqrt{17}\right)^{1/3}$$

23. Consider one side AB of a square ABCD, (read in order) on the line $y = 2x - 17$, and the other two vertices C, D on the parabola $y = x^2$.

(a) Find the minimum intercept of the line CD on y-axis.

(b) Find the maximum possible area of the square ABCD.

(c) Find the area enclosed by the line CD with minimum y-intercept and the parabola $y = x^2$

माना वर्ग ABCD (क्रम में) की एक भुजा AB, सरल रेखा $y = 2x - 17$ पर है तथा अन्य दो शीर्ष C, D परवलय $y = x^2$ पर

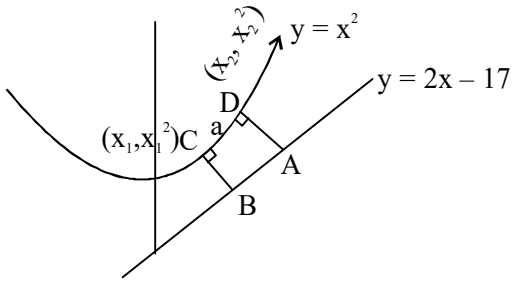
(a) सरल रेखा CD का न्यूनतम y-अन्तःखण्ड ज्ञात कीजिए।

(b) वर्ग ABCD का अधिकतम सम्भव क्षेत्रफल ज्ञात कीजिए।

(c) वह क्षेत्रफल ज्ञात कीजिए जो कि रेखा CD का न्यूनतम y-अन्तःखण्ड और परवलय $y = x^2$ से परिबद्ध है

**Ans.** (a) 3 (b) 1280 (c) $\frac{32}{3}$

**Sol.**




(a) As ABCD is a squre $\therefore$ AB || CD

Equation of line CD $\equiv y = 2x + c$

$$\text{distance between AB \& CD} = \frac{|c+17|}{\sqrt{2^2+1}} = \frac{|c+17|}{\sqrt{5}}$$

be 'a'

To find x coordinates of C and D solve

$y = x^2$ and $y = 2x + c$

$x^2 = 2x + c$

$x^2 - 2x - c = 0 \Leftarrow x_1, x_2$

$\Rightarrow x_1 + x_2 = 2$

Now length of CD

$$= \sqrt{(x_2 - x_1)^2 + (x_2^2 - x_1^2)^2}$$

$$= \sqrt{(x_2 - x_1)^2 + (x_2 - x_1)^2 (x_2 + x_1)^2}$$

$$= \sqrt{(x_2 - x_1)^2 \{1 + 4\}}$$

$$= \sqrt{5}\,|x_2 - x_1|$$

Now $\quad |x_2 - x_1| = \dfrac{\sqrt{D}}{|a'|}$

$$= \sqrt{4 + 4c}$$

$$= 2\sqrt{1 + c}$$

$$\therefore \qquad a = 2\sqrt{5}\sqrt{1 + c} = \frac{|c + 17|}{\sqrt{5}}$$

$4.5^2 (1 + c) = (c + 17)^2$

$100(1 + c) = c^2 + 34c + 289$

$100 + 100\ c = c^2 + 34c + 289$

$c^2 - 66C + 189 = 0$

$c = 63, 3$

$\therefore\ c_{min} = 3$

$c_{max} = 63$

(b) Maximum area $= a^2$

$$= \left(2\sqrt{5}\sqrt{1 + c}\right)^2$$

$$= 4.5(1 + c)$$

$$= 20(1 + 63)$$

$$= 20 \times 64$$

$$= 1280$$

(c) Equation of line CD with min y intercept i.e. c = 3

i.e. c = 3

$y = 2x + 3$ and $y = x^2$

Point of intersection

$x^2 = 2x + 3$

$\therefore\ x^2 - 2x - 3 = 0$

$x = 3, -1$

$$\therefore \quad \int_{-1}^{3} \left( (2x+3) - x^2 \right) dx$$

$$-\frac{x^3}{3} + x^2 + 3x \Big|_{-1}^{3}$$

$$\Rightarrow \frac{32}{3}$$

## COMPREHENSION TYPE QUESTIONS

**Paragraph for Question Nos. 24 to 26**

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of f(x) and let $t = |s|$.

माना $f(x) = 1 + 2x + 3x^2 + 4x^3$ एक बहुपदीय फलन है तथा S, f(x) के सभी वास्तविक भिन्न–भिन्न मूलों का योग है और माना $t = |s|$.

24. The real number s lies in the interval : **[JEE-2010]**

वास्तविक संख्या s अन्तराल में होगी : **[JEE-2010]**

(A) $\left(-\frac{1}{4}, 0\right)$ (B) $\left(-11, -\frac{3}{4}\right)$ (C) $\left(-\frac{3}{4}, -\frac{1}{2}\right)$ (D) $\left(0, \frac{1}{4}\right)$

**Ans.** C

**Sol.** $\because f'(x) = 2 + 6x + 12x2 > 0 \ \forall \ x \in R$

$\therefore$ f(x) is strictly increasing in R

$\because f(0) = 1, f(-1) = -2, f\left(-\frac{1}{2}\right) = \frac{1}{4}$ and $f\left(-\frac{3}{4}\right) = -\frac{1}{2}$

$\therefore$ f(x) = 0 has only one real root lying in $\left(-\frac{3}{4}, -\frac{1}{2}\right)$

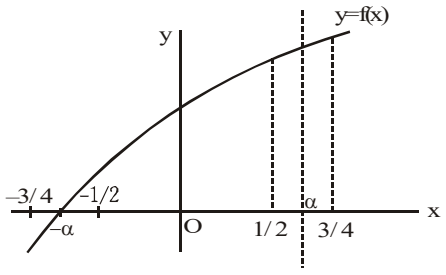
25. The area bounded by the curve y = f(x) and the lines x = 0, y = 0 and x = t, lies in the interval : **[JEE-2010]**

y = f(x) सरल रेखाओं x = 0, y = 0 और x = t से परिबद्ध क्षेत्रफल किस अन्तराल में होगा : **[JEE-2010]**

(A) $\left(\frac{3}{4}, 3\right)$ (B) $\left(\frac{21}{64}, \frac{11}{16}\right)$ (C) (9, 10) (D) $\left(0, \frac{21}{64}\right)$

**Ans.** A

**Sol.**




Let real root is $-\alpha$

$\Rightarrow \quad t = |s| = \alpha$

Required area

$$A = \int_0^{\alpha} f(x)dx \quad \text{and} \quad \int_0^{1/2} f(x)dx < A < \int_0^{3/4} f(x)dx$$

$$\Rightarrow \left|x + x^2 + x^3 + x^4\right|_0^{1/2} < A < \left|x + x^2 + x^3 + x^4\right|_0^{3/4} < \left|4x\right|_0^{3/4}$$

$$\Rightarrow \frac{15}{16} < A < 3$$

26. The function $f'(x)$ is : **[JEE-2010]**

(A) increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$

(B) decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$

(C) increasing in $(-t, t)$

(D) decreasing in $(-t, t)$

फलन $f'(x)$ है : **[JEE-2010]**

(A) $\left(-t, -\frac{1}{4}\right)$ में वर्धमान और $\left(-\frac{1}{4}, t\right)$ में ह्रासमान

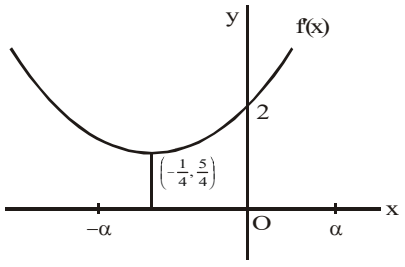
(B) $\left(-t, -\frac{1}{4}\right)$ में ह्रासमान और $\left(-\frac{1}{4}, t\right)$ में वर्धमान

(C) ; अन्यत्र वर्धमान

(D) ; अन्यत्र ह्रासमान

**Ans.** B

**Sol.**




$f'(x) = 2(6x2 + 3x + 1)$

$\Rightarrow$ $f'(x)$ is decreasing in $\left(-\alpha, -\frac{1}{4}\right)$ increasing in $\left(-\frac{1}{4}, \alpha\right)$

or $f'(x)$ is decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$

**Paragraph for Question nos. 27 to 29**

Consider three real-valued functions f, g and h defined on R ( the set of real numbers). Let $f(x) = 2x^3 + 3\left(1-\frac{3a}{2}\right)x^2 + 3(a^2-a)x + b$ where $a, b \in R$ and $g(x) = \frac{f'(x)}{6}$. Also h(x) is such that $h''(x) = 6x - 4$ and h(x) has a local minimum value 5 at x = 1.

वास्तविक मान वाले फलनों f, g तथा h जो कि R ( वास्तविक संख्याओं के समुच्चय) पर परिभाषित है, माना $f(x) = 2x^3 + 3\left(1-\frac{3a}{2}\right)x^2 + 3(a^2-a)x + b$ जहाँ $a, b \in R$ तथा $g(x) = \frac{f'(x)}{6}$. तथा h(x) इस प्रकार है कि $h''(x) = 6x - 4$ तथा h(x) का, x = 1 पर स्थानीय निम्निष्ठ 5 है।

27. The true set of values of a for which f(x) has negative point of local minimum, is :

a का वास्तविक समुच्चय जिनके लिये f(x) के पास स्थानीय निम्निष्ठ का ऋणात्मक बिन्दु है :

(A) $(-\infty, 0)$ (B) $(1, \infty)$ (C) $(0, 1)$ (D) $(1, \infty) - \{2\}$

**Ans.** A

**Sol.** We have

$f(x) = 2x3 + 3\left(1-\frac{3a}{2}\right)x2 + 3(a2-a)x + b$

So, $f'(x) = 6x2 + 6\left(1-\frac{3a}{2}\right)x + 3(a2-a)$

Now, for f(x) to have negative point of local minimum, we must have both roots of the equation f'(x) = 0 real (unequal) and negative.

$\therefore$ Discriminant $> 0 \Rightarrow 36\left[\left(1-\frac{3a}{2}\right)^2 - 2(a^2-a)\right] > 0 \Rightarrow \left(\frac{a}{2}-1\right)^2 > 0 \Rightarrow a \in R - \{2\}$ .....(1)

Also, sum of roots $< 0 \Rightarrow \frac{-6\left(1-\frac{3a}{2}\right)}{6} < 0 \Rightarrow \frac{3a}{2} - 1 < 0 \Rightarrow a < \frac{2}{3}$ .....(2)

And product of roots $> 0 \Rightarrow \frac{3(a^2-a)}{6} > 0 \Rightarrow a(a-1) > 0 \Rightarrow a \in (-\infty, 0) \cup (1, \infty)$ .....(3)

$\therefore$ From (1) $\cap$ (2) $\cap$ (3), we get $a \in (-\infty, 0)$

28. The complete set of values of a for which vertex of parabola $y = g(x)$ has negative ordinate, is :

a के वास्तविक मानो का समुच्चय जिनके लिये, परवलय $y = g(x)$ का शीर्ष ऋणात्मक कोटि का है :

(A) $\left(\frac{1}{2},1\right)$ (B) $(0, \infty)$ (C) $R - \{2\}$ (D) $(1, \infty) - \{2\}$

**Ans.** C

**Sol.** We have

$$g(x) = \frac{f'(x)}{6} = x^2 + \left(1 - \frac{3a}{2}\right)x + \left(\frac{a^2 - a}{2}\right)$$

As discriminant of $g(x) = \left(\frac{a}{2} - 1\right)^2 > 0 \; \forall \; a \in R - \{2\}$

$\Rightarrow$ Ordinate of the vertex of parabola $y = g(x)$ lies below x-axis.

So, $a \in R - \{2\}$

29. The area bounded by $y = h(x)$ between $x = 0$ and $x = 2$, is :

$y = h(x)$, से $x = 0$ एवं $x = 2$ के मध्य का क्षेत्रफल है :

(A) $\frac{23}{3}$ (B) $\frac{20}{3}$ (C) $\frac{40}{3}$ (D) $\frac{32}{3}$

**Ans.** D

**Sol.** We have

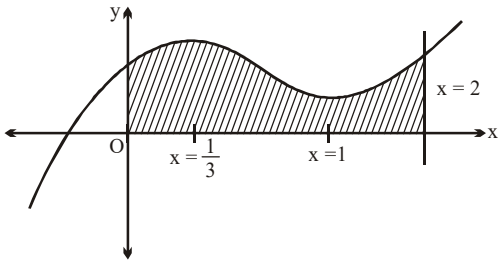
$h''(x) = 6x - 4 \Rightarrow h'(x) = 3x2 - 4x + c$

As $h'(1) = 0$, so $0 = 3 - 4 + c \Rightarrow c = 1$

$\therefore$ $h'(x) = 3x2 - 4x + 1 \Rightarrow h(x) = x3 - 2x2 + x + k$

As $h(1) = 5$, so $5 = 1 - 2 + 1 + k \Rightarrow k = 5$.

Hence, $h(x) = x3 - 2x2 + x + 5$




Clearly, required area

$$= \int_0^2 h(x)dx = \int_0^2 \left(x^3 - 2x^2 + x + 5\right)dx = \left(\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} + 5x\right)_0^2 = \frac{32}{3} \text{ (square units).}$$

**Paragraph for Question Nos. 30 to 32**

Consider the functions defined implicitly by the equation $y^3 - 3y + x = 0$ on various intervals in the real line. If $x \in (-\infty, -2) \cup (2, \infty)$, the equation implicitly defines a unique real valued differentiable function $y = f(x)$. If $x \in (-2, 2)$, the equation implicitly defines a unique real valued differentiable function $y = g(x)$ satisfying $g(0) = 0$.

माना एक फलन अस्पष्ट रूप में समीकरण $y^3 - 3y + x = 0$ द्वारा वास्तविक संख्या रेखा के अन्तरालों के लिये परिभाषित है यदि $x \in (-\infty, -2) \cup (2, \infty)$ तब समीकरण एक अद्वितीय वास्तविक मान वाला अवकलनीय फलन $y = f(x)$ द्वारा परिभाषित है और यदि $x \in (-2, 2)$ तब समीकरण, अद्वितीय वास्तविक अवकलनीय फलन $y = g(x)$, $g(0) = 0$, द्वारा परिभाषित है।

30. If $f(-10\sqrt{2}) = 2\sqrt{2}$, then $f''(-10\sqrt{2}) =$

यदि $f(-10\sqrt{2}) = 2\sqrt{2}$, तो $f''(-10\sqrt{2}) =$

(A) $\frac{4\sqrt{2}}{7^3 3^2}$ (B) $-\frac{4\sqrt{2}}{7^3 3^2}$ (C) $\frac{4\sqrt{2}}{7^3 3}$ (D) $-\frac{4\sqrt{2}}{7^3 3}$

**Ans.** B

**Sol.** $y3 - 3y + x = 0$

$3y2y' - 3y' + 1 = 0$ $\qquad y' = \frac{-1}{3(y^2 - 1)}$

$f(-10\sqrt{2}) = 2\sqrt{2}$

$f'(-10\sqrt{2}) = -\frac{1}{3(7)} = -\frac{1}{21}$

$6y(y')2 + 3y2y'' - 3y'' = 0$

$y'' = -\frac{2y(y')^2}{y^2 - 1}$

$f''(-10\sqrt{2}) = \frac{-2(2\sqrt{2})}{441 \times 7} = \frac{-4\sqrt{2}}{7^3 3^2}$

31. The area of the region bounded by the curve $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$, where $-\infty < a < b < -2$, is :

वक्र $y = f(x)$, x-अक्ष और रेखाओं $x = a$ और $x = b$ जहाँ $-\infty < a < b < -2$, द्वारा परिबद्ध क्षेत्रफल है :

(A) $\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx + b f(b) - a f(a)$ (B) $-\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx + b f(b) - a f(a)$

(C) $\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx - b f(b) + a f(a)$ (D) $-\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx - b f(b) + a f(a)$

**Ans.** A

**Sol.** $\int_a^b f(x)dx = [xf(x)]ab - \int_a^b xf'(x)dx$

$= bf(b) - af(a) + \int_a^b \frac{x}{3\left[(f(x))^2 - 1\right]} dx$

$= \int_a^b \frac{x}{3\left[(f(x))^2 - 1\right]} dx + bf(b) - af(a)$

32. $\int_{-1}^{1} g'(x)\,dx =$

(A) $2g(-1)$ (B) 0 (C) $-2\,g(1)$ (D) $2\,g(1)$

**Ans.** D

**Sol.** $y^3 - 3y + x = 0$

when $x \in (-2, 2)$ $y = g(x)$ and $g(0) = 0$

$(g(x))^3 - 3g(x) + x = 0$

$(g(-x))^3 - 3g(-x) - x = 0$

$+ \quad + \quad +$

---

$(g(x))^3 + (g(-x))^3 - 3(g(x) + g(-x)) = 0$

$\{g(x) + g(-x)\}^2 \{(g(x)^2 + (g(-x))^2 - g(x).g(-x) - 3\} = 0$

$\Rightarrow g(x) + g(-x) = 0$

$\Rightarrow g(x) = -g(-x)$ _____(A)

$\therefore$ $g(x)$ is odd

Now $\int_{-1}^{1} g'(x)\,dx = g(x)\Big|_{-1}^{1}$

$= g(1) - g(-1)$ _____(1)

Put $x = 1$ in A

$g(1) = -g(-1)$

$g(-1) = -g(1)$

$\therefore$ Equation becomes $g(1) - (-g(1))$

$\Rightarrow$ $2g(1)$

## ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

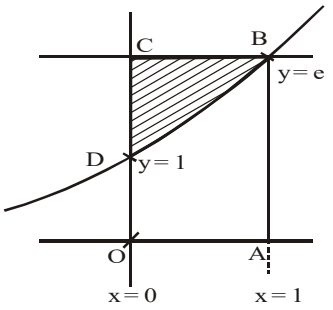
33. Area of the region bounded by the curve $y = e^x$ and lines $x = 0$ and $y = e$ is : **[JEE-2009]**

वक्र $y = e^x$ और रेखाओं $x = 0$ और $y = e$ द्वारा परिबद्ध क्षेत्रफल है : **[JEE-2009]**

(A) $e - 1$ (B) $\int_1^e \ln(e+1-y)dy$ (C) $e - \int_0^1 e^x dx$ (D) $\int_1^e \ln y \, dy$

**Ans.** BCD

**Sol.** $A = \int_1^e \ln y \, dy$




Apply

$= \int_1^e \ln(e+1-y)dy$

$A = ar(OABC) - ar(OABD)$

$= e - \int_1^e e^x dx$ .

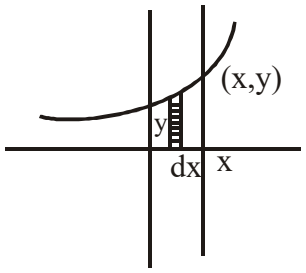
34. The area bounded by a curve, the axis of co-ordinates & the ordinate of some point of the curve is equal to the length of the corresponding arc of the curve . If the curve passes through the point $P(0, 1)$ then the equation of this curve can be :

वक्र, निर्देशी अक्षों तथा वक्र के किसी बिन्दु की कोटि द्वारा घिरा क्षेत्रफल, वक्र की संगत चाप की लम्बाई के बराबर है तथा वक्र बिन्दु $P(0, 1)$ से गुजरता है तब वक्र का समीकरण होगा :

(A) $y = \frac{1}{2}(e^x - e^{-x} + 2)$ (B) $y = \frac{1}{2}(e^x + e^{-x})$ (C) $y = 1$ (D) $y = \frac{2}{e^x + e^{-x}}$

**Ans.** BC

**Sol.**




$A = \int_0^x y dx = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

$$y^2 = 1 + \left(\frac{dy}{dx}\right)^2$$

Case-I $y \neq 1$ Case-II $y = 1$

$$\int \frac{dy}{\sqrt{y^2-1}} = \pm \int dx \qquad \because y \neq 1$$

$$\ln | y + \sqrt{y^2-1} | = x + C$$

$$(0,1) \Rightarrow C = 0$$

$$\ell n \left| y + \sqrt{y^2-1} \right| = x$$

$$y + \sqrt{y^2-1} > 0$$

$$y + \sqrt{y^2-1} = e^x$$

$$y - \sqrt{y^2-1} = e^{-x}$$

add $y = \frac{e^x + e^{-x}}{2}$

35. Let T be the triangle with vertices (0, 0), (0, $c^2$) and (c, $c^2$) and let R be the region between $y = cx$ and $y = x^2$ where $c > 0$ then :

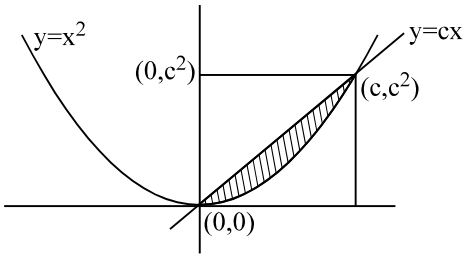
यदि T एक त्रिभुज है जिसके शीर्ष (0, 0), (0, $c^2$) तथा (c, $c^2$) है तथा यदि R, $y = cx$ एवं $y = x^2$ द्वारा घिरा क्षेत्र है तो : (जहाँ $c > 0$)

(A) Area (R) = $\frac{c^3}{6}$ (B) Area of R = $\frac{c^3}{3}$

(C) $\lim\limits_{c \to 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = 3$ (D) $\lim\limits_{c \to 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = \frac{3}{2}$

**Ans.** AC

**Sol.** Area (T) = $\frac{c \cdot c^2}{2} = \frac{c^3}{2}$




$$\text{Area (R)} = \frac{c^3}{2} - \int_0^c x^2\,dx$$

$$= \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6}$$

$$\therefore \quad \lim_{c\to 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = \lim_{c\to 0^+} \frac{c^3}{2}\cdot\frac{6}{c^3} = 3$$

36. Suppose $f$ is defined from $R \to [-1, 1]$ as $f(x) = \frac{x^2-1}{x^2+1}$ where R is the set of real number. Then the statement which does not hold is :

(A) $f$ is many one onto

(B) $f$ increases for $x > 0$ and decrease for $x < 0$

(C) minimum value is not attained even though f is bounded

(D) the area included by the curve $y = f(x)$ and the line $y = 1$ is $\pi$ sq. units

माना $f$, $R \to [-1, 1]$ में परिभाषित है $f(x) = \frac{x^2-1}{x^2+1}$ जहाँ R वास्तविक सख्याएं है तब कौनसा कथन असत्य है :

(A) $f$ बहुएकैकी आच्छादक है

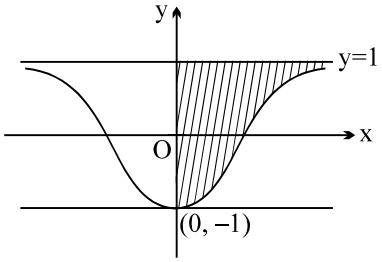
(B) $f$, $x > 0$ में वर्धमान तथा $x < 0$ में ह्रासमान

(C) यदि f परिबद्ध है तो भी न्यूनतम मान गृहण नहीं करता

(D) वक्र $y = f(x)$ तथा रेखा $y = 1$ द्वारा परिबद्ध क्षेत्रफल $\pi$ वर्ग ईकाई है

**Ans.** ACD

**Sol.** $y = f(x) = \frac{x^2-1}{x^2+1} = 1 - \frac{2}{x^2+1}$




$$f'(x) = \frac{4x}{(x^2+1)^2}$$

$x > 0$, f is increasing and $x < 0$ f is decreasing $\Rightarrow$ (B) is true

range is $[-1, 1)$ $\Rightarrow$ into $\Rightarrow$ (A) is false;

minimum value occurs at $x = 0$ and $f(0) = -1$ $\Rightarrow$ (C) is false

$$A = 2\int_0^\infty \left(1 - \frac{x^2-1}{x^2+1}\right)dx = 4\int_0^\infty \frac{dx}{x^2+1} = 4\cdot\tan^{-1}\Big|_0^\infty = 4\cdot\frac{\pi}{2} = 2\pi \quad \Rightarrow \quad \text{(D) is false]}$$

## MATCH THE COLUMN TYPE QUESTION(S)

37. Match the following column :

| **Column I** | **Column II** |
|---|---|
| (A) The area between the curves $y = 2x^4 - x^2$, the x-axis and the ordinates of two minimum of the curve is | (P) 2 sin 1 |
| (B) The area bounded by the curve $x = at^2$, $y = 2at$, and the x-axis $(1 \le t \le 3)$ is | (Q) $\frac{7}{120}$ |
| (C) The area of a circle centred at (1, 2) and passing through (4, 6) is | (R) $\frac{104a^2}{3}$ |
| (D) The area bounded by $y = \cos x$, $y = 0$, $\lvert x \rvert = 1$ is given by | (S) $25\pi$ |

निम्न कॉलम को सुमेलित कीजिए :

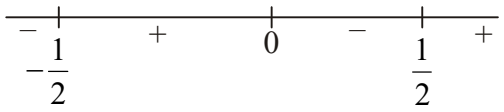
| **Column I** | **Column II** |
|---|---|
| (A) वक्र $y = 2x^4 - x^2$ के दो निम्निष्ठों के अक्ष निर्देशांक $x = x_1$, $x_2$ है, तो वक्र, ग. अक्ष तथा रेखाओं $x = x_1$ व $x = x_2$ से घिरा हुआ क्षेत्रफल होगा | (P) 2 sin 1 |
| (B) वक्र $x = at^2$, $y = 2at$ तथा x-अक्ष $(1 \le t \le 3)$ के मध्य का क्षेत्रफल होगा | (Q) $\frac{7}{120}$ |
| (C) वृत का क्षेत्रफल जिसका केन्द्र (1, 2) तथा जो (4, 6) से गुजरता है | (R) $\frac{104a^2}{3}$ |
| (D) वक्रों $y = \cos x$, $y = 0$ तथा $\lvert x \rvert = 1$ से घिरा हुआ क्षेत्रफल होगा | (S) $25\pi$ |

**Ans.** A - Q, B - R, C - S, D - P

**Sol.** (a) $y = 2x^4 - x^2$

$y' = 8x^3 - 2x$

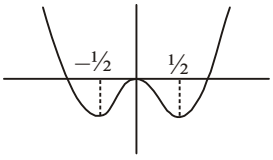
$= 2x(4x^2 - 1)$




$\min \text{ at} = -\frac{1}{2}, \frac{1}{2}$

$$A = -2\int_0^{\frac{1}{2}} \left(2x^4 - x^2\right)$$

$= \frac{7}{120}$




(b) Eliminating t, we obtain the equation of the curve as $y^2 = 4ax$

$t = 1 \Rightarrow x = a$ and $t = 3 \Rightarrow x = 9a$

Hence required area

$$= \int_a^{9a} y\,dx = \int_a^{9a} 2\sqrt{a}\sqrt{x}\,dx$$

$$= 2\sqrt{a}\left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_a^{9a} = \frac{4}{3}\sqrt{2}\left[(9a)^{\frac{3}{2}} - a^{\frac{3}{2}}\right]$$

$$= \frac{4}{3}\sqrt{a}\left[27a^{\frac{3}{2}} - a^{\frac{3}{2}}\right] = \frac{4}{3}\sqrt{a}\left[26a^{\frac{3}{2}}\right]$$

$$\frac{104}{3}a^2$$

(c) Given that the centre of the circle is (1,2)

Radius of the circle = $\sqrt{(4-1)^2 + (6-2)^2}$

$= \sqrt{9+16}$

$= 5$

So, the area of the circle = $\pi r^2$

$= \pi \times (5)^2$

$= 25\pi$

38. Match the following column :

| **Column I** | **Column II** |
|---|---|
| (A) The area bounded by the curves $y = 2\cos x$ and the x-axis from $x = 0$ to $x = 2\pi$ is | (P) $\frac{(\pi^3 - 8)}{4}$ |
| (B) The area in the first quadrant between $x^2 + y^2 = \pi^2$ and $y = \sin x$ is | (Q) 1 |
| (C) The measurement of the area bounded by the coordinate axes and the curve $y = \log_e x$ is | (R) 2a |
| (D) The area of the loop between the curve $y = a\sin x$ and x-axis is | (S) 8 |

निम्न कॉलम को सुमेलित कीजिए :

| **Column I** | **Column II** |
|---|---|
| (A) वक्रों $y = 2\cos x$, x- अक्ष तथा $x = 0,\ x = 2\pi$ के मध्य का क्षेत्रफल होगा | (P) $\frac{(\pi^3 - 8)}{4}$ |
| (B) वक्रों $x^2 + y^2 = \pi^2$ तथा $y = \sin x$ के मध्य प्रथम चुतुर्थांश में क्षेत्रफल होगा | (Q) 1 |
| (C) निर्देशांक अक्षों तथा $y = \log_e x$ के मध्य क्षेत्रफल होगा | (R) 2a |
| (D) वक्र $y = a\sin x$ तथा x-अक्ष के मध्य के घेरे (loop) का क्षेत्रफल होगा | (S) 8 |

**Ans.** A - S, B - P, C - Q, D - R

**Sol.** (A) $A = 4\times\int_0^{\frac{\pi}{2}} 2\cos x$

$= 4\times 2\sin x\Big|_0^{\frac{\pi}{2}}$

$= 8$

(B) $x^2 + y^2 = \pi^2$

$y = \sin x$

$\text{Required Area} = \frac{1}{4}\pi r^2 - \int_0^{\pi}\sin x dx$

$= \frac{1}{4}\pi^3 - 2 = \frac{\pi^3 - 8}{4}$

(C) $A = -\int_0^1 \log_e x dx$

$= -\left(x\log_e x - x\right)_0^1$

$= -(-1)$

$= 1$

(D) $A = \int_0^{\pi} a\sin x dx$

$= -a\cos x\Big|_0^{\pi}$

$= 2a$

## Exercise J-M

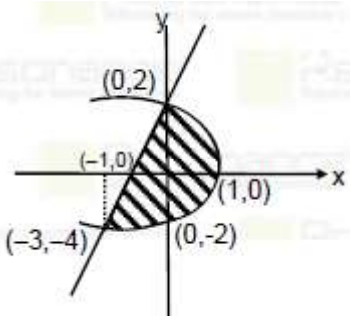
1. The area enclosed by the curves $y^2 + 4x = 4$ and $y - 2x = 2$ is : **[JEE(Main)Jan.2023]**

   वक्रो $y^2 + 4x = 4$ तथा $y - 2x = 2$ से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

   (A) $\frac{25}{3}$ (B) 9 (C) $\frac{23}{3}$ (D) $\frac{22}{3}$

**Ans.** B

**Sol.**




Required area $= \int_{-3}^{0} \sqrt{4-4x}dx - \frac{1}{2}(2)(4) + \frac{1}{2}(1)(2) + 2\int_{0}^{1} \sqrt{4-4x}dx$

$= 2\int_{-3}^{0} \sqrt{1-x}dx + 4\int_{0}^{1} \sqrt{1-x}dx - 3$

$= -2 \times \frac{2}{3}\left[(1-x)^{3/2}\right]_{-3}^{0} - 4\frac{2}{3}\left[(1-x)^{3/2}\right] - 3$

$= -\frac{4}{3}(1-8) - \frac{8}{3}(0-1) - 3$

$= \frac{28}{3} + \frac{8}{3} - 3 = 9$

2. Let T and C respectively be the transverse and conjugate axes of the hyperbola $16x^2 - y^2 + 64x + 4y + 44 = 0$. Then the area of the region above the parabola $x^2 = y + 4$, below the transverse axis T and on the right of the conjugate axis C is : **[JEE(Main)Jan.2023]**

   माना अतिपरवलय $16x^2 - y^2 + 64x + 4y + 44 = 0$ के अनुप्रस्थ तथा संयुग्मी अक्ष क्रमशः T तथा C है। तो परवलय $x^2 = y + 4$ के Åij]vuqLFkkv {kT के नीचे तथा संयुग्मी अक्ष की दांयी तरफ के क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

   (A) $4\sqrt{6} + \frac{44}{3}$ (B) $4\sqrt{6} + \frac{28}{3}$ (C) $4\sqrt{6} - \frac{44}{3}$ (D) $4\sqrt{6} - \frac{28}{3}$

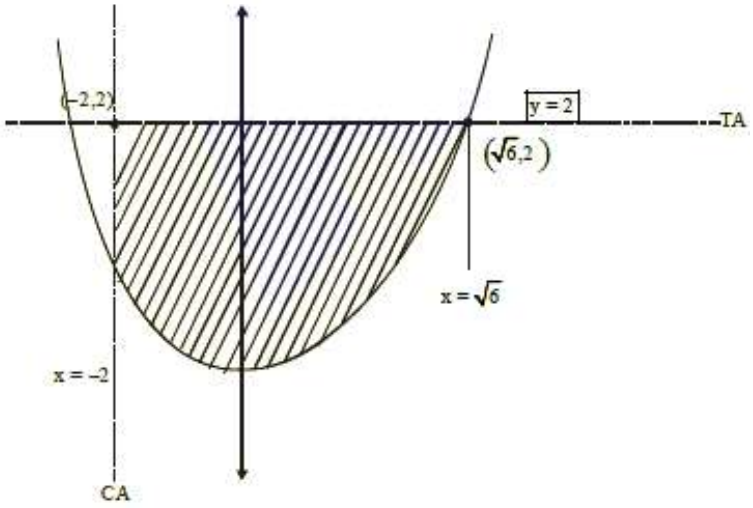
**Ans.** B

**Sol.** $16(x^2 + 4x) - (y^2 - 4y) + 44 = 0$

$16(x + 2)^2 - 64 - (y - 2)^2 + 4 + 44 = 0$

$16(x + 2)^2 - (y - 2)^2 = 16$

$$\frac{(x+2)^2}{1}-\frac{(y-2)^2}{16}=1$$




$$A=\int_{-2}^{\sqrt{6}}\left(2-\left(x^2-4\right)\right)dx$$

$$A=\int_{-2}^{\sqrt{6}}\left(6-x^2\right)dx=\left(6x-\frac{x^3}{3}\right)_{-2}^{\sqrt{6}}$$

$$A=\left(6\sqrt{6}-\frac{6\sqrt{6}}{3}\right)-\left(-12+\frac{8}{3}\right)$$

$$A=\frac{12\sqrt{6}}{3}+\frac{28}{3}$$

$$A=4\sqrt{6}+\frac{28}{3}$$

3. Let $A=\left\{(x,y)\in IR^2 : y\geq 0, 2x\leq y\leq \sqrt{4-(x-1)^2}\right.$ and

$B=\left\{(x,y)\in IR\times IR : 0\leq y\leq \min\left\{2x,\sqrt{4-(x-1)^2}\right\}\right\}$.

Then the ratio of the area of A to the area of B is : **[JEE(Main)Jan.2023]**

माना $A=\left\{(x,y)\in IR^2 : y\geq 0, 2x\leq y\leq \sqrt{4-(x-1)^2}\right.$ और

$B=\left\{(x,y)\in IR\times IR : 0\leq y\leq \min\left\{2x,\sqrt{4-(x-1)^2}\right\}\right\}$ है।

तब A के क्षेत्रफल तथा B के क्षेत्रफल का अनुपात है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi+1}{\pi-1}$ (B) $\frac{\pi}{\pi+1}$ (C) $\frac{\pi}{\pi-1}$ (D) $\frac{\pi-1}{\pi+1}$

**Ans.** D

**Sol.** $2x \le y \le \sqrt{4-(x-1)^2} : y \ge 0$

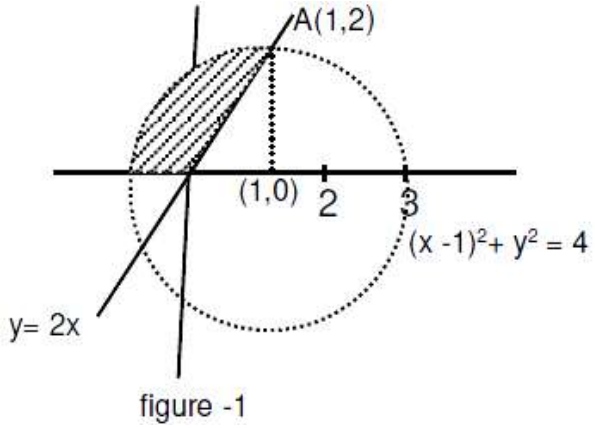


figure -1

area of A : area of shaded Region as show in figure–1

$\therefore$ area of $A = \dfrac{\pi \times 4}{2} - \left(\dfrac{1}{2} \times 1 \times 2\right)$

$= \pi - 1$

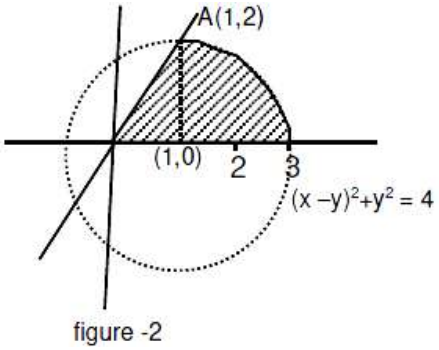


figure -2

area of B : area of shaded Region in as show in above figure–2

$\therefore$ area of $B = \dfrac{1}{2} \times 1 \times 2 + \dfrac{\pi \times 4}{4} = (1+\pi)$

$\dfrac{A}{B} = \dfrac{\text{area of A}}{\text{area of B}} = \dfrac{\pi - 1}{\pi + 1}$

4. Let $\Delta$ be the area of the region $\left\{(x,y) \in IR^2 : x^2 + y^2 \le 21, y^2 \le 4x, x \ge 1\right\}$.

Then $\dfrac{1}{2}\left(\Delta - 21\sin^{-1}\dfrac{2}{\sqrt{7}}\right)$ is equal to : **[JEE(Main)Jan.2023]**

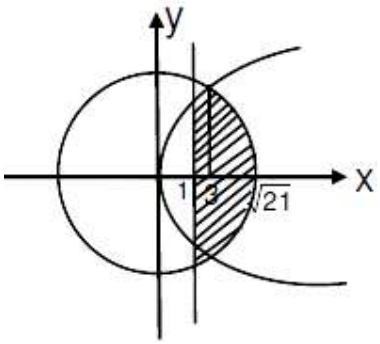
माना क्षेत्र $\left\{(x,y) \in IR^2 : x^2 + y^2 \le 21, y^2 \le 4x, x \ge 1\right\}$ का क्षेत्रफल $\Delta$ है, तब $\dfrac{1}{2}\left(\Delta - 21\sin^{-1}\dfrac{2}{\sqrt{7}}\right)$ बराबर है :

**[JEE(Main)Jan.2023]**

(A) $2\sqrt{3}-\frac{2}{3}$ (B) $\sqrt{3}-\frac{2}{3}$ (C) $2\sqrt{3}-\frac{1}{3}$ (D) $\sqrt{3}-\frac{4}{3}$

**Ans.** D

**Sol.**




$$\text{Required area} = 2\left(\int_1^2 2\sqrt{x}dx + \int_3^{\sqrt{21}}\sqrt{21-x^2}dx\right) = 2\left(2\left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_1^3 + \left[\frac{x}{2}\sqrt{21-x^2}+\frac{21}{2}\sin^{-1}\left(\frac{x}{\sqrt{21}}\right)\right]_3^{\sqrt{21}}\right)$$

$$2\left(\frac{4}{3}\left(3\sqrt{3}-1\right)+\frac{21\pi}{4}-\frac{3}{2}\times 2\sqrt{3}-\frac{21}{2}\sin^{-1}\left(\sqrt{\frac{3}{7}}\right)\right) = 8\sqrt{3}-\frac{8}{3}+\frac{21\pi}{2}-6\sqrt{3}-21\sin^{-1}\left(\sqrt{\frac{3}{7}}\right)$$

$$\therefore \Delta = 2\sqrt{3}+\frac{21\pi}{2}-\frac{8}{3}-21\sin^{-1}\left(\sqrt{\frac{3}{7}}\right)$$

$$\therefore \frac{1}{2}\left(\Delta - 21\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)\right)$$

$$\frac{1}{2}\left[2\sqrt{3}-\frac{8}{3}+\frac{21\pi}{2}-21\left(\sin^{-1}\left(\frac{\sqrt{3}}{7}\right)+\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)\right)\right] \qquad \because \sin^{-1}\left(\frac{\sqrt{3}}{\sqrt{7}}\right)+\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)=\frac{\pi}{2}$$

$$\frac{1}{2}\left(\Delta - 21\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)\right) = \frac{1}{2}\left[2\sqrt{3}-\frac{8}{3}+\frac{21\pi}{2}-21\left(\frac{\pi}{2}\right)\right] = \sqrt{3}-\frac{4}{3}$$

5. The area of the region $A=\left\{(x,y):|\cos x-\sin x|\le y\le \sin x, 0\le x\le \frac{\pi}{2}\right\}$ is : **[JEE(Main)Jan.2023]**

क्षेत्र $A=\left\{(x,y):|\cos x-\sin x|\le y\le \sin x, 0\le x\le \frac{\pi}{2}\right\}$ का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

(A) $\sqrt{5}-2\sqrt{2}+1$ (B) $1-\frac{3}{\sqrt{2}}+\frac{4}{\sqrt{5}}$

(C) $\frac{3}{\sqrt{5}}-\frac{3}{\sqrt{2}}+1$ (D) $\sqrt{5}+2\sqrt{2}-4.5$

**Ans.** A

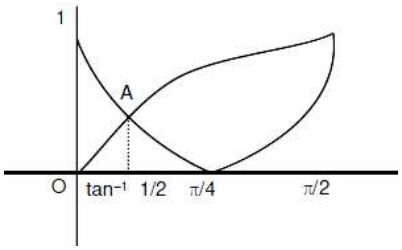
**Sol.** $$y=\begin{cases}\cos x-\sin x & : \quad 0\le x\le\frac{\pi}{4}\\ \sin x-\cos x & : \quad \frac{\pi}{4}<x\le\frac{\pi}{2}\end{cases}$$

$$y^1=\begin{cases}-\sin x-\cos x & : \quad 0\le x\le\frac{\pi}{4}\\ \cos x+\sin x & : \quad \frac{\pi}{4}<x\le\frac{\pi}{2}\end{cases}$$

For pt. A

$\cos x-\sin x=\sin x$

$\tan x=\frac{1}{2}$




Let $x=\tan^{-1}\frac{1}{2}$

Area $=\int_{\tan^{-1}1/2}^{\pi/4}(\sin x-(\cos x-\sin x))dx+\int_{\pi/4}^{\pi/2}(\sin x-\sin x+\cos x)dx$

$=\left(-2\cos x-\sin x\right)_x^{\pi/4}+\left(\sin x\right)_{\pi/4}^{\pi/2}$

$=\left(\frac{-2}{\sqrt{2}}-\frac{1}{\sqrt{2}}\right)-\left(-2\cos x-\sin x\right)+\left(1-\frac{1}{\sqrt{2}}\right)$

$=-\sqrt{2}-\frac{1}{\sqrt{2}}+2\cos x+\sin x+1-\frac{1}{\sqrt{2}}=-2\sqrt{2}+1+\frac{2\times2}{\sqrt{5}}+\frac{1}{\sqrt{5}}$

$=\sqrt{5}+1-2\sqrt{2}$

6. Let q be the maximum integral value of p in [0,10] for which the roots of the equation $x^2-px+\frac{5}{4}p=0$ are rational. Then the area of the region $\left\{(x,y):0\le y\le(x-q)^2,0\le x\le q\right\}$ is : **[JEE(Main)Jan.2023]**

माना [0,10] में p का अधिकतम पूर्णांक मान, जिसके लिए समीकरण $x^2 - px + \frac{5}{4}p = 0$ के मूल परिमेय q है। तब क्षेत्र $\{(x,y) : 0 \le y \le (x-q)^2, 0 \le x \le q\}$ का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

(A) 25 (B) $\frac{125}{3}$ (C) 164 (D) 243

**Ans.** D

**Sol.** $D = p^2 - 5p$ = Perfect square of a rational number

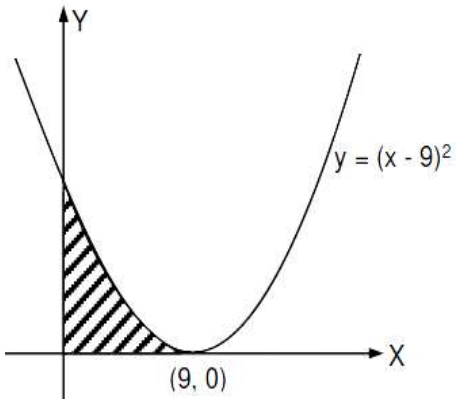
$\Rightarrow D = p(p-5)$

Noe put values in reverse orde as q is maximum value of p

p = 10 reject

p = 9 select

$\Rightarrow q = 9$




So area of shaded region $= \int_0^9 (x-9)^2 dx$

$= \left[\frac{(x-9)^3}{3}\right]_0^9$

$\Rightarrow$ 243 sq. unit

7. The area enclosed by the closed curve C given by the differential equation $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$, $y(1) = 0$ is $4\pi$. Let P and Q be the points of intersection of the curve C and the y-axis. If normals at P and Q on the curve C intersect x-axis at points R and S respectively, then the length of the line segment RS is : **[JEE(Main)Jan.2023]**

अवकल समीकरण $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$, $y(1) = 0$ से प्राप्त संवृत्त वक्र C से घिरे क्षेत्र का क्षेत्रफल $4\pi$ है। माना वक्र C व y अक्ष के प्रतिच्छेदन बिंदु P व Q है। यदि वक्र C के बिंदु P व Q पर अभिलम्ब x अक्ष को क्रमशः बिंदु R व S पर मिलते है, तब रेखा खण्ड RS की y Ecbz S **[JEE(Main)Jan.2023]**

(A) $\frac{2\sqrt{3}}{3}$ (B) 2 (C) $2\sqrt{3}$ (D) $\frac{4\sqrt{3}}{3}$

**Ans.** D

**Sol.** $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$

$\frac{dy}{dx} = \frac{x+a}{2-y}$

$(2-y)\,dy = (x+a)\,dx$

$2y - \frac{y^2}{2} = \frac{x^2}{2} + ax + c$

$a+c=-\frac{1}{2}$ as $y(1)=0$

$X^2+Y^2+2ax-4y-1-2a=0$

$\pi r^2=4\pi$

$r^2=4$

$4=a^2+4+1+2a$

$(a+1)^2=0$

P or $Q=\left(0,2\pm\sqrt{3}\right)$

Equation of normal at P, Q are $y-2=\sqrt{3}(x-1)$

$y-2=-\sqrt{3}(x-1)$

$R=\left(1-\frac{2}{\sqrt{3}},0\right)$

$S=\left(1+\frac{2}{\sqrt{3}},0\right)$

$RS=\frac{4}{\sqrt{3}}=4\frac{\sqrt{3}}{3}$

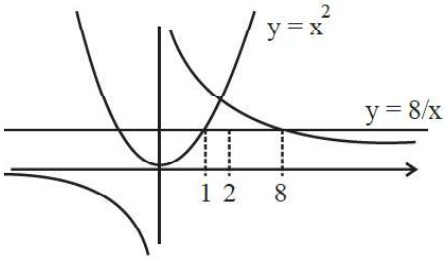
8. The area of the region given by $\left\{(x,y):xy\le 8,1\le y\le x^2\right\}$ is : **[JEE(Main)Jan.2023]**

$\left\{(x,y):xy\le 8,1\le y\le x^2\right\}$ द्वारा दिए क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

(A) $8\log_e 2+\frac{7}{6}$ (B) $16\log_e 2-\frac{14}{3}$ (C) $8\log_e 2-\frac{13}{3}$ (D) $16\log_e 2+\frac{7}{3}$

**Ans.** B

**Sol.** $xy\le 8$ $1\le y\le x^2$




Area $=\int_1^2\left(x^2-1\right)dx+\int_2^8\left(\frac{8}{x}-1\right)dx$

$=\left(\frac{x^3}{3}\right)_1^2+8\left(\ell nx\right)_2^8-\left(x\right)_1^8$

$$=\frac{7}{3}+8\left(2\ell n2\right)-7$$

$$=16\ell n2-\frac{14}{3}$$

9. If the area of the region bounded by the curves $y^2 - 2y = -x$, $x + y = 0$ is A, then 8 A is equal to _____.

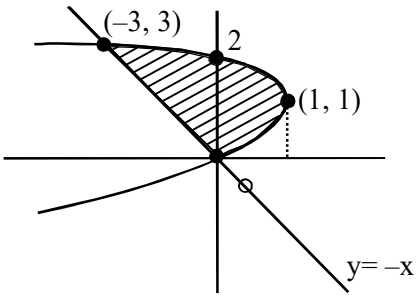
**[JEE(Main)Jan.2023]**

यदि वक्रों $y^2 - 2y = -x$, $x + y = 0$ से घिरे क्षेत्र का क्षेत्रफल A है, तो 8 A बराबर है _____

**[JEE(Main)Jan.2023]**

**Ans.** 36

**Sol.**




$$A=\int_0^3\left[\left(2y-y^2\right)-(-y)\right]dy=\int_0^3\left(3y-y^2\right)dy=\left[\frac{3y^2}{2}-\frac{y^3}{3}\right]_0^3$$

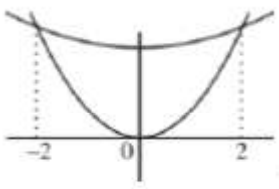
$$=\frac{3\times9}{2}-\frac{27}{3}=\frac{9}{2}$$ So, 8A = 36

10. If the area enclosed by the parabolas $P_1 : 2y = 5x^2$ and $P_2 : x^2 - y + 6 = 0$ is equal to the area enclosed by $P_1$ and $y = \alpha x$, $\alpha > 0$, then $\alpha^3$ is equal to _____. **[JEE(Main)Jan.2023]**

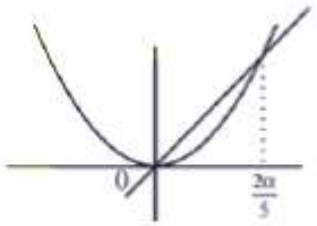
यदि परवलयों $P_1 : 2y = 5x^2$ तथा $P_2 : x^2 - y + 6 = 0$ से घिरे क्षेत्र का क्षेत्रफल, परवलय $P_1$ तथा $y = \alpha x$, $\alpha > 0$ से घिरे क्षेत्र के {kslkQy d scjkcj g\$r ks$\alpha^3$ है_____ **[JEE(Main)Jan.2023]**

**Ans.** 600

**Sol.**




Abscissa of point of intersection of $2y = 5x^2$
and $y = x^2 + 6$ is $\pm 2$



$$\text{Area } = 2\int_0^2\left(x^2+6-\frac{5x^2}{2}\right)dx = \int_0^{\frac{2\alpha}{5}}\left(\alpha x-\frac{5x^2}{2}\right)dx$$

$$\Rightarrow \int_0^{\frac{2\alpha}{5}}\left(\alpha x-\frac{5x^2}{2}\right)dx = 16$$

$$\Rightarrow \alpha^3 = 600$$

11. Let $\alpha$ be the area of the larger region bounded by the curve $y^2 = 8x$ and the lines $y = x$ and $x = 2$, which lies in the first quadrant. Then the value of $3\alpha$ is equal to _____. **[JEE(Main)Jan.2023]**

माना प्रथम चतुर्थांश में वक्र $y^2 = 8x$ और रेखाओं $y = x$ एवं $x = 2$ से घिरे बड़े क्षेत्र का क्षेत्रफल $\alpha$ है। तो $3\alpha$ का मान बराबर है _____ **[JEE(Main)Jan.2023]**

**Ans.** 22

**Sol.** On solving $y^2 = 8x$ and $y = x$, we get

$x = 0, 8$

Now shaded area $= \int_2^8\left(2\sqrt{2}\cdot\sqrt{x}-x\right)dx$

$$= 2\sqrt{2}\int_2^8\sqrt{x}\cdot dx-\int_2^8 x\,dx$$

$$= \frac{4\sqrt{2}}{3}\left[x^{\frac{3}{2}}\right]_2^8-\frac{1}{2}\left[x^2\right]_2^8$$

$$= \frac{4\sqrt{2}}{3}\left[8^{\frac{3}{2}}-2^{\frac{3}{2}}\right]-\frac{1}{2}\left[8^2-2\right]$$

$$= \frac{4\sqrt{2}\times 2\sqrt{2}}{3}[8-1]-\frac{1}{2}\times 60$$

$$= \frac{8\times 2\times 7}{3}-30 = \frac{22}{3} = \alpha$$

$$3\alpha = 22$$

12. Let A be the area of the region $\left\{(x,y): y\ge x^2, y\ge(1-x)^2, y\le 2x(1-x)\right\}$. Then 540 A is equal to _____. **[JEE(Main)Jan.2023]**

माना क्षेत्र $\left\{(x,y): y \geq x^2, y \geq (1-x)^2, y \leq 2x(1-x)\right\}$ का क्षेत्रफल A है। तब 540 A का मान बराबर है_____

**[JEE(Main)Jan.2023]**

**Ans.** 25

**Sol.** $A = 2\int_{\frac{1}{3}}^{\frac{1}{2}} \left(2x - 2x^2 - (1-x)^2\right)dx$

$= 2\left[2x^2 - x^3 - x\right]_{1/3}^{1/2}$

So $A = \frac{5}{108} \Rightarrow 540A = \frac{5}{108} \times 540 = 25$

13. Let for $x \in R$ $f(x) = \frac{x+|x|}{2}$ and $g(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$. The area bounded by the curve $y = (fog)(x)$ and the lines $y = 0$, $2y - x = 15$ is equal to _____.

**[JEE(Main)Jan.2023]**

माना $x \in R$ के लिए $f(x) = \frac{x+|x|}{2}$ तथा $g(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ है, तो वक्र $y = (fog)(x)$ तथा रेखाओं $y = 0$, $2y - x = 15$ से घिरे क्षेत्र का क्षेत्रफल बराबर है_____

**[JEE(Main)Jan.2023]**

**Ans.** 72

**Sol.** $f(x) = \frac{x+|x|}{2} = \begin{bmatrix} x & x \geq 0 \\ 0 & x < 0 \end{bmatrix}$

$g(x) = \begin{bmatrix} x^2 & x \geq 0 \\ x & x < 0 \end{bmatrix}$

$fog(x) = f[g(x)] = \begin{bmatrix} g(x) & g(x) \geq 0 \\ 0 & g(x) < 0 \end{bmatrix}$

$fog(x) = \begin{bmatrix} x^2 & x \geq 0 \\ 0 & x < 0 \end{bmatrix}$

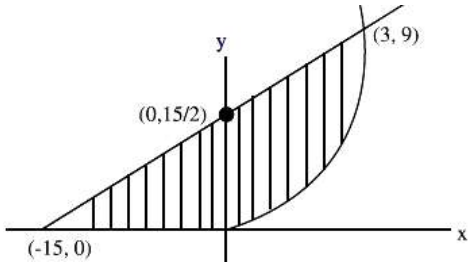
$2y - x = 15$

$A = \int_0^3 \left(\frac{x+15}{2} - x^2\right)dx + \frac{1}{2} \times \frac{15}{2} \times 15$

$\frac{x^2}{4} + \frac{15x}{2} - \frac{x^3}{3}\Big|_0^3 + \frac{225}{4}$

$$= \frac{9}{4} + \frac{45}{2} - 9 + \frac{225}{4} = \frac{99 - 36 + 225}{4}$$

$$= \frac{288}{4} = 72$$




14. Let the area of the region $\left\{(x, y) : |2x - 1| \le y \le \left|x^2 - x\right|, 0 \le x \le 1\right\}$ be A. Then $(6A + 11)^2$ is equal to _____ .

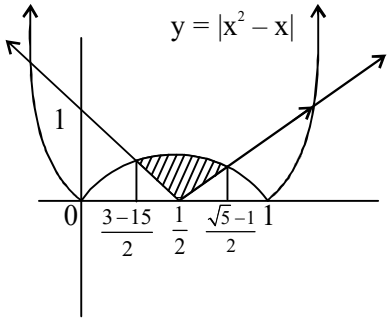
**[JEE(Main)Jan.2023]**

माना क्षेत्र $\left\{(x, y) : |2x - 1| \le y \le \left|x^2 - x\right|, 0 \le x \le 1\right\}$ का क्षेत्रफल A है। तो $(6A + 11)^2$ बराबर है_____

**[JEE(Main)Jan.2023]**

**Ans.** 125

**Sol.** $y \ge |2x - 1|$ $y \le |x^2 - x|$




Both curve are symmetric about $x = \frac{1}{2}$ Hence

$$A = 2\int_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}} \left(\left(x - x^2\right) - \left(1 - 2x\right)\right) dx$$

$$A = 2\int_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}} \left(-x^2 + 3x - 1\right) dx = 2\left(\frac{-x^3}{3} + \frac{3}{2}x^2 - x\right)_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}}$$

on solving $6A + = 5\sqrt{5}$

$(6A + 11)^2 = 125$

15. Let A be the area bounded by the curve $y = x|x-3|$, the x-axis and the ordinates $x = -1$ and $x = 2$. Then 12A is equal to _____. **[JEE(Main)Jan.2023]**

माना वक्र $y = x|x-3|$, x-अक्ष तथा कोटियों $x = -1$ व $x = 2$ से घिरे क्षेत्र का क्षेत्रफल A है। तब 12A का मान है_____ **[JEE(Main)Jan.2023]**

**Ans.** 62

**Sol.** $A = \int_{-1}^{0} \left(x^2 - 3x\right)dx + \int_{0}^{2} \left(3x - x^2\right)dx$

$$\Rightarrow A = \frac{x^3}{3} - \frac{3x^2}{2}\Big|_{-1}^{0} + \frac{3x^2}{2} - \frac{x^3}{3}\Big|_{0}^{2}$$

$$\Rightarrow A = \frac{11}{6} + \frac{10}{3} = \frac{31}{6}$$

$\therefore$ $12A = 62$

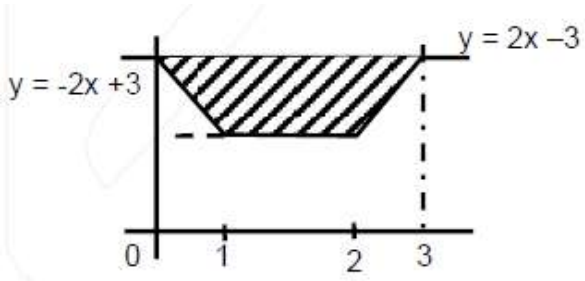
16. The area bounded by the curves $y = |x-1| + |x-2|$ and $y = 3$ is equal to : **[JEE(Main)April,2023]**

वक्रों $y = |x-1| + |x-2|$ तथा $y = 3$ से परिबद्ध क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) 4 (B) 3 (C) 6 (D) 5

**Ans.** A

**Sol.**




$$\text{area} = \frac{1}{2} \times (1+3) \times 2 = 4$$

17. The area of the region $\left\{(x, y) : x^2 \le y \le 8 - x^2, y \le 7\right\}$ is : **[JEE(Main)April,2023]**

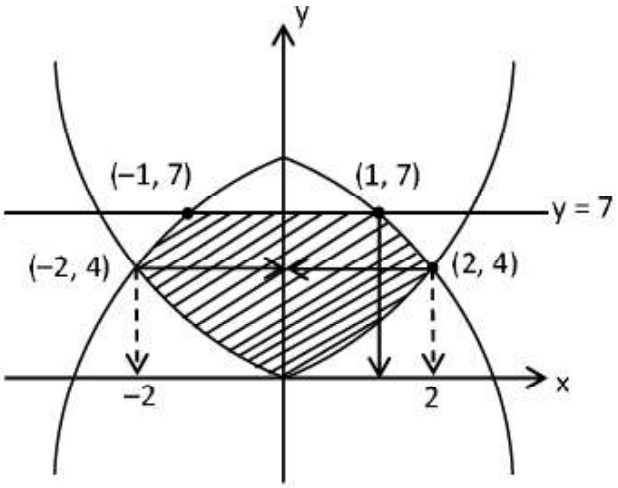
क्षेत्र $\left\{(x, y) : x^2 \le y \le 8 - x^2, y \le 7\right\}$ का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) 18 (B) 24 (C) 20 (D) 21

**Ans.** C

**Sol.** $y \ge x^2 \quad y \le 8 - x^2 \quad y \le 7$

$x^2 = 8 - x^2$




$x^2 = 4$

$x = \pm 2$

$2\left(1.7 + \int_1^2 \left(8 - 2x^2\right)dx\right) - 2\int_0^1 \left(x^2\right)dx$

$= 2\left[7 + \left(8x - \frac{2x^3}{3}\right)_1^2\right] - 2\left(\frac{x^3}{3}\right)_0^1$

$= 2\left[7 + \left(16 - \frac{16}{3}\right) - \left(8 - \frac{2}{3}\right)\right] - 2\left(\frac{1}{3}\right)$

$= 2\left[7 + \frac{32}{3} - \frac{22}{3}\right] = 2\left[7 + \frac{10}{3}\right]\frac{-2}{3}$

$= \frac{60}{3} = 20$

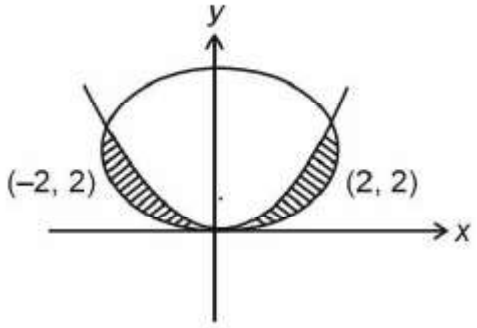
18. Area of the region $\left\{(x, y) : x^2 + (y-2)^2 \le 4, x^2 \ge 2y\right\}$ is : **[JEE(Main)April,2023]**

क्षेत्र $\left\{(x, y) : x^2 + (y-2)^2 \le 4, x^2 \ge 2y\right\}$ का क्षेत्रफल : **[JEE(Main)April,2023]**

(A) $2\pi - \frac{16}{3}$ (B) $\pi + \frac{8}{3}$ (C) $\pi - \frac{8}{3}$ (D) $2\pi + \frac{16}{3}$

**Ans.** A

**Sol.** $x^2 + (y-2)^2 \le 4, x^2 \ge 2y$




Area of required region $= 2\left[\frac{1}{4}\pi(4) - \int_0^2 \sqrt{2}\cdot\sqrt{y}dy\right]$

$\Rightarrow 2\left[\pi - \frac{\sqrt{2}\cdot y^{3/2}}{3/2}\Big|_0^2\right]$

$\Rightarrow 2\left[\pi - \frac{2\sqrt{2}}{3}\cdot 2\sqrt{2}\right] = 2\left[\pi - \frac{8}{3}\right] = 2\pi - \frac{16}{3}$

19. The area of the region enclosed by the curve $y = x^3$ and its tangent at the point $(-1, -1)$ is :

**[JEE(Main)April,2023]**

वक्र $y = x^3$ तथा इसके बिंदु $(-1, -1)$ पर स्पर्श रेखा से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2023]**

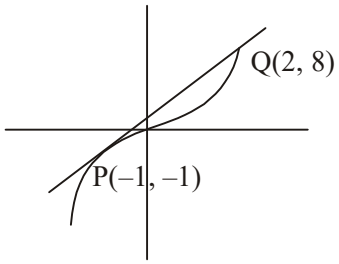
(A) $\frac{23}{4}$ (B) $\frac{19}{4}$ (C) $\frac{27}{4}$ (D) $\frac{31}{4}$

**Ans.** C

**Sol.** Given $y = x^3$ _____(1)

$\Rightarrow \frac{dy}{dx} = 3x^2$

$\left(\frac{dy}{dx}\right)_{(-1,-1)} = 3$




Equation of tangent at $(-1, -1)$

$(y + 1) = 3(x + 1)$

$y = 3x + 2$ _____(2)

Solving (1) and (2)

$x^3 = 3x + 2$

$x^3 - 3x - 2 = 0$ : $-1$, $-1$, $\alpha$

$-2 + \alpha = 0$

$\alpha = 2$

$Q \equiv (2, 8)$

Required area $= \int_{-1}^{2} \left(3x + 2 - x^3\right) dx$

$= \frac{3}{2}(4-1) + 2(2+1) - \frac{1}{4}(16-1)$

$= \frac{9}{2} + 6 - \frac{15}{4} = \frac{18+24-15}{4} = \frac{27}{4}$

20. The area of the region enclosed by the curve $f(x) = \max\{\sin x, \cos x\}, -\pi \le x \le \pi$ and the x-axis is :

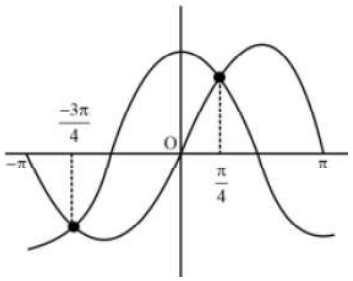
**[JEE(Main)April,2023]**

वक्र $f(x) = \max\{\sin x, \cos x\}, -\pi \le x \le \pi$ तथा x-अक्ष से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) 4 (B) $2(\sqrt{2}+1)$ (C) $2\sqrt{2}(\sqrt{2}+1)$ (D) $4(\sqrt{2})$

**Ans.** A

**Sol.**




Area =

$$\left|\int_{-\pi}^{\frac{-3\pi}{4}} \sin x\,dx\right| + \left|\int_{\frac{-3\pi}{4}}^{\frac{-\pi}{2}} \cos x\,dx\right| + \int_{\frac{-\pi}{2}}^{\frac{\pi}{4}} \cos x\,dx + \int_{\frac{\pi}{4}}^{\pi} \sin x\,dx = 4$$

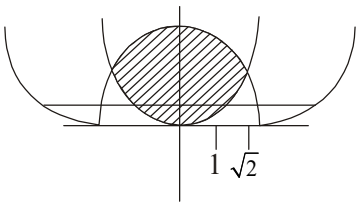
21. The area of the region $\left\{(x,y): x^2 \le y \le |x^2-4|, y \ge 1\right\}$ is : **[JEE(Main)April,2023]**

क्षेत्र $\left\{(x,y): x^2 \le y \le |x^2-4|, y \ge 1\right\}$ का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) $\frac{3}{4}(4\sqrt{2}-1)$ (B) $\frac{4}{3}(4\sqrt{2}+1)$ (C) $\frac{3}{4}(4\sqrt{2}+1)$ (D) $\frac{4}{3}(4\sqrt{2}-1)$

**Ans.** D

**Sol.**




$$A = 2\left(\int_0^1 \left(3-x^2\right)dx + \int_1^{\sqrt{2}} \left(4-2x^2\right)dx\right) = 2\left(\left(3x-\frac{x^3}{3}\right)_0^1 + \left(4x-\frac{2x^3}{3}\right)_1^{\sqrt{2}}\right)$$

$$= 2\left(3-\frac{1}{3}+\left(4\sqrt{2}-\frac{4\sqrt{2}}{3}-4+\frac{2}{3}\right)\right) = 2\left(\frac{-2}{3}+\frac{2}{3}\left(4\sqrt{2}\right)\right) = \frac{4\left(4\sqrt{2}-1\right)}{3}$$

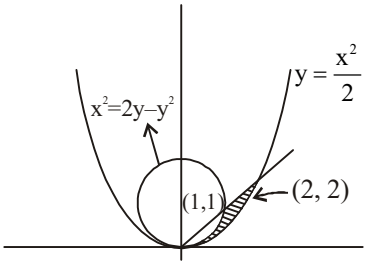
22. If the area of the region $S = \left\{(x,y): 2y-y^2 \le x^2 \le 2y, x \ge y\right\}$ is equal to $\frac{n+2}{n+1} - \frac{\pi}{n-1}$, then the natural number n is equal to _____. **[JEE(Main)April,2023]**

यदि क्षेत्र $S = \left\{(x,y): 2y-y^2 \le x^2 \le 2y, x \ge y\right\}$ का क्षेत्रफल $\frac{n+2}{n+1} - \frac{\pi}{n-1}$ के बराबर है, तब प्राकृतिक संख्या n बराबर है

_____ **[JEE(Main)April,2023]**

**Ans.** 5

**Sol.** $2y - y^2 \le x^2 \le 2y, x \ge y$




$$\text{Area} = \int_1^2 (\sqrt{2y} - y)dy + \int_0^1 \left(\sqrt{2y} - \sqrt{2y - y^2}\right)dy$$

$$= \frac{7 - 2^{\frac{5}{2}}}{6} + \frac{2^{\frac{7}{2}} - 3\pi}{12}$$

$$= \frac{14 - 2^{\frac{7}{2}} + 2^{\frac{7}{2}} - 3\pi}{12} = \frac{7}{6} - \frac{\pi}{4}$$

$n = 5$

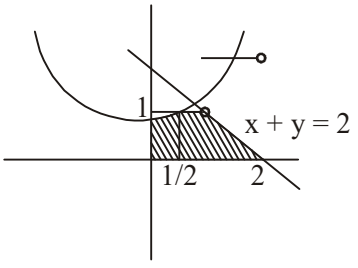
23. Let the area enclosed by the lines $x + y = 2$, $y = 0$, $x = 0$ and the curve $f(x) = \min\left\{x^2 + \frac{3}{4}, 1 + [x]\right\}$ where $[x]$ denotes the greatest integer $\le x$, be A. Then the value of 12A is _____. **[JEE(Main)April,2023]**

माना $[x]$ महत्तम पूर्णांक $\le x$ है। माना रेखाओं $x + y = 2$, $y = 0$, $x = 0$ तथा वक्र $f(x) = \min\left\{x^2 + \frac{3}{4}, 1 + [x]\right\}$ से घिरे क्षेत्र का {ksk Qy A है। तो 12 A का मान है : **[JEE(Main)April,2023]**

**Ans.** 17

**Sol.** $$A = \int_0^{\frac{1}{2}} \left(x^2 + \frac{3}{4}\right)dx + \frac{1}{2}\left(\frac{3}{2} + \frac{1}{2}\right) \times 2$$

$$\frac{1}{3}.\frac{1}{8} + \frac{3}{4}.\frac{1}{2} + 1$$



$$12A = \frac{1}{2} + \frac{9}{2} + 12 = 17$$

24. Let y = p(x) be the parabola passing through the points (–1, 0), (0, 1) and (1, 0). If the area of the region $\left\{(x, y) : (x+1)^2 + (y-1)^2 \le 1, y \le p(x)\right\}$ is A, then $12(\pi - 4A)$ is equal to _____.

**[JEE(Main)April,2023]**

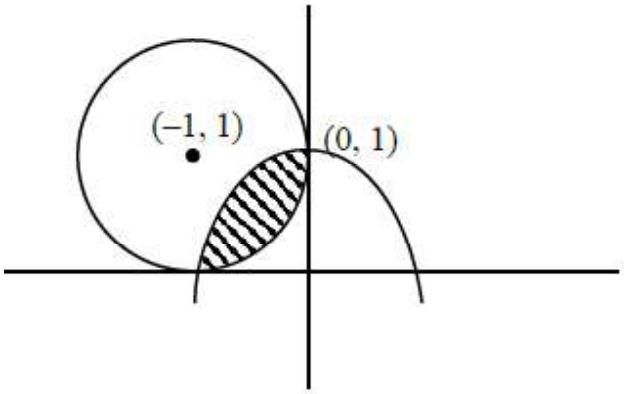
माना बिंदुओं (–1, 0), (0, 1) तथा (1, 0) से होकर जाने वाला परवलय y = p(x) है। यदि क्षेत्र $\left\{(x, y) : (x+1)^2 + (y-1)^2 \le 1, y \le p(x)\right\}$ का क्षेत्रफल A है, तो $12(\pi - 4A)$ बराबर है _____

**[JEE(Main)April,2023]**

**Ans.** 16

**Sol.** There can be infinitely many parabolas through given points.

Let parabola $x^2 = -4a(y-1)$




Passes through (1, 0)

$$\therefore b = -4a(-1) \Rightarrow a = \frac{1}{4}$$

$$\therefore x^2 = -(y-1)$$

Now area covered by parabola $= \int_{-1}^{0} \left(1 - x^2\right) dx$

$$= \left(x - \frac{x^3}{3}\right)_1^b = (0-0) - \left\{-1 + \frac{1}{3}\right\}$$

$$= \frac{2}{3}$$

Required Area = Area of sector – {Area of square – Area covered by Parabola}

$$= \frac{\pi}{4} - \left\{1 - \frac{2}{3}\right\}$$

$$= \frac{\pi}{4} - \frac{1}{3}$$

$$\therefore 12(\pi-4A)=12\left[\pi-4\left(\frac{\pi}{4}-\frac{1}{3}\right)\right]$$

$$=12\left[\pi-\pi+\frac{4}{3}\right]$$

$$=16$$

25. If the area of the region $\left\{(x,y):\left|x^2-2\right|\le y\le x\right\}$ is A, then $6A+16\sqrt{2}$ is equal to _____.

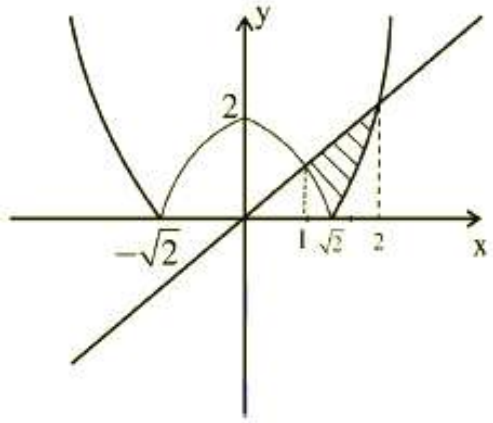
**[JEE(Main)April,2023]**

यदि क्षेत्र $\left\{(x,y):\left|x^2-2\right|\le y\le x\right\}$ का क्षेत्रफल A है, तो $6A+16\sqrt{2}$ बराबर है _____

**[JEE(Main)April,2023]**

**Ans.** 27

**Sol.** $\left|x^2-2\right|\le y\le x$




$$A=\int_1^{\sqrt{2}}\left(x-\left(2-x^2\right)\right)dx+\int_{\sqrt{2}}^{2}\left(x-\left(x^2-2\right)\right)dx$$

$$=\left(1-2\sqrt{2}+\frac{2\sqrt{2}}{3}\right)-\left(\frac{1}{2}-2+\frac{1}{3}\right)+\left(2-\frac{8}{3}+4\right)-\left(1-\frac{2\sqrt{2}}{3}+2\sqrt{2}\right)$$

$$=-4\sqrt{2}+\frac{4\sqrt{2}}{3}+\frac{7}{6}+\frac{10}{3}=\frac{-8\sqrt{2}}{3}+\frac{9}{2}$$

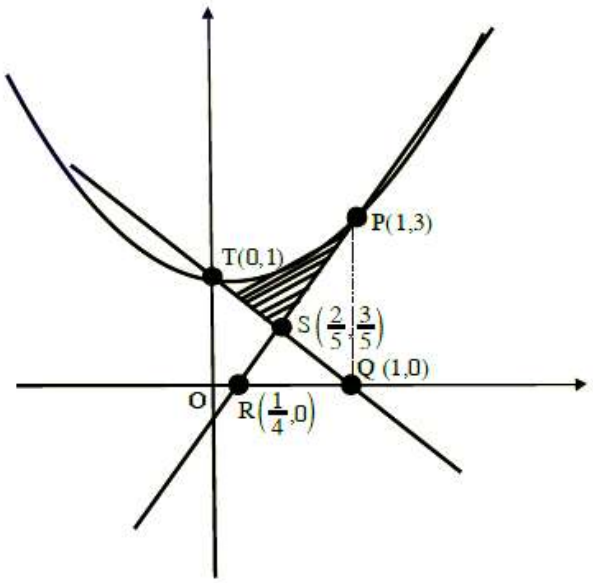
$$6A=-16\sqrt{2}+27\ \therefore 6A+16\sqrt{2}=27$$

26. If A is the area in the first quadrant enclosed by the curve $C: 2x^2-y+1=0$, the tangent to C at the point (1, 3) and the line $x+y=1$, then the value of 60A is _____. **[JEE(Main)April,2023]**

यदि वक्र $C: 2x^2-y+1=0$, C के बिंदु (1, 3) पर स्पर्श रेखा तथा रेखा $x+y=1$ से घिरे, प्रथम चतुर्थांश में, क्षेत्र का क्षेत्रफल A है, तो 60A बराबर है _____ **[JEE(Main)April,2023]**

**Ans.** 16

**Sol.**




$y = 2x^2 + 1$

Tangent at (1, 3)

$y = 4x - 1$

$$A = \int_0^1 \left(2x^2 + 1\right)dx - \text{area of}(\Delta QOT) - \text{are of}(\Delta PQR) + \text{area of}(\Delta QRS)$$

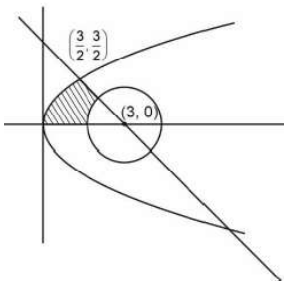
$$A = \left(\frac{2}{3} + 1\right) - \frac{1}{2} - \frac{9}{8} + \frac{9}{40} = \frac{16}{60}$$

27. If the area bounded by the curve $2y^2 = 3x$, lines $x + y = 3$, $y = 0$ and outside the circle $(x - 3)^2 + y^2 = 2$ is A, then $4(\pi + 4A)$ is equal to _____. **[JEE(Main)April,2023]**

यदि वक्र $2y^2 = 3x$, रेखाओं $x + y = 3$, $y = 0$ तथा वृत्त $(x - 3)^2 + y^2 = 2$ के बाहर गिरा हुआ क्षेत्रफल A है, तो $4(\pi + 4A)$ मान gkxk_____ **[JEE(Main)April,2023]**

**Ans.** 42

**Sol.**




$$\int_0^{3/2}\left((3-y) - \frac{2y^2}{3}\right)dy - \pi\left(\sqrt{2}\right)^2 \cdot \frac{1}{8}$$

$$= \frac{36 - 9 - 6}{8} - \frac{\pi}{4} = \frac{21}{8} - \frac{\pi}{4}$$

$4(\pi + 4A) = 42$

28. The area (in square units) of the region bounded by the parabola $y^2 = 4(x-2)$ and the line $y = 2x - 8$, is :

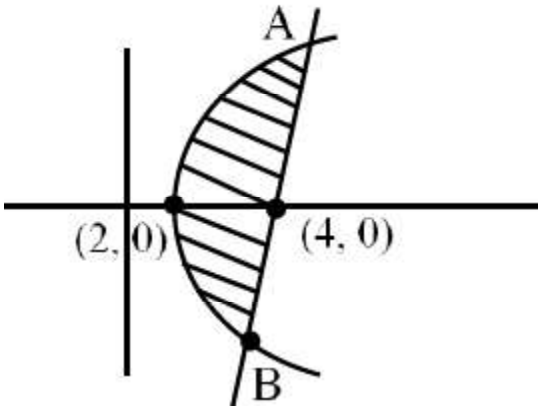
**[JEE(Main)Jan.2024]**

परवलय $y^2 = 4(x-2)$ तथा रेखा $y = 2x - 8$ से घिरे क्षेत्र का क्षेत्रफल (वर्ग इकाई में) है : **[JEE(Main)Jan.2024]**

(A) 9 (B) 8 (C) 7 (D) 6

**Ans.** A

**Sol.**




$y^2 = 4x - 8$

$y = 2x - 8$

$\Rightarrow y^2 = 2(y+8) - 8$

$\Rightarrow y^2 - 2y - 8 = 0$

$y = 4 \qquad y = -2$

$$\int_{-2}^{4}\left(\frac{y+8}{2} - \frac{y^2+8}{4}\right)dy$$

$$\Rightarrow \int_{-2}^{4}\frac{y}{2}dy - \int_{-2}^{4}\frac{y^2}{4}dy + \int_{-2}^{4}2dy$$

$$\Rightarrow \left[\frac{y^2}{4}\right]_{-2}^{4} - \frac{1}{4}\left[\frac{y^3}{3}\right]_{-2}^{4} + 2[y]_{-2}^{4} = 9$$

29. The area of the region $\left\{(x,y): y^2 \le 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \ne 3\right\}$ is :

**[JEE(Main)Jan.2024]**

क्षेत्र $\left\{(x,y): y^2 \le 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \ne 3\right\}$ का क्षेत्रफल है : **[JEE(Main)Jan.2024]**

(A) $\frac{32}{3}$ (B) $\frac{16}{3}$ (C) $\frac{64}{3}$ (D) $\frac{8}{3}$

**Ans.** A

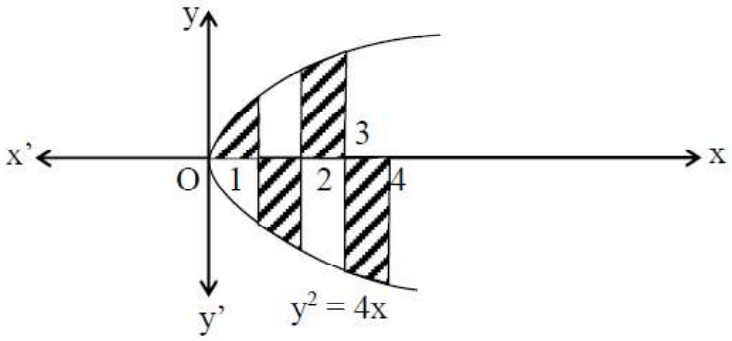
**Sol.** $\left\{(x,y); y^2 \le 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \ne 3\right\}$

$y^2 \le 4x,\ x < 4,\ \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0$

Case (i) $y > 0$ $\Rightarrow \frac{x(x-1)(x-2)}{(x-3)(x-4)} < 0$

$x \in (1,2) \cup (3,4)$

$$\text{Area} = \int_0^1 2\sqrt{x}\,dx + \left|\int_1^2 2\sqrt{x}\,dx\right| + \int_2^3 2\sqrt{x}\,dx + \left|\int_3^4 2\sqrt{x}\,dx\right|$$




$$= \frac{4}{3}\left(\left[x^{\frac{3}{2}}\right]_0^1 + \left|\left[x^{\frac{3}{2}}\right]_1^2\right| + \left[x^{\frac{3}{2}}\right]_2^3 + \left|\left(x^{\frac{3}{2}}\right)_3^4\right|\right)$$

$$A = \frac{4}{3}\left(1 + 2\sqrt{2} - 1 + 3\sqrt{3} - 2\sqrt{2} + 4\sqrt{4} - 3\sqrt{3}\right) = \frac{4}{3} \times 8 = \frac{32}{3}$$

30. The area of the region enclosed by the parabolas $y = 4x - x^2$ and $3y = (x-4)^2$ is equal to :

**[JEE(Main)Jan.2024]**

परवलयों $y = 4x - x^2$ तथा $3y = (x-4)^2$ से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2024]**

(A) 32/9 (B) 4 (C) 6 (D)14/3

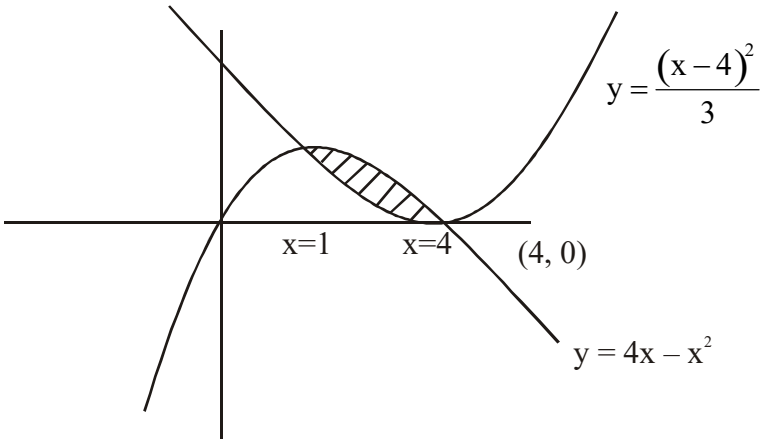
**Ans.** C

**Sol.** Solving $3y = (x-4)^2$ and $y = 4x - x^2$

$12x - 3x^2 = x^2 - 8x + 16$

$\Rightarrow 4x^2 - 20x + 16 = 0$

$(x-1)(x-4) = 0 \Rightarrow x = 1, 4$




$$\text{Area} = \int_1^4 \left( \left(4x - x^2\right) - \frac{(x-4)^2}{3} \right) dx$$

$$= \left( 2x^2 - \frac{x^3}{3} - \frac{(x-4)^3}{9} \right)_1^4 = 2(16-1) - \frac{1}{3}(3)(21) - \frac{1}{9}(0+27) = 30 - 21 - 3 = 6$$

31. The area enclosed by the curves $xy + 4y = 16$ and $x + y = 6$ is equal to : **[JEE(Main)Jan.2024]**

वक्रों $xy + 4y = 16$ तथा $x + y = 6$ से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)Jan.2024]**

(A) $28 - 30 \log_e 2$ (B) $32 - 30 \log_e 2$ (C) $30 - 28\log_e 2$ (D) $30 - 32 \log_e 2$

**Ans.** D

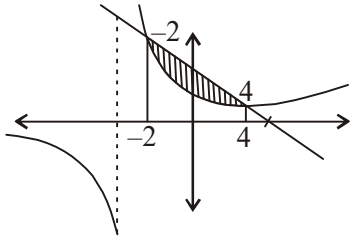
**Sol.** $y(x + 4) = 16$ & $y = 6 - x$

solve both curves,

$(6 - x)(x + 4) = 16$

$6x - x^2 + 24 - 4x = 16$

$x^2 - 2x - 8 = 0 \Rightarrow x = -2, 4$




$$\int_{-2}^{4} \left( 6 - x - \left( \frac{16}{x+4} \right) \right) dx$$

$$\Rightarrow \left[ 6x - \frac{x^2}{2} - 16\ell n|x+4| \right]_{-2}^{4}$$

$$\Rightarrow (24 - 8 - 16\ell n8) - (-12 - 2 - 16\ell n2)$$

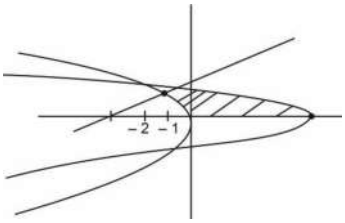
$$\Rightarrow 30 - 32\ell n2$$

32. Let the area of the region $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ be $\frac{m}{n}$, where m and n are coprime numbers. Then m + n is equal to _____. **[JEE(Main)Jan.2024]**

माना क्षेत्र $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ का क्षेत्रफल $\frac{m}{n}$ है, जहाँ m तथा n असहभाज्य संख्याएँ हैं। तो m + n बराबर है _____। **[JEE(Main)Jan.2024]**

**Ans.** 119

**Sol.**




Area between these curve will be

$$\int_{-2}^{-1}\left(-\sqrt{\frac{-x}{2}} + \frac{x+4}{2}\right)dx + \int_{-1}^{0}\left(\sqrt{\frac{8-x}{4}} - \sqrt{\frac{-x}{2}}\right)dx + \int_{0}^{8}\sqrt{\frac{8-x}{4}}dx = \frac{107}{12} = \frac{m}{n}$$

$$\Rightarrow m + n = 119$$

33. If the area of the region $\left\{(x, y) : 0 \leq y \leq \min\left\{2x, 6x - x^2\right\}\right\}$ is A, then 12 A is equal to _____.

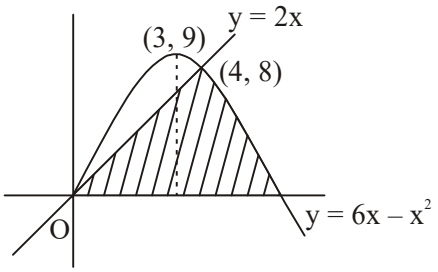
**[JEE(Main)Jan.2024]**

यदि क्षेत्र $\left\{(x, y) : 0 \leq y \leq \min\left\{2x, 6x - x^2\right\}\right\}$ का क्षेत्रफल A है, तो 12 A बराबर है _____

**[JEE(Main)Jan.2024]**

**Ans.** 304

**Sol.**




Solving $y = 2x$ and $y = 6x - x^2$

$2x = 6x - x^2$

$x^2 - 4x = 0$

$y = x = 0, x = 4, y = 8$

$A = \frac{1}{2}4.8 + \int_4^6 \left(6x - x^2\right)dx$

$= 16 + \left(3x^2 - \frac{x^3}{3}\right)_4^6$

$A = 16 + 3(36 - 16) - \frac{1}{3}(216 - 64)$

$A = 16 + 60 - \frac{152}{3}$

$A = 16 + \frac{28}{3}$

$12A = 304$

34. The area (in sq. units) of the part of the circle $x^2 + y^2 = 169$ which is below the line $5x - y = 13$ is $\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta}\sin^{-1}\left(\frac{12}{13}\right)$, where $\alpha$, $\beta$ are coprime number. Then $\alpha + \beta$ is equal to _____.

**[JEE(Main)Jan.2024]**

वृत्त $x^2 + y^2 = 169$ के रेखा $5x - y = 13$ से नीचे के भाग का क्षेत्रफल (वर्ग इकाई में) $\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta}\sin^{-1}\left(\frac{12}{13}\right)$ है, जहां $\alpha$, $\beta$ असहभाज्य संख्याऐं है। तो $\alpha + \beta$ बराबर है_____. **[JEE(Main)Jan.2024]**

**Ans.** 171

**Sol.** Solving line and circle $x = 0, 5$

$\tan\alpha = \frac{12}{5}$

$\sin\alpha = \frac{12}{13}$

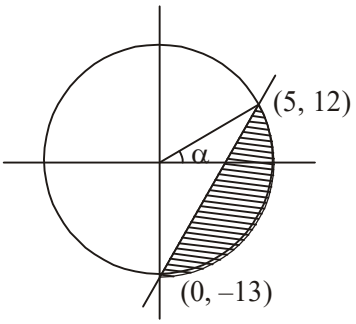
$\sin\left(\frac{\pi}{2} + \alpha\right) = \cos\alpha = \frac{5}{13}$

Now area of segment $= \frac{1}{2}\left(r^2\right)(\theta - \sin\theta)$

$$= \frac{1}{2} \times 169 \times \left[ \frac{\pi}{2} + \sin^{-1} \alpha - \sin\left( \frac{\pi}{2} + \alpha \right) \right]$$

$$= \frac{169}{2} \left[ \frac{\pi}{2} + \sin^{-1} \frac{12}{13} - \frac{5}{13} \right]$$

$$= \frac{169\pi}{4} + \frac{169}{2} \sin^{-1} \frac{12}{13} - \frac{65}{2}$$



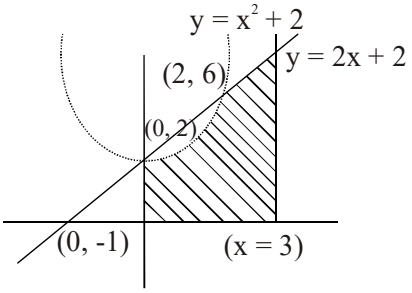

$\alpha = 169, \beta = 2$

35. Let the area of the region $\{(x, y) : 0 \le x \le 3, 0 \le y \le \min \{x^2 + 2, 2x + 2\}\}$ be A. Then 12A is equal to :
**[JEE(Main)Jan.2024]**

माना क्षेत्र $\{(x, y) : 0 \le x \le 3, 0 \le y \le \min \{x^2 + 2, 2x + 2\}\}$ का क्षेत्रफल A है। तो 12A बराबर है।
**[JEE(Main)Jan.2024]**

**Ans.** 164

**Sol.**




$$A = \int_0^2 \left( x^2 + 2 \right) dx + \int_2^3 (2x + 2) dx$$

$$A = \left( \frac{x^3}{3} + 2x \right)_0^2 + \left( x^2 + 2x \right)_2^3$$

$$A = \frac{41}{3} \Rightarrow 12A = 164$$

36. The area of the region enclosed by the parabola $(y - 2)^2 = x - 1$, the line $x - 2y + 4 = 0$ and the positive coordinate axes is _____.
**[JEE(Main)Jan.2024]**

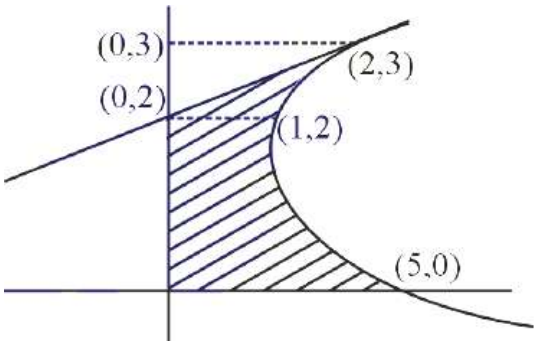
परवलय $(y - 2)^2 = x - 1$, रेखा $x - 2y + 4 = 0$ तथा धनात्मक निर्देशांक अक्षों से घिरे क्षेत्र का क्षेत्रफल _____
**[JEE(Main)Jan.2024]**

**Ans.** 5

**Sol.** Solving the equations

$(y - 2)^2 = x - 1$ and $x - 2y + 4 = 0$

$x = 2(y - 2)$




$\frac{x^2}{4} = x - 1$

$x^2 - 4x + 4 = 0$

$(x - 2)^2 = 0$

$x = 2$

Exclose area (w.r.t. y-axis) $= \int_0^3 x\,dy - \text{Area of } \Delta$

$= \int_0^3 \left((y-2)^2 + 1\right) dy - \frac{1}{2} \times 1 \times 2$

$= \int_0^3 \left(y^2 - 4y + 5\right) dy - 1$

$= \left[\frac{y^3}{3} - 2y^2 + 5y\right]_0^3 - 1$

$= 9 - 18 + 15 - 1 = 5$

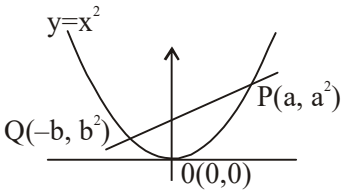
37. Three points O(0, 0), P(a, $a^2$), Q(–b, $b^2$), a > 0, b > 0, are on the parabola $y = x^2$. Let $S_1$ be the area of the region bounded by the line PQ and the parabola, and $S_2$ be the area of the triangle OPQ. If the minimum value of $\frac{S_1}{S_2}$ is $\frac{m}{n}$, gcd(m, n) = 1, then m + n is equal to _____. **[JEE(Main)Jan.2024]**

परवलय $y = x^2$ पर तीन बिंदु O(0, 0), P(a, $a^2$), Q(–b, $b^2$), a > 0, b > 0 हैं। माना रेखा PQ तथा परवलय से घिरे क्षेत्र का क्षेत्रफल $S_1$ है तथा त्रिभुज OPQ का क्षेत्रफल $S_2$ है। यदि $\frac{S_1}{S_2}$ का न्यूनतम मान $\frac{m}{n}$, gcd(m, n) = 1 हैं, तो m + n बराबर है _____

**[JEE(Main)Jan.2024]**

**Ans.** 7

**Sol.**




Equation of PQ

$$y+b=\frac{a^2-b^2}{a+b}\left(x-b^2\right)$$

$$\Rightarrow y=(a-b)x+ab$$

$$S_1=\int_{-b}^{a}\left((a-b)x+ab-x^2\right)dx$$

$$=\left(\frac{(a-b)}{2}x^2+abx-\frac{x^3}{3}\right)_{-b}^{a}$$

$$=\frac{1}{6}(a+b)^3$$

$$S_2=\frac{1}{2}\begin{vmatrix}-b & b^2 & 1\\ 0 & 0 & 1\\ a & a^2 & 1\end{vmatrix}=\frac{1}{2}ab(a+b)$$

$$\frac{S_1}{S_2}=\frac{\frac{1}{6}(a+b)^3}{\frac{1}{2}\cdot ab(a+b)}$$

$$=\frac{1}{3}\frac{(a+b)^2}{ab}=\frac{1}{3}\left(\frac{a}{b}+\frac{b}{a}+2\right)$$

$$\because \frac{b}{a}+\frac{a}{b}\geq 2$$

$$\Rightarrow\left(\frac{S_1}{S_2}\right)_{min}=\frac{4}{3}=\frac{m}{n}$$

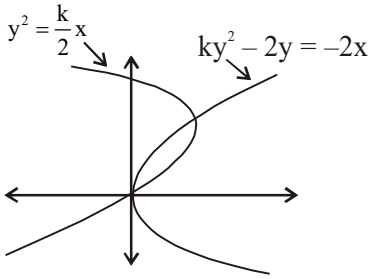
$$\Rightarrow m+n=7$$

38. The sum of squares of all possible values of k, for which area of the region bounded by the parabolas $2y^2 = kx$ and $ky^2 = 2(y - x)$ is maximum, is equal to _____. **[JEE(Main)Jan.2024]**

k के सभी संभव मानों, जिनके लिए परवलयों $2y^2 = kx$ तथा $ky^2 = 2(y - x)$ से घिरे क्षेत्र का क्षेत्रफल अधिकतम है, के वर्गों का योग है _____ **[JEE(Main)Jan.2024]**

**Ans.** 8

**Sol.**




$$\text{Area} = A = \int_0^{\frac{2k}{k^2+4}} \left[ \left( \frac{ky^2 - 2y}{-2} \right) - \left( \frac{2y^2}{k} \right) \right] dy$$

$$\text{Area} = \left[ \left( \frac{-k}{2} - \frac{2}{k} \right) \frac{y^3}{3} + \frac{y^2}{2} \right]_0^{\frac{2k}{k^2+4}}$$

$$\text{Area} = \frac{-\left(k^2+4\right)}{2k} \times \left( \frac{2k}{k^2+4} \right)^3 \times \frac{1}{3} + \left( \frac{2k}{k^2+4} \right)^2 \times \frac{1}{2}$$

$$= -\left( \frac{2k}{k^2+4} \right)^2 \times \frac{1}{3} + \left( \frac{2k}{k^2+4} \right)^2 \times \frac{1}{2}$$

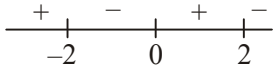
$$= \frac{1}{6} \times \frac{4k^2}{\left(k^2+4\right)^2}$$

$$A(k) = \frac{2}{3} \frac{k^2}{\left(k^2+4\right)^2}$$

$$A'(K) = \frac{2}{3} \frac{2K\left(K^2+4\right)^2 - 2K^2\left(K^2+4\right)(2K)\Big]}{\left(K^2+4\right)^4} = 0$$

$$\Rightarrow \frac{2}{3}(2K)\left(K^2+4\right)\left[K^2+4-2K^2\right] = 0$$

K = 0 or K = 2 or K = –2




$\therefore$ K = 0 is local minima

K = –2 and 2 are local maxima

$\therefore (-2)^2 + 2^2 = 8$

39. One of the points of intersection of the curves $y = 1 + 3x - 2x^2$ and $y = \frac{1}{x}$ is $\left(\frac{1}{2}, 2\right)$. Let the area of the region enclosed by the curves be $\frac{1}{24}\left(\ell\sqrt{5} + m\right) - n\log_e\left(1+\sqrt{5}\right)$, where $\ell, m, n \in N$. Then $\ell + m + n$ is equal to :

वक्रों $y = 1 + 3x - 2x^2$ तथा $y=\frac{1}{x}$ का एक प्रतिच्छेदन $\left(\frac{1}{2},2\right)$ है। माना इन वक्रों से घिरे का क्षेत्रफल $\frac{1}{24}\left(\ell\sqrt{5}+m\right)-n\log_e\left(1+\sqrt{5}\right)$ है, जहाँ $\ell, m, n \in N$ है, तो $\ell + m + n$ का मान होगा :

**[JEE(Main)April,2024]**

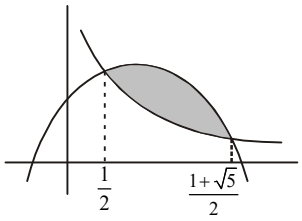
(A) 32 (B) 29 (C) 31 (D) 30

**Ans.** D

**Sol.** $2x^3 - 3x^2 - x + 1 = 0$

$\Rightarrow (2x - 1)(x^2 - x - 1) = 0$

$\Rightarrow x=\frac{1}{2}, x=\frac{1+\sqrt{5}}{2}$




$$\text{Area} = \int_{\frac{1}{2}}^{\frac{1+\sqrt{5}}{2}} \left(1+3x-2x^2-\frac{1}{x}\right)dx$$

$$=\left[x++\frac{3x^2}{2}-\frac{2x^3}{3}-\ell nx\right]$$

$$=\frac{\sqrt{5}+1}{2}+\frac{3}{8}\left(\sqrt{5}+1\right)^2-\frac{1}{12}\left(\sqrt{5}+1\right)^3-\ell n\left(\frac{\sqrt{5}+1}{2}\right)-\left(\frac{1}{2}+\frac{3}{8}-\frac{1}{12}-\ell n\frac{1}{2}\right)$$

$$=\frac{1}{24}\left[12\left(\sqrt{5}+1\right)+9\left(\sqrt{5}+1\right)^2-2\left(\sqrt{5}+1\right)^3-12-9+2\right]-\ell n\left(\frac{\sqrt{5}+1}{2}\times 2\right)$$

$$=\frac{1}{24}\left[12\left(\sqrt{5}+1\right)+9\left(6+2\sqrt{5}\right)-2\left(5\sqrt{5}+1+3\sqrt{5}\left(\sqrt{5}+1\right)-19\right)\right]-\ell n\left(\sqrt{5}+1\right)$$

$$=\frac{1}{24}\left[14\sqrt{5}+15\right]-\ell n\left(\sqrt{5}+1\right)$$

$\therefore l = 14, m = 15, n = 1$

$l + m + n = 30$

40. The are (in sq. units) of the region described by $\{(x, y) : y^2 \le 2x, \text{ and } y \ge 4x - 1\}$ is :

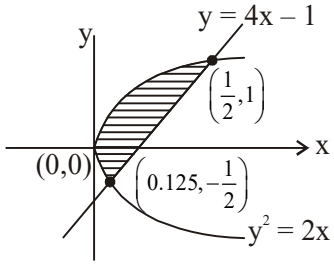
**[JEE(Main)April,2024]**

क्षेत्र $\{(x, y) : y^2 \le 2x$ तथा $y \ge 4x - 1\}$ का क्षेत्रफल (वर्ग इकाईयों में) है : **[JEE(Main)April,2024]**

(A) $\frac{8}{9}$ (B) $\frac{9}{32}$ (C) $\frac{11}{12}$ (D) $\frac{11}{32}$

**Ans.** B

**Sol.** $\text{Area} = \int_{-\frac{1}{2}}^{1}\left(\frac{y+1}{4}-\frac{y^2}{2}\right)dy$




$$=\left[\frac{y^2}{8}+\frac{y}{4}-\frac{y^3}{6}\right]_{-\frac{1}{2}}^{1}$$

$$=\left(\frac{1}{8}+\frac{1}{4}-\frac{1}{6}\right)-\left(\frac{1}{32}-\frac{1}{8}+\frac{1}{48}\right)$$

$$=\frac{5}{24}+\frac{7}{96}$$

$$=\frac{27}{96}$$

$$=\frac{9}{32}$$

41. The area enclosed between the curves $y = x|x|$ and $y = x - |x|$ is : **[JEE(Main)April,2024]**

वक्रों $y = x|x|$ तथा $y = x - |x|$ के बीच घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2024]**

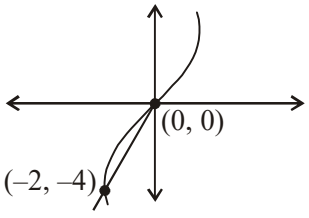
(A) 2/3 (B) 4/3 (C) 8/3 (D) 1

**Ans.** B

**Sol.** $y = x|x|$ & $y = x - |x|$

$$y = x|x| = \begin{cases} x^2 & , x > 0 \\ -x^2 & , x < 0 \end{cases}$$

$$y = x - |x| = \begin{cases} 0 & , x > 0 \\ 2x & , x < 0 \end{cases}$$



point of intersections are (0 , 0) & (–2, 4).

$\therefore$ Area $= \int_{-2}^{0} \left(-x^2 - 2x\right) dx$

$= \left[\frac{-x^3}{3} - \frac{2x^2}{2}\right]_{-2}^{0}$

$= -\left[\frac{8}{3} - 4\right] = \frac{4}{3}$ sq. unit

42. Let the area of the region enclosed by the curves $y = 3x$, $2y = 27 - 3x$ and $y = 3x - x\sqrt{x}$ be A. Then 10A is equal to : **[JEE(Main)April,2024]**

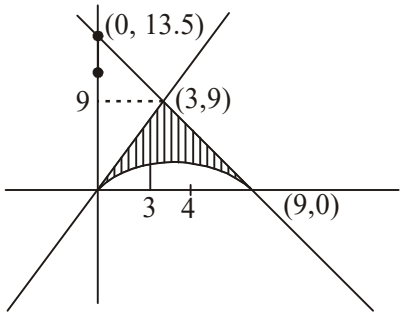
माना वक्रों $y = 3x$, $2y = 27 - 3x$ और $y = 3x - x\sqrt{x}$ से घिरे क्षेत्र का क्षेत्रफल A है, तो 10A बराबर है :

**[JEE(Main)April,2024]**

(A) 184 (B) 172 (C) 154 (D) 162

**Ans.** D

**Sol.**




$y = 3x - x^{\frac{3}{2}}$ $\qquad y' = 3 - \frac{3}{2}\sqrt{x}$

$y' = 3\left(1 - \frac{\sqrt{x}}{2}\right)$

$f'(x)$ + –

Area of shaded region

$= \int_0^3 \left(3x - \left(3x - x^{\frac{3}{2}}\right)\right) dx + \int_3^9 \frac{27 - 3x}{2} - \left(3x - x^{\frac{3}{2}}\right) dx$

$= \int_0^3 x^{\frac{3}{2}} dx + \frac{1}{2}\int_3^9 \left(27 - 9x + 2x^{\frac{3}{2}}\right) dx = \left(\frac{2}{5}x^{\frac{5}{2}}\right)_0^3 + \frac{1}{2}\left[27x - \frac{9x^2}{2} + 2 \times \frac{2}{5}x^{\frac{5}{2}}\right)_3^9$

$$=\frac{2}{5}\times 9\sqrt{3}+\frac{1}{2}\left[243-\frac{729}{2}+\frac{4}{5}\times 81\times 3-81+\frac{81}{2}-\frac{4}{5}\times 9\sqrt{3}\right]$$

$$=\frac{1}{2}\left[\frac{486-729-81}{2}+\frac{972}{5}\right]=\frac{81}{5}$$

$\Rightarrow 10A = 162$

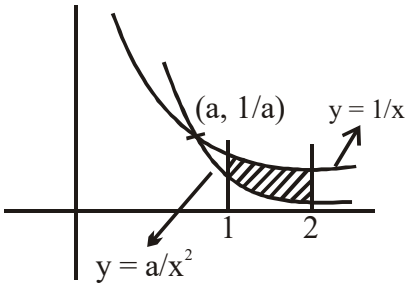
43. If the area of the region $\left\{(x,y):\frac{a}{x^2}\le y\le\frac{1}{x},1\le x\le 2,0<a<1\right\}$ is $\left(\log_e 2\right)-\frac{1}{7}$ then the value of $7a-3$ is equal to : **[JEE(Main)April,2024]**

यदि क्षेत्र $\left\{(x,y):\frac{a}{x^2}\le y\le\frac{1}{x},1\le x\le 2,0<a<1\right\}$ का क्षेत्रफल $\left(\log_e 2\right)-\frac{1}{7}$ है, तो $7a-3$ का मान बराबर है : **[JEE(Main)April,2024]**

(A) 1 (B) 2 (C) 0 (D) –1

**Ans.** D

**Sol.** Required Area $=\int_1^2\left(\frac{1}{x}-\frac{a}{x^2}\right)dx=\left(\ln x+\frac{a}{x^2}\right)=\ln 2+\frac{a}{2}-a$




Now, $\ln 2+\frac{a}{2}-a=\ln 2-\frac{1}{7}$

$\frac{a}{2}=\frac{1}{7}$ $\Rightarrow$ $7a = 2$ and $7a - 3 = -1$

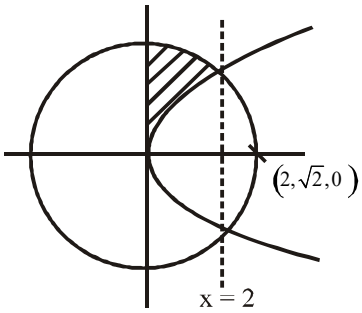
44. The area of the region in the first quadrant inside the circle $x^2+y^2=8$ and outside the parabola $y^2=2x$ is equal to: **[JEE(Main)April,2024]**

प्रथम चतुर्थांश में वृत्त $x^2 + y^2 = 8$ के अन्दर तथा परवलय $y^2 = 2x$ के बाहर के क्षेत्र का क्षेत्रफल है : **[JEE(Main)April,2024]**

(A) $\pi-\frac{2}{3}$ (B) $\pi-\frac{1}{3}$ (C) $\frac{\pi}{2}-\frac{1}{3}$ (D) $\frac{\pi}{2}-\frac{2}{3}$

**Ans.** A

**Sol.**




Requried Area = Ar(circle from 0 to 2) – ar(para from 0 to 2)

$$=\int_0^2 \sqrt{8-x^2}\,dx-\int_0^2 \sqrt{2x}\,dx$$

$$=\left[\frac{x}{2}\sqrt{8-x^2}+\frac{8}{2}\sin^{-1}\frac{x}{2\sqrt{2}}\right]_0^2-\sqrt{2}\left[\frac{x\sqrt{x}}{\frac{3}{2}}\right]_0^2$$

$$=\frac{2}{2}\sqrt{8-4}+\frac{8}{2}\sin^{-1}\frac{2}{2\sqrt{2}}-\frac{2\sqrt{2}}{3}\left(2\sqrt{2}-0\right)$$

$$\Rightarrow 2+4\cdot\frac{\pi}{4}-\frac{8}{3}=\pi-\frac{2}{3}$$

45. The parabola $y^2 = 4x$ divides the area of the circle $x^2 + y^2 = 5$ in two parts. The area of the smaller part is equal to:

**[JEE(Main)April,2024]**

परवलय $y^2 = 4x$, वृत्त $x^2 + y^2 = 5$ के क्षेत्रफल को दो भागों में बाँटता है। छोटे भाग का क्षेत्रफल है :

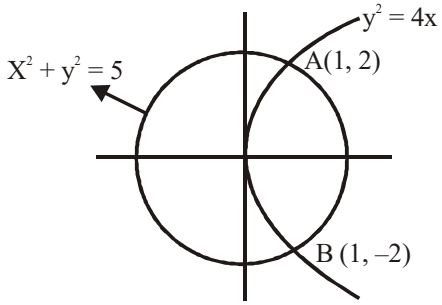
**[JEE(Main)April,2024]**

(A) $\frac{1}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (B) $\frac{1}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

(C) $\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (D) $\frac{2}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

**Ans.** C

**Sol.**




$$\text{Area of smaller region} = \int_{-2}^{2}\left(\sqrt{5-y^2}-\frac{y^2}{4}\right)dy$$

$$= 2\int_{0}^{2}\left(\sqrt{5-y^2}-\frac{y^2}{4}\right)dy$$

$$= 2\left[\left(\frac{y}{2}\sqrt{5-y^2}+\frac{5}{2}\sin^{-1}\left(\frac{y}{\sqrt{5}}\right)\right)_0^2-\frac{1}{12}\left(y^3\right)_0^2\right]$$

$$= 2\left[1+\frac{5}{2}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)-\frac{2}{3}\right]=\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$$

46. The area of the region enclosed by the parabolas $y = x^2 - 5x$ and $y = 7x - x^2$ is _____.

**[JEE(Main)April,2024]**

परवलयों $y = x^2 - 5x$ तथा $y = 7x - x^2$ से घिरे क्षेत्र का क्षेत्रफल है _____ **[JEE(Main)April,2024]**

**Ans.** 198

Answer by Matrix is (72)

**Sol.** $y = x^2 - 5x$

$y = 7x - x^2$

Now $x^2 - 5x = 7x - x^2$

$x = 0, 6$

$$\text{So Area} = \int_0^6\left(x^2-5x-7x+x^2\right)dx$$

$$= 2\frac{x^3}{3}-12\frac{x^2}{2}\Big]_0^6$$

$$= \left[=2\times\frac{6^3}{3}-6\times36\right)0$$

$$= \frac{2\times6\times6\times6}{3} - 6\times36$$

$$= 36 \times 4 - 6 \times 36$$

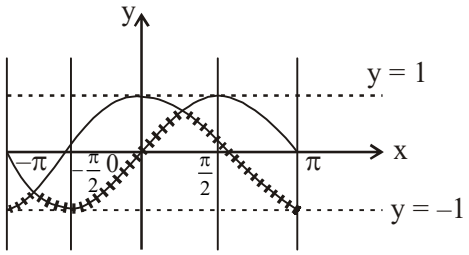
$$= 36(-2) = 72$$

47. Let the area of the region enclosed by the curve $y = \min\{\sin x, \cos x\}$ and the x-axis between $x = -\pi$ to $x = \pi$ be A. Then $A^2$ is equal to _____. **[JEE(Main)April,2024]**

वक्र $y = \min\{\sin x, \cos x\}$ तथा x-अक्ष के बीच परिबद्ध क्षेत्र का क्षेत्रफल A है। जहाँ x-अक्ष $x = -\pi$ तथा $x = \pi$ के बीच है, तो $A^2$ का मान होगा _____ **[JEE(Main)April,2024]**

**Ans.** 16

**Sol.** $y = f(x) = \min\{\sin x, \cos x\}$




$$A = -\int_{-\pi}^{\frac{-3\pi}{4}} \cos x dx - \int_{\frac{-3\pi}{4}}^{0} \sin x dx + \int_{0}^{\frac{\pi}{4}} \sin x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x dx$$

$$-\int_{\frac{\pi}{2}}^{\pi} \cos x dx$$

$A = 4$

$A^2 = 16$

## Exercise J - A

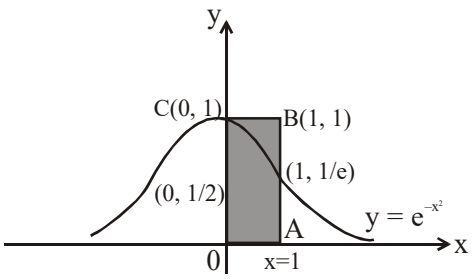
1. Let S be the area of the region enclosed by $y = e^{-x^2}, y = 0, x = 0$ and $x = 1$. Then : **[JEE-2012]**

यदि S वक्रों $y = e^{-x^2}, y = 0, x = 0$ तथा $x = 1$ के मध्य घिरा हुआ क्षेत्रफल हो तो : **[JEE-2012]**

(A) $S \geq \frac{1}{e}$ (B) $S \geq 1 - \frac{1}{e}$ (C) $S \leq \frac{1}{4}\left(1 + \frac{1}{\sqrt{e}}\right)$ (D) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}}\left(1 - \frac{1}{\sqrt{2}}\right)$

**Ans.** ABD

**Sol.** Area (OABC) = 1



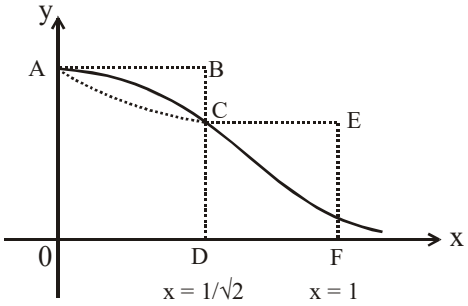

Shaded area is S.

Clearly S < 1

and $\int_0^1 e^{-x^2} dx > \int_0^1 e^{-x} dx$

$\Rightarrow \quad S > 1 - \frac{1}{e}$ ($\therefore$ (B) is correct

Again S ≥ Area (trapezium ACDO)




$\Rightarrow \quad S \geq \frac{1}{2}\left(1 + \frac{1}{\sqrt{e}}\right)\left(\frac{1}{\sqrt{2}}\right)$

$\Rightarrow \quad S \geq \frac{1}{2\sqrt{2}}\left(1 + \frac{1}{\sqrt{e}}\right)$

$\therefore \quad$ C is wrong

Also S ≤ Sum of areas of rectangles ABDO & CEFD

$\Rightarrow \quad S \leq \frac{1}{\sqrt{2}} \times 1 + \left(1 - \frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{e}}\right)$

$\Rightarrow \quad S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}}\left(1 - \frac{1}{\sqrt{2}}\right)$

($\therefore$ (D) is correct)

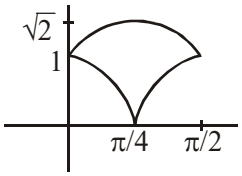
2. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is : **[JEE(Adv.)2013]**

अन्तराल $\left[0,\frac{\pi}{2}\right]$ पर वक्रों $y = \sin x + \cos x$ तथा $y = |\cos x - \sin x|$ द्वारा परिबद्ध क्षेत्रफल है : **[JEE(Adv.)2013]**

(A) $4\left(\sqrt{2}-1\right)$ (B) $2\sqrt{2}\left(\sqrt{2}-1\right)$ (C) $2\left(\sqrt{2}+1\right)$ (D) $2\sqrt{2}\left(\sqrt{2}+1\right)$

**Ans.** B

**Sol.** $y = \sin x + \cos x = \sqrt{2}\sin\left(x+\frac{\pi}{4}\right)$




$$y = |\cos x - \sin x| = \sqrt{2}\left(\cos\left(x+\frac{\pi}{4}\right)\right)$$

Area

$$= \int_0^{\pi/4}\left[\left(\sin x+\cos x\right)-\left(\cos x-\sin x\right)\right]dx$$

$$+\int_{\pi/4}^{\pi/2}\left[\left(\sin x+\cos x\right)-\left(\sin x-\cos x\right)\right]dx$$

$$= \int_0^{\pi/4} 2\sin x\,dx + \int_{\pi/4}^{\pi/2} 2\cos x\,dx$$

$$= \left[-2\cos x\right]_0^{\pi/4} + \left[2\sin x\right]_{\pi/4}^{\pi/2}$$

$$= 2\sqrt{2}\left(\sqrt{2}-1\right)$$

3. Let $f:\left[\frac{1}{2},1\right]\to R$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right)=1$. Then the value of $\int_{1/2}^{1} f(x)\,dx$ lies in the interval : **[JEE(Adv.)2013]**

माना कि $f:\left[\frac{1}{2},1\right]\to R$ (सभी वास्तविक संख्याओं का समुच्चय) एक धनात्मक अचरेतर (non constant) तथा अवकलनीय फलन gSft 1 ds fy ; sf'(x) < 2 f(x) तथा $f\left(\frac{1}{2}\right)=1$ है, तब $\int_{1/2}^{1} f(x)\,dx$ का मान निम्न अन्तराल में है : **[JEE(Adv.)2013]**

(A) $(2e-1, 2e)$ (B) $(e-1, 2e-1)$ (C) $\left(\frac{e-1}{2}, e-1\right)$ (D) $\left(0, \frac{e-1}{2}\right)$

**Ans.** D

**Sol.** $f'(x) - 2f(x) < 0$

Multiply both side by $e^{-2x}$

$$e^{-2x} f'(x) - 2e^{-2x} f(x) < 0$$

$$\frac{d}{dx}\left(e^{-2x} f(x)\right) < 0$$

Now, $g(x) = e^{-2x} f(x)$

$\therefore$ g(x) is a decreasing function.

$$x > \frac{1}{2}$$

$$g(x) < g\left(\frac{1}{2}\right)$$

$$\Rightarrow \quad e^{-2x} f(x) < \frac{1}{e}$$

$$\Rightarrow \quad f(x) < e^{2x-1}$$

$$\Rightarrow \quad \int_{1/2}^{1} f(x)dx < \frac{1}{e}\int_{1/2}^{1} e^{2x}dx$$

$$= \left[\frac{1}{2e}e^{2x}\right]_{1/2}^{1} = \frac{1}{2e}\left(e^2 - e\right) = \frac{1}{2}\left(e-1\right)$$

$$\Rightarrow \quad \int_{1/2}^{1} f(x)dx < \frac{e-1}{2}$$

obviously $f(x)$ is positive

$$\therefore \quad \int_{1/2}^{1} f(x)dx > 0$$

4. Let $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 t\, dt$ for all $x \in R$ and $f: \left[0, \frac{1}{2}\right] \to [0, \infty)$ be a continuous function. For $a \in \left[0, \frac{1}{2}\right]$, if $F'(a) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$, then $f(0)$ is. **[JEE(Adv.)2015]**

माना कि सभी $x \in R$ के लिए, $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 t\,dt$ तथा $f: \left[0, \frac{1}{2}\right] \to [0, \infty)$ एक संतत फलन है। यदि उन सभी $a \in \left[0, \frac{1}{2}\right]$ के लिए $F'(a) + 2$ उस क्षेत्र का क्षेत्रफल है, जो कि $x = 0$, $y = 0$, $y = f(x)$ और $x = a$ से घिरा हुआ है, तब $f(0)$ का मान है।

**[JEE(Adv.)2015]**

**Ans.** 3

**Sol.** $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 + dt$

$$F'(x) = 2\left(\cos\left(x^2 + \frac{\pi}{6}\right)\right)^2 2x - 2\cos^2 x$$

$$\because F'(a) + 2 = \int_0^a f(x)dx$$

$$2\left(\cos\left(a^2 + \frac{\pi}{6}\right)\right)^2 2a - 2\cos^2 a + 2 = \int_0^a f(x)dx$$

$$\Rightarrow 4\cos^2\left(a^2 + \frac{\pi}{6}\right) + 4a2\cos\left(a^2 + \frac{\pi}{6}\right).\left(-\sin\left(a^2 + \frac{\pi}{6}\right)\right) \times 2a + 4\cos a \sin a = f(a)$$

$$\therefore f(0) = 4\left(\frac{\sqrt{3}}{2}\right)^2 = 3$$

5. Match the following column : **[JEE(Adv.)-2015]**

| **Column I** | **Column II** |
|---|---|
| (A) In a triangle $\Delta XYZ$, let a, b and c be the lengths of the sides opposite to the angles X, Y and Z, respectively. If $2(a^2 - b^2) = c^2$ and $\lambda = \frac{\sin(X-Y)}{\sin Z}$, then possible values of n for which $\cos(n\pi\lambda) = 0$ is (are) | (P) 1 |
| (B) In a triangle $\Delta XYZ$, let a, b and c be the lengths of the sides opposite to the angles X, Y and Z, respectively. If $1 + \cos 2X - 2\cos 2Y = 2 \sin X \sin Y$, then possible value(s) of a/b is (are) | (Q) 2 |
| (C) In $R^2$, let $\sqrt{3}\,\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\,\hat{j}$ and $\beta\,\hat{i} + (1-\beta)\hat{j}$ be the position vectors of X, Y and Z with respect to the origin O, respectively. If the distance of Z | (R) 3 |

from the bisector of the acute angle of $\overrightarrow{OX}$ with $\overrightarrow{OY}$ is $\frac{3}{\sqrt{2}}$, then possible value(s) of $|\beta|$ is (are)

(D) Suppose that $F(\alpha)$ denotes the area of the region bounded by $x = 0$, $x = 2$, $y^2 = 4x$ and $y = |\alpha x - 1| + |\alpha x - 2| + \alpha x$, where $\alpha \in \{0, 1\}$. Then the value(s) of $F(\alpha) + \frac{8}{3}\sqrt{2}$, when $\alpha = 0$ and $\alpha = 1$, is (are) (S) 5

(T) 6

निम्न कॉलम कों सुमेहिलत कीजिए : **[JEE(Adv.)-2015]**

| **Column I** | **Column II** |
|---|---|
| (A) माना कि एक त्रिभुज $\Delta XYZ$ में कोणों X, Y और Z के सामने की भुजाओं की लम्बाईयाँ क्रमशः a, b और c है। मानाकि $2(a^2 - b^2) = c^2$ और $\lambda = \frac{\sin(X-Y)}{\sin Z}$ है। यदि $\cos(n\pi\lambda) = 0$ तब n के संभव मान है। | (P) 1 |
| (B) माना कि एक त्रिभुज $\Delta XYZ$ मे कोणों X, Y और Z के सामने की भुजाओं की लम्बाईयाँ क्रमशः a, b और c है। यदि $1 + \cos 2X - 2\cos 2Y = 2 \sin X \sin Y$ तब a/b के संभव मान है | (Q) 2 |
| (C) माना कि $R^2$ में, मूल बिन्दु O के सापेक्ष $\sqrt{3}\,\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\,\hat{j}$ और $\beta\,\hat{i} + (1-\beta)\hat{j}$ क्रमशः र एल और Z के स्थिति सदिश है। यदि $\overrightarrow{OX}$ और $\overrightarrow{OY}$ के न्यून कोण के समद्विभाजक से Z की दूरी $\frac{3}{\sqrt{2}}$ हो, तो $\|\beta\|$ का संभव मान है। | (R) 3 |
| (D) माना कि $F(\alpha)$ उस क्षेत्र के क्षेत्रफल को दर्शाता है जो $x = 0$, $x = 2$, $y^2 = 4x$ और $\psi = \|\alpha x - 1\| + \|\alpha x - 2\| + \alpha x$, से घिरा हुआ है, जहाँ $\alpha \in \{0, 1\}$ है। $\alpha = 0$ और $\alpha = 1$ के लिए $F(\alpha) + \frac{8}{3}\sqrt{2}$ का मान है। | (S) 5 |
| | (T) 6 |

**Ans.** A -PRS , B -P , C - PQ, D - ST

**Sol.** (A) $2(a^2 - b^2) = c^2$

$2(4R^2(\sin^2 x \times - \sin^2 y)) = 4R^2 \sin^2 z$

$2(\sin(x-y)\sin(x+y)) = \sin^2 z$

$\frac{\sin(x-y)}{\sin z} = \frac{1}{2} = \lambda$

$$\cos(n\pi\lambda) = \cos\left(\frac{n\pi}{2}\right) = 0$$

n = odd integer

(B) $1 + \cos 2x = 2\cos 2y = 2\sin x \sin y$

$$\sin^2 x + \sin x \sin y - 2\sin^2 y = 0$$

$$(\sin x - \sin y)(\sin x + 2\sin y) = 0$$

$\sin x = \sin y$

a = b

$$\frac{a}{b} = 1$$

(C) distance of z from bisector of $\overrightarrow{OX}$ & $\overrightarrow{OY}$ = $\frac{3}{\sqrt{2}}$

$$\left(\beta - \frac{1}{2}\right)^2 + \left(\beta - \frac{1}{2}\right)^2 = \frac{9}{2}$$

$\beta = 2, -1$

$|\beta| = 2, 1$

(D) when $\alpha = 0$

$$A = 6 - \int_0^2 2\sqrt{x}\,dx$$

$$= 6 - \frac{8\sqrt{2}}{3}$$

when $\alpha = 1$

$$A = \int_0^1 \left(3 - x - 2\sqrt{x}\right)dx + \int_1^2 \left(x + 1 - 2\sqrt{x}\right)dx$$

$$= \left|3x - \frac{x^2}{2} - \frac{4}{3}x^{\frac{3}{2}}\right|_0^1 + \left|\frac{x^2}{2} + x - \frac{4}{3}x^{\frac{3}{2}}\right|_1^2$$

$$= 5 - \frac{8\sqrt{2}}{3}$$

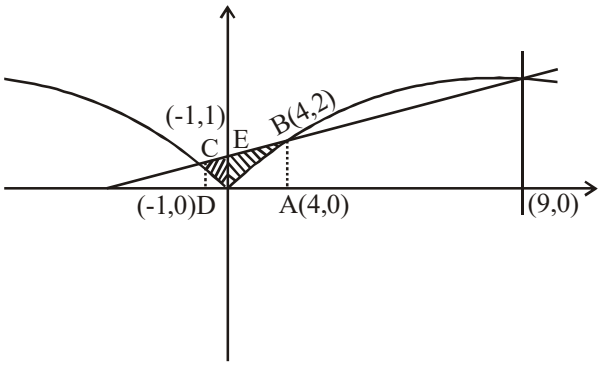
6. Area of the region $\left\{(x, y) \in R^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\right\}$ is equal to : **[JEE(Adv.)2016]**

क्षेत्र $\left\{(x, y) \in R^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\right\}$ का क्षेत्रफल है : **[JEE(Adv.)2016]**

(A) $\frac{1}{6}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{3}$

**Ans.** C

**Sol.** Shifting origin to (–3, 0)




Area $\left\{(x, y) \in R^2 : y \geq \sqrt{|x|}, 5y \leq x+6 \leq 15\right\}$

Desired Area = Area of trapezium ABCDA – [Area DCOD + Area OBAO]

$$= \frac{1}{2} \times (1+2) \times 5 - \left( \int_{-1}^{0} \sqrt{-x}\ dx + \int_{0}^{4} \sqrt{x}\ dx \right)$$

$$= \frac{15}{2} - \left( \frac{2}{3} + \frac{16}{3} \right)$$

$$= \frac{3}{2}$$

7. If the line $x = \alpha$ divides the area of region $R = \{(x, y) \in R^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ into two equal parts, then : **[JEE(Adv.)-2017]**

यदि रेखा $x = \alpha$ क्षेत्र $R = \{(x, y) \in R^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ के क्षेत्रफल को दो बराबर भागों में विभाजित करती है, तब : **[JEE(Adv.)-2017]**

(A) $0 < \alpha \le \frac{1}{2}$ (B) $2\alpha^4 - 4\alpha^2 + 1 = 0$ (C) $\alpha^4 + 4\alpha^2 - 1 = 0$ (D) $\frac{1}{2} < \alpha < 1$

**Ans.** BD

**Sol.** $y = x^3$

$$\int_0^1 \left(x - x^3\right)dx = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\int_0^\alpha \left(x - x^3\right)dx = \frac{1}{8}$$

$$4\alpha^2 - 2\alpha^4 = 1$$

$$2\alpha^4 - 4\alpha^2 + 1 = 0$$

$2t^2 - 4t + 1 = 0$ (taking $t = \alpha^2$)

$$t = \frac{4 \pm \sqrt{16-8}}{4}$$

$$t = \frac{4 \pm 2\sqrt{2}}{4}$$

$$t = \alpha^2 = 1 \pm \frac{1}{\sqrt{2}}$$

$$\therefore \alpha^2 = 1 - \frac{1}{\sqrt{2}} \qquad \Rightarrow \frac{1}{2} < \alpha < 1$$

8. Let $f: [0, \infty) \to R$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt$ for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE ? **[JEE(Adv.)2018]**

(A) The curve $y = f(x)$ passes through the point $(1, 2)$

(B) The curve $y = f(x)$ passes through the point $(2, -1)$

(C) The area of the region $\left\{(x, y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ is $\frac{\pi - 2}{4}$

(D) The area of the region $\left\{(x, y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ is $\frac{\pi - 1}{4}$

माना कि $f: [0, \infty) \to R$ एक ऐसा संतत फलन है कि $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt$ सभी $x \in [0, \infty)$ के लिये। तब निम्नलिखित esa lsa dkSu l k(से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) वक्र $y = f(x)$ बिन्दु (1, 2) से गुजरता है

(B) वक्र $y = f(x)$ बिन्दु (2, –1) से गुजरता है

(C) क्षेत्र $\left\{(x,y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ का क्षेत्रफल $\frac{\pi-2}{4}$ है

(D) क्षेत्र $\left\{(x,y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ का क्षेत्रफल $\frac{\pi-1}{4}$ है

**Ans.** BC

**Sol.** $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)dt \Rightarrow f(x).e^{-x} = (1-2x).e^{x} + \int_0^x e^{-t} f(t)dt$

$\Rightarrow f'(x)e^{-x} - e^{-x} \cdot f(x) = -2 \cdot e^{-x} - (1-2x) \cdot e^{-x} + e^{-x} \cdot f(x)$

$\Rightarrow f'(x) - 2f(x) = (2x-3)$

I.F. $= e^{-2x}$

$\therefore y.e^{-x} = \int \underset{|}{(2x-3)}.\underset{||}{e^{-2x}}dx$

$\Rightarrow y.e^{-2x} = (2x-3).\frac{e^{-2x}}{-2} - 2\int \frac{e^{-2x}}{-2}dx \qquad \Rightarrow y.e^{-2x} = -\frac{(2x-3)e^{-2x}}{2} - \frac{e^{-2x}}{2} + c$

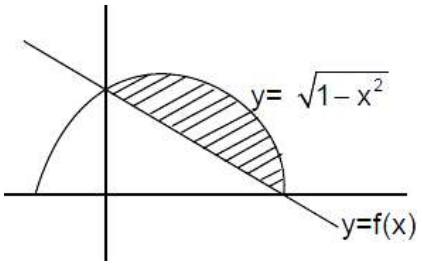
$\Rightarrow y.e^{-2x} = -\frac{(2x-3)-1}{2} + c.e^{2x} \qquad \Rightarrow y = (1-x) + c.e^{2x}$

$\Rightarrow y = (1-x) + c.e^{2x}$

put $x = 0$

$1 = 1 + c \qquad \Rightarrow c = 0$

$\therefore y = 1 - x$ which passes through point (2, –1)




Now

required area $= \frac{1}{4}.\pi.(1)^2 - \frac{1}{2}.1.1 = \frac{\pi}{4} - \frac{1}{2}$

9. A farmer $F_1$ has a land in the shape of a triangle with vertices at P(0,0), Q(1,1) and R(2,0). From this land, a neighbouring farmer $F_2$ takes away the region which lies between the side PQ and a curve of the form $y = x^n$ $(n > 1)$. If the area of the region taken away by the farmer $F_2$ is exactly 30% of the area of $\Delta PQR$, then the value of n is _____ . **[JEE(Adv.)2018]**

एक किसान $F_1$ के पास एक त्रिभुजाकार भूमि है जिसके शीर्ष P(0,0), Q(1,1) तथा R(2,0) पर हैं। एक पड़ोसी किसान $F_2$ इस भूमि से उस क्षेत्र को ले लेता है जो कि भुजा PQ और $y = x^n$ $(n > 1)$ के रूप वाले वक्र के बीच स्थित है। यदि किसान $F_2$ द्वारा लिये गये क्षेत्र का क्षेत्रफल $\Delta PQR$ के क्षेत्रफल का ठीक 30% है, तब n का मान है। **[JEE(Adv.)2018]**

**Ans.** 4

**Sol.** $\frac{30}{100}A = \frac{3}{10}$

$$A = \int_0^1 \left(x - x^n\right) dx = \frac{3}{10}$$

$$\frac{x^2}{2} - \frac{x^{n+1}}{n+1}\Bigg|_0^1 = \frac{3}{10}$$

$$\frac{1}{2} - \frac{1}{n+1} = \frac{3}{10}$$

$n = 4$

10. The area of the region $\{(x,y): xy \le 8, 1 \le y \le x^2\}$ is : **[JEE(Adv.)2019]**

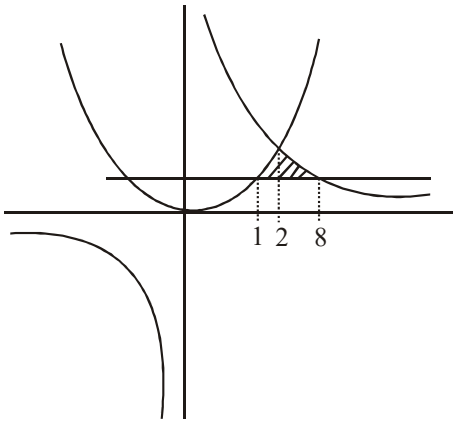
क्षेत्र $\{(x,y): xy \le 8, 1 \le y \le x^2\}$ का क्षेत्रफल है : **[JEE(Adv.)2019]**

(A) $16\log_e 2 - 6$ (B) $8\log_e 2 - \frac{14}{3}$ (C) $16\log_e 2 - \frac{14}{3}$ (D) $8\log_e 2 - \frac{7}{3}$

**Ans.** C

**Sol.** $x^3 = 8$

$x = 2$




$$\text{Area} = \int_1^2 \left(x^2 - 1\right) dx + \int_2^8 \left(\frac{8}{x} - 1\right) dx$$

$$= \left(\frac{x^3}{3} - x\right)_1^2 + \left(8\ln x - x\right)_2^8$$

$$= \left(\frac{8}{3} - 2 - \frac{1}{3} + 1\right) + \left[8\ln 8 - 8\ln 2 - 8 + 2\right]$$

$$= \left(\frac{7}{3} - 1\right) + 8\ln 4 - 6$$

$$= 16\ln 2 + \frac{7}{3} - 7$$

$$= 16\ln 2 - \frac{14}{3}$$

11. Let the functions $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$. Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is :

**[JEE(Adv.)2020]**

माना कि फलनों $f: \mathbb{R} \to \mathbb{R}$ एवं $g: \mathbb{R} \to \mathbb{R}$ को $f(x) = e^{x-1} - e^{-|x-1|}$ एवं $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$ के द्वारा परिभाषित किया जाता

gSr c i Ze pr dkZkeksa Øks $y = f(x)$, $y = g(x)$ एवं $x = 0$ के द्वारा प्रतिबद्ध क्षेत्र का क्षेत्रफल है? **[JEE(Adv.)2020]**

(A) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

(B) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

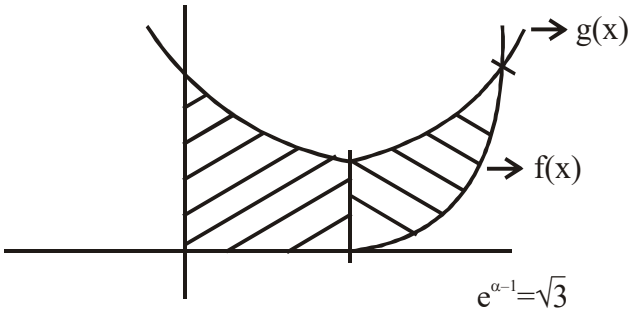
(C) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

(D) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

**Ans.** A

**Sol.** $f(x) \begin{cases} \to e^{x-1} - e^{1-x} & x \geq 1 \\ \to 0 & x < 1 \end{cases}$

$g(x) = \frac{1}{2}\left(e^{x-1} + e^{1-x}\right)$




$P\left(1+ln\sqrt{3}, \frac{2}{\sqrt{3}}\right) \equiv P\left(\alpha, \frac{2}{\sqrt{3}}\right)$

$\text{Required Area} = \int_0^1 \frac{e^{x-1}+e^{1-x}}{2}\,dx + \int_1^\alpha f(x) - f(x)\,dx$

$= \frac{1}{2}\left(e-\frac{1}{e}\right) + \int_1^\alpha \left(\frac{e^{x-1}+e^{1-x}}{2} - \left(e^{x-1}-e^{1-x}\right)\right)dx$

$= \frac{1}{2}\left(e-\frac{1}{e}\right) + \int_1^\alpha \left(\frac{3}{2}e^{1-x} - \frac{e^{x-1}}{2}\right)dx = \left(2-\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

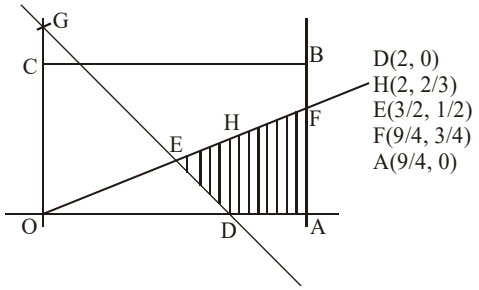
12. The area of the region $\left\{(x,y): 0 \leq x \leq \frac{9}{4},\quad 0 \leq y \leq 1,\quad x \geq 3y,\quad x+y \geq 2\right\}$ is : **[JEE(Adv.)2021]**

क्षेत्र $\left\{(x,y): 0 \leq x \leq \frac{9}{4},\quad 0 \leq y \leq 1,\quad x \geq 3y,\quad x+y \geq 2\right\}$ का क्षेत्रफल है : **[JEE(Adv.)2021]**

(A) $\frac{11}{32}$ (B) $\frac{35}{96}$ (C) $\frac{37}{96}$ (D) $\frac{13}{32}$

**Ans.** A

**Sol.**




$AB \equiv x = \frac{9}{4}$

$BC \equiv y = 1$

$DG \equiv x + y = 2$

$OF \equiv x = 3y$

$DG \equiv x = 2$

we will divide shaded region in two parts

required area = Area of $\Delta DEH$ + area of trapezium AFHD

$$= \frac{1}{2}\left(\frac{2}{3}\right)\left(\frac{1}{2}\right) + \frac{1}{2}\left(\frac{2}{3} + \frac{3}{4}\right)\left(\frac{1}{4}\right) = \frac{11}{32}$$

13. Consider the functions $f, g : R \to R$ defined by $f(x) = x^2 + \frac{5}{12}$ and $g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right) & |x| \le \frac{3}{4}, \\ 0, & |x| > \frac{3}{4} \end{cases}$. If $\alpha$ is the area of the region $\left\{(x, y) \in R \times R : |x| \le \frac{3}{4}, 0 \le y \le \min\{f(x), g(x)\}\right\}$ then the value of $9\alpha$ is ____. **[JEE(Adv.)2022]**

फलनों $f, g : R \to R$ पर विचार कीजिए जो $f(x) = x^2 + \frac{5}{12}$ एवं $g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right) & |x| \le \frac{3}{4}, \\ 0, & |x| > \frac{3}{4} \end{cases}$ द्वारा परिभाषित है।

यदि क्षेत्र $\left\{(x, y) \in R \times R : |x| \le \frac{3}{4}, 0 \le y \le \min\{f(x), g(x)\}\right\}$ का क्षेत्रफल $\alpha$ है, तब $9\alpha$ का मान _____ है।

**[JEE(Adv.)2022]**

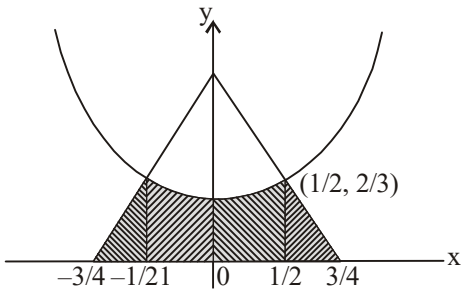
**Ans.** 6

Sol. $f(x)=x^2+\frac{5}{12}$

$g(x)=2\left(1-\frac{4}{3}|x|\right)$

$|x|\le\frac{3}{4}$ and $0\le y\le(f(x),g(x))$

for $x\ge 0$




$x^2+\frac{5}{12}=2-\frac{8}{3}x$

$\Rightarrow 12x^2+32x-19=0$

$12x^2+38x-6x-19=0\Rightarrow x=\frac{1}{2}$

Required area $=2\left(\int_0^{1/2}\left(x^2+\frac{5}{12}\right)dx+\frac{1}{2}\times\frac{1}{4}\times\left(2-\frac{4}{3}\right)\right)$

$\Rightarrow\alpha=2\left(\frac{1}{24}+\frac{5}{24}+\frac{2}{24}\right)=\frac{2}{3}$ sq.units $\Rightarrow 9\alpha=9\times\frac{2}{3}=6$

14. Let $f:[0,1]\to[0,1]$ be the function defined by $f(x)=\frac{x^3}{3}-x^2+\frac{5}{9}x+\frac{17}{36}$. Consider the square region $S=[0,1]\times[0,1]$. Let $G=\{(x,y)\in S:y>f(x)\}$ be called the green region and $R=\{(x,y)\in S:y<f(x)\}$ be called the red region. Let $L_h=\{(x,h)\in S:x\in[0,1]\}$ be the horizontal line drawn at a height $h\in[0,1]$. Then which of the following statements is(are) true? **[JEE(Adv.)2023]**

(A) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line $L_h$ equals the area of the green region below the line $L_h$

(B) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line $L_h$ equals the area of the red region below the line $L_h$

(C) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line $L_h$ equals the area of the red region below the line $L_h$

(D) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line $L_h$ equals the area of the green region below the line $L_h$

माना कि फलन $f:[0,1]\to[0,1]$, $f(x)=\frac{x^3}{3}-x^2+\frac{5}{9}x+\frac{17}{36}$ से परिभाषित है। वर्गाकार क्षेत्र(square region) $S=[0,1]\times[0,1]$ पर विचार कीजिए। माना कि $G=\{(x,y)\in S: y>f(x)\}$ हरित क्षेत्र(green region) एवं $R=\{(x,y)\in S: y<f(x)\}$ लाल क्षेत्र (red region) कहलाता है। मान लीजिये की $h\in[0,1]$ की ऊँचाई (height) पर खींची गई {kSfr t jsf[kk(horizontal line) $L_h=\{(x,h)\in S: x\in[0,1]\}$ है। तब निम्न में से कौनसा (से) कथन सत्य है(हैं)?

**[JEE(Adv.)2023]**

(A) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा $L_h$ के ऊपर के लाल क्षेत्र का क्षेत्रफल रेखा $L_h$ के नीचे के हरित क्षेत्र के क्षेत्रफल के बराबर है

(B) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा $L_h$ के ऊपर के हरित क्षेत्र का क्षेत्रफल रेखा $L_h$ के नीचे के लाल क्षेत्र के क्षेत्रफल के बराबर है

(C) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा $L_h$ के ऊपर के लाल क्षेत्र का क्षेत्रफल रेखा $L_h$ के नीचे के लाल क्षेत्र के क्षेत्रफल के बराबर है

(D) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा $L_h$ के ऊपर के हरित क्षेत्र का क्षेत्रफल रेखा $L_h$ के नीचे के हरित क्षेत्र के क्षेत्रफल के बराबर है

Ans. BCD

Sol. $f(x)=\frac{x^3}{3}-x^2+\frac{5x}{9}+\frac{17}{36}$

$f'(x)=x^2-2x+\frac{5}{9}$

$f'(x)=0$ at $x=\frac{1}{3}$ in $[0,1]$

$A_R=$ Area of Red region

$A_G=$ Area of Green region

$A_R=\int_0^1 f(x)\,dx=\frac{1}{2}$

Total area = 1

$\Rightarrow A_G=\frac{1}{2}$

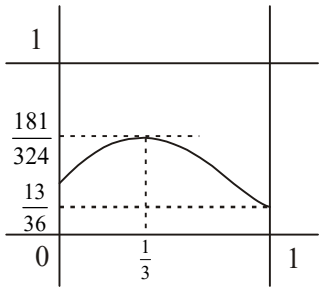
$\int_0^1 f(x)\,dx=\frac{1}{2}$

$A_G=A_R$

$f(0)=\frac{17}{36}$

$$f(1)=\frac{13}{36}\approx 0.36$$

$$f\left(\frac{1}{3}\right)=\frac{181}{324}\approx 0.558$$




(A) Given statement would be correct when $h=\frac{3}{4}$ but $h\in\left[\frac{1}{4},\frac{2}{3}\right]$

$\Rightarrow$ (A) is incorrect

(B) Given statement would be correct when $h=\frac{1}{4}$

$\Rightarrow$ (B) is correct

(C) When $h=\frac{181}{324}, A_R=\frac{1}{2}, A_G<\frac{1}{2}$

$h=\frac{13}{36}, A_R<\frac{1}{2}, A_G=\frac{1}{2}$

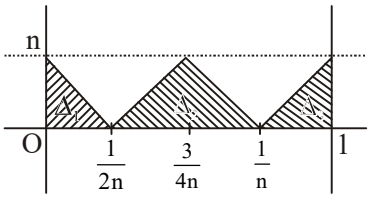
$\Rightarrow A_R=A_G$ for some $h\in\left(\frac{13}{36},\frac{181}{324}\right)$

$\Rightarrow$ (C) is correct

(D) Option (D) is remaining coloured part of option (C), hence option (D) is also correct.

15. Let $n\geq 2$ be a natural number and $f:[0,1]\to R$ be the function defined by

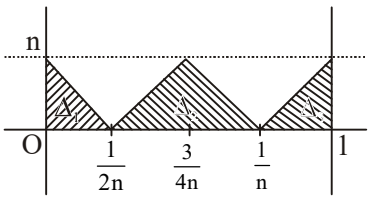
$$f(x)=\begin{cases} n(1-2nx) & \text{if } 0\leq x\leq\frac{1}{2n} \\ 2n(2nx-1) & \text{if } \frac{1}{2n}\leq x\leq\frac{3}{4n} \\ 4n(1-nx) & \text{if } \frac{3}{4n}\leq x\leq\frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \text{if } \frac{1}{n}\leq x\leq 1 \end{cases}$$



If n is such that the area of the region bounded by the curves $x = 0$, $x = 1$, $y = 0$ and $y = f(x)$ is 4 , then the maximum value of the function f is _____. **[JEE(Adv.)2023]**

माना कि $n \geq 2$ एक प्राकृत संख्या (natural number) है एवं फलन $f : [0,1] \to R$ इस प्रकार परिभाषित है

$$f(x) = \begin{cases} n(1-2nx) & \text{if } 0 \leq x \leq \dfrac{1}{2n} \\ 2n(2nx-1) & \text{if } \dfrac{1}{2n} \leq x \leq \dfrac{3}{4n} \\ 4n(1-nx) & \text{if } \dfrac{3}{4n} \leq x \leq \dfrac{1}{n} \\ \dfrac{n}{n-1}(nx-1) & \text{if } \dfrac{1}{n} \leq x \leq 1 \end{cases}$$




यदि n इस प्रकार है कि वक्रों $x = 0$, $x = 1$, $y = 0$ एवं $y = f(x)$ से घिरे क्षेत्र का क्षेत्रफल 4 है तब फलन f का महत्तम मान (maximum value) है _____. **[JEE(Adv.)2023]**

Ans. 8

Sol. $$f(x) = \begin{cases} n(1-2nx) & 0 \leq n \leq \dfrac{1}{2n} \\ 2n(2nx-1) & \dfrac{1}{2n} \leq x \leq \dfrac{3}{4n} \\ 4n(1-nx) & \dfrac{3}{4n} \leq x \leq \dfrac{1}{n} \\ \dfrac{n}{n-1}(nx-1) & \dfrac{1}{n} \leq x \leq 1 \end{cases}$$

$\Delta_1 + \Delta_2 + \Delta_3 = 4$

$$\frac{1}{2} n \times \frac{1}{2n} + \frac{1}{2} \times n \times \frac{1}{2n} + \frac{1}{2} \times n \times \left(1 - \frac{1}{n}\right) = 4$$

$$4 = \frac{n}{2}$$

$n = 8$

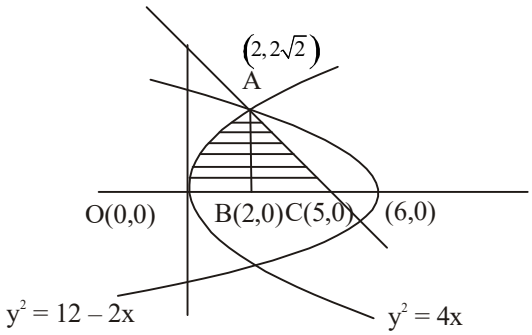
16. Let $S = \left\{(x, y) \in R \times R : x \geq 0,\ y \geq 0,\ y^2 \leq 4x,\ y^2 \leq 12 - 2x \text{ and } 3y + \sqrt{8}x \leq 5\sqrt{8}\right\}$ If the area of the region S is $\alpha\sqrt{2}$ , then $\alpha$ is eqaul to : **[JEE(Adv.)2024]**

माना कि $S = \left\{(x, y) \in R \times R : x \geq 0,\ y \geq 0,\ y^2 \leq 4x,\ y^2 \leq 12 - 2x \text{ और } 3y + \sqrt{8}x \leq 5\sqrt{8}\right\}$ है। यदि क्षेत्र (region) S का {ksl Qy $\alpha\sqrt{2}$ है, तब $\alpha$ बराबर है : **[JEE(Adv.)2024]**

(A) $\frac{17}{2}$ (B) $\frac{17}{3}$ (C) $\frac{17}{4}$ (D) $\frac{17}{5}$

**Ans.** B

**Sol.** $3y+\sqrt{8}x=5\sqrt{8}$




$4x = 12 - 2x \Rightarrow 6x = 12 \Rightarrow x = 2$

$x = 2 \Rightarrow y^2 = 8 \Rightarrow y = \pm 2\sqrt{2}$

line $3y+\sqrt{8}x=5\sqrt{8}$ passes through $A\left(2,2\sqrt{2}\right)$ & C(5,0)

Required area = Area of OAB + Area of $\Delta ABC$

$$=\int_0^2 2\sqrt{x}dx+\frac{1}{2}(5-2)\left(2\sqrt{2}\right)$$

$$=\frac{4}{3}\left(x^{\frac{3}{2}}\right)_0^2+3\sqrt{2}$$

$$=\frac{8}{3}\sqrt{2}+3\sqrt{2}$$

$$=\frac{17}{3}\sqrt{2}$$

17. Let **R** denote the set of all real numbers. Then the area of the region

$$\left\{(x,y)\in \mathbf{R}\times\mathbf{R}: x>0, y>\frac{1}{x}, 5x-4y-1>0, 4x+4y-17<0\right\}$$

is: **[JEE(Adv.)2025]**

मान लीजिए कि **R** सभी वास्तविक संख्याओं (real numbers) के समुच्चय (set) को दर्शाता है। तब क्षेत्र (region)

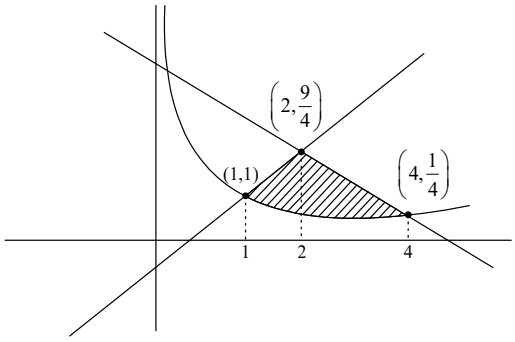
$$\left\{(x,y)\in \mathbf{R}\times\mathbf{R} : x>0, y>\frac{1}{x}, 5x-4y-1>0, 4x+4y-17<0\right\}$$

का क्षेत्रफल (area) है : **[JEE(Adv.)2025]**

(A) $\frac{17}{16}-\log_e 4$ (B) $\frac{33}{8}-\log_e 4$ (C) $\frac{57}{8}-\log_e 4$ (D) $\frac{17}{2}-\log_e 4$

**Ans.** B

**Sol.**




$$A=\frac{1}{2}\times\left(1+\frac{9}{4}\right)\times 1+\frac{1}{2}\times\left(\frac{9}{4}+\frac{1}{4}\right)\times 2-\int_1^4\frac{1}{x}dx$$

$$A=\frac{33}{8}-\ell n4$$

18. Let S denote the locus of the mid-points of those chords of the parabola $y^2 = x$, such that the area of the region enclossed between the parabola and the chord is $\frac{4}{3}$. Let R denote the region lying in the first quadrant, enclosed by the parabola $y^2 = x$, the curve S, and the lines $x = 1$ and $x = 4$.

Then which of the following statements is (are) TRUE? **[JEE(Adv.)2025]**

(A) $\left(4,\sqrt{3}\right)\in S$ (B) $\left(5,\sqrt{2}\right)\in S$

(C) Area of R is $\frac{14}{3}-2\sqrt{3}$ (D) Area of R is $\frac{14}{3}-\sqrt{3}$

मान लीजिए कि S, परवलय (parabola) $y^2 = x$ की उन सभी जीवाओं (chord) के मध्य–बिन्दुओं (mid-points) का बिन्दुपथ (locus) है, जिनके लिए परवलय एवं जीवा द्वारा घिरे क्षेत्र (region enclosed) का क्षेत्रफल (area) $\frac{4}{3}$ है। मान लीजिए कि R प्रथम चतुर्थांश (first quadrant) में उस क्षेत्र को दर्शाता है, जो परवलय $y^2 = x$, वक्र (curve) S, और रेखाओं $x = 1$ तथा $x = 4$ द्वारा घिरा हुआ है।

r c fuEufy f[ kr d Hkqkesa sd kS l k¼ s1 R, gS½2 **[JEE(Adv.)2025]**

(A) $\left(4,\sqrt{3}\right)\in S$

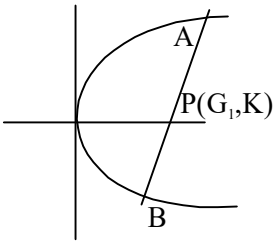
(B) $\left(5,\sqrt{2}\right)\in S$

(C) $\tau$ का क्षेत्रफल $\frac{14}{3}-2\sqrt{3}$ है

(D) R का क्षेत्रफल $\frac{14}{3}-\sqrt{3}$ है

**Ans.** AC

**Sol.** Equation of chord AB $T = S_1$




$$ky-\left(\frac{x+h}{2}\right)=k^2-h$$

$$x-2ky+2k^2-h=0$$

$$\Rightarrow h-k^2<0$$

For area interchange x & y

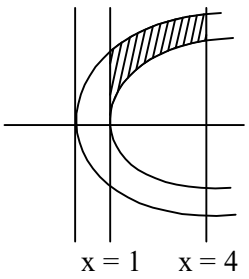
$$y-2kx+2k^2-h=0 \;\&\; y=x^2$$

$$x^2-2kx+\left(2k^2-h\right)=0\left\langle\begin{matrix}\alpha\\ \beta\end{matrix}\right.$$

$$|\alpha-\beta|=\sqrt{4k^2-4\left(2k^2-h\right)}=\sqrt{4h-4k^2}$$

$$A=\int_{\alpha}^{\beta}\left(\left(2kx+h-2k^2\right)-x^2\right)dx=\frac{4}{3}$$

$$\frac{\left(4h-4k^2\right)^{\frac{3}{2}}}{6}=\frac{4}{3}$$




$$\left(4h-4k^2\right)^{\frac{3}{2}}=8$$

$$\Rightarrow (4h-4k^2)=4$$

$$h-k^2=1$$

$$\left(4,\sqrt{3}\right)\in S$$

$$A=\int_{1}^{4}\left(\sqrt{x}-\sqrt{x-1}\right)dx=\frac{2}{3}\left[\left(x^{\frac{3}{2}}-\left(x-1\right)^{\frac{3}{2}}\right)\right]_1^4$$

$$=\frac{2}{3}\left(8-3\sqrt{3}-1\right)=\frac{14}{3}-2\sqrt{3}$$

**Question Stem for Question Nos. 19 and 20**

Consider the curve $C_1$ given by $y = e^{-x}$ for $x \in [0, 10\pi]$ and the curve $C_2$ given by $y = e^{-x}(\sin x + \cos x)$ for $x \in [0, 10\pi]$

Let n be the total number of points of intersection of the curves $C_1$ and $C_2$

Suppose that $\alpha_1, \alpha_2, \ldots\ldots\ldots, \alpha_n \in [0, 10\pi]$ are the x-coordinates of the points of intersection of the curves $C_1$ and $C_2$ such that $\alpha_1 < \alpha_2 < \ldots < \alpha_n$

वक्र $C_1$ जो $y = e^{-x}$ सभी $x \in [0, 10\pi]$ के लिए, द्वारा दिया गया है, और वक्र $C_2$ जो $y = e^{-x}(\sin x + \cos x)$ सभी $x \in [0, 10\pi]$ के लिए, द्वारा दिया गया है, पर विचार कीजिए।

मान लीजिए कि वक्रों $C_1$ और $C_2$ के कुल प्रतिच्छेद बिंदुओं की संख्या n है।

मान लीजिए कि $\alpha_1, \alpha_2, \ldots\ldots\ldots, \alpha_n \in [0, 10\pi]$, वक्रों $C_1$ और $C_2$ के प्रतिच्छेद बिंदुओं के x-निर्देशांक इस प्रकार हैं कि $\alpha_1 < \alpha_2 < \ldots < \alpha_n$ है।

19. The value of n is _____. **[JEE(Adv.)2026]**

n का मान _____ है। **[JEE(Adv.)2026]**

**Ans.** 11

**Sol.** To find the points of intersection $e^{-x} = e^{-x}(\sin x + \cos x)$

$1 = \sin x + \cos x$

$$\cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$x - \frac{\pi}{4} = 2k\pi + \frac{\pi}{4} \Rightarrow x = 2k\pi + \frac{\pi}{2} \text{ or } x - \frac{\pi}{4} = 2k\pi - \frac{\pi}{4} \Rightarrow x = 2k\pi \quad \text{where } k \in \mathbb{Z}$$

Now, we find the values of x that lie within the given domain $x \in [0, 10\pi]$

Φρομ $\xi = 2\kappa\pi$: $x = 0, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi$ (6 points)

From $x = 2k\pi + \frac{\pi}{2}$: $x = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \frac{13\pi}{2}, \frac{17\pi}{2}$ (5 points)

$$\alpha_1 = 0, \alpha_2 = \frac{\pi}{2}, \alpha_3 = 2\pi, \alpha_4 = \frac{5\pi}{2}, \ldots, \alpha_{11} = 10\pi$$

Thus, the total number of intersection points is $n = 11$

20. Let $\beta$ be the area of the region enclosed between the curves $C_1$, $C_2$ and the lines $x = \alpha_1$ and $x = \alpha_4$. Then the value

of $-\frac{1}{\pi}\log_e\left(\beta-2e^{-\frac{\pi}{2}}\right)$ is _____. **[JEE(Adv.)2026]**

मान लीजिए कि वक्रों $C_1$, $C_2$, और रेखाओं $x=\alpha_1$ और $x=\alpha_4$ से परिबद्ध क्षेत्र का क्षेत्रफल $\beta$ है। तब $-\frac{1}{\pi}\log_e\left(\beta-2e^{-\frac{\pi}{2}}\right)$ का मान _____ है। **[JEE(Adv.)2026]**

**Ans.** 2.5

**Sol.** $\alpha_1=0$ and $\alpha_4=\frac{5\pi}{2}$

The enclosed area $\beta$ is given by

$$\beta=\int_0^{\frac{5\pi}{2}}|y_{C_1}-y_{C_2}|dx=\int_0^{\frac{5\pi}{2}}e^{-x}\,|1-\sin x-\cos x|dx$$

First, let's determine the indefinite integral without absolute values

$$I=\int e^{-x}(\sin x+\cos x-1)dx=F(x)$$

$$F(x)=e^{-x}(1-\cos x)$$

Since the expression inside the absolute value changes sign only at the roots $(\alpha_1,\alpha_2,\alpha_3,\alpha_4)$, the total area $\beta$ is the sum of the absolute differences of F(x) at consecutive boundaries

$$\beta=|F(\alpha_2)-F(\alpha_1)|+|F(\alpha_3)-F(\alpha_2)|+|F(\alpha_4)-F(\alpha_3)|$$

Substituting these values into the area formula

$$\beta=\left|e^{-\frac{\pi}{2}}-0\right|+\left|0-e^{-\frac{\pi}{2}}\right|+\left|e^{-\frac{5\pi}{2}}-0\right|$$

$$\beta=e^{-\frac{\pi}{2}}+e^{-\frac{\pi}{2}}+e^{-\frac{5\pi}{2}}=2e^{-\frac{\pi}{2}}+e^{-\frac{5\pi}{2}}$$

$$\beta-2e^{-\frac{\pi}{2}}=e^{-\frac{5\pi}{2}}$$

$$\log_e\left(\beta-2e^{-\frac{\pi}{2}}\right)=\log_e\left(e^{-\frac{5\pi}{2}}\right)=-\frac{5\pi}{2}$$

**Question Stem for Question Nos. 21 and 22**

Consider the ellipses given by : $x^2 + 4y^2 = 1$ and $4x^2 + y^2 = 1$

दीर्घवृत्तों : $x^2 + 4y^2 = 1$ और $4x^2 + y^2 = 1$ पर विचार कीजिए।

21. Let P be the point in the first quadrant where the given ellipses intersect. If $\theta$ is the acute angle between the tangents to the given ellipses at the point P, then the value of $4\tan\theta$ is _____. **[JEE(Adv.)2026]**

मान लीजिए कि P, प्रथम चतुर्थांश में वह बिन्दु है जहाँ दिए गये दीर्घवृत्त प्रतिच्छेद करते हैं। यदि $\theta$, बिन्दु P पर दिए गये दीर्घवृत्तों की स्पर्श रेखाओं के बीच का न्यूनकोण है, तब $4\tan\theta$ का मान _____ होगा। **[JEE(Adv.)2026]**

**Ans.** 7.5

**Sol.** Given the equations of the two ellipses:

$x^2 + 4y^2 = 1$ and $4x^2 + y^2 = 1$

Subtracting equation (1) from equation (2):

$(4x^2 + y^2) - (x^2 + 4y^2) = 1 - 1$

$3x^2 - 3y^2 = 0 \quad \Rightarrow x^2 = y^2$

Since the point P lies in the first quadrant, both $x > 0$ and $y > 0$, so we have $x = y$.

$x^2 + 4x^2 = 1 \quad \Rightarrow 5x^2 = 1 \quad \Rightarrow x = \frac{1}{\sqrt{5}}$

Since $y = x$, the coordinates of point P are: $P = \left(\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$

Differentiating both equations with respect to x to find the slopes ($m_1$ and $m_2$):

For the first ellipse ($x^2 + 4y^2 = 1$): $2x + 8y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{4y}$

At $P\left(\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$, the slope $m_1$ is: $m_1 = -\frac{\frac{1}{\sqrt{5}}}{4\left(\frac{1}{\sqrt{5}}\right)} = -\frac{1}{4}$

For the second ellipse ($4x^2 + y^2 = 1$): $8x + 2y\frac{dy}{dx} = 0 \quad \Rightarrow \frac{dy}{dx} = -\frac{4x}{y}$

At $P\left(\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$, the slope $m_2$ is: $m_2 = -\frac{4\left(\frac{1}{\sqrt{5}}\right)}{\frac{1}{\sqrt{5}}} = -4$

The angle $\theta$ between two lines with slopes $m_1$ and $m_2$ is given by:

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| \Rightarrow \quad \tan\theta = \left|\frac{-\frac{1}{4} - (-4)}{1 + \left(-\frac{1}{4}\right)(-4)}\right| = \left|\frac{-\frac{1}{4} + 4}{1 + 1}\right| = \left|\frac{\frac{15}{4}}{2}\right| = \frac{15}{8}$$

$$4\tan\theta = 4 \times \frac{15}{8} = \frac{15}{2} = 7.5$$

22. If $\alpha$ is the area of the common region that lies inside both the given ellipses, then the value of $\cot\alpha$ is _____.

**[JEE(Adv.)2026]**

यदि $\alpha$ दिए गये दोनों दीर्घवृत्तों के अंदर के उभयनिष्ठ क्षेत्र का क्षेत्रफल है, तब $\cot\alpha$ का मान _____होगा।

**[JEE(Adv.)2026]**

**Ans.** 0.75

**Sol.** The given ellipses are highly symmetrical across all four quadrants and about the line $y = x$. We can convert their equations into polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$) to easily integrate the bounded area.

First Ellipse : $r^2\cos^2\theta + 4r^2\sin^2\theta = 1 \qquad \Rightarrow r_1^2 = \frac{1}{1 + 3\sin^2\theta}$

Second Ellipse: $4r^2\cos^2\theta + r^2\sin^2\theta = 1 \qquad \Rightarrow r_2^2 = \frac{1}{4 - 3\sin^2\theta}$

In the first quadrant, the two curves intersect at $\theta = \frac{\pi}{4}$. Due to symmetry, the total area $\alpha$ is 8 times the area bounded from $\theta = 0$ to $\theta = \frac{\pi}{4}$ by the inner curve ($r_2$) :

$$\theta = 8 \times \left(\frac{1}{2}\int_0^{\pi/4} r_2^2\, d\theta\right) = 4\int_0^{\pi/4} \frac{1}{4 - 3\sin^2\theta} d\theta$$

$$\alpha = 4\int_0^{\pi/4} \frac{\sec^2\theta}{4\sec^2\theta - 3\tan^2\theta} d\theta \Rightarrow \alpha = 4\int_0^{\pi/4} \frac{\sec^2\theta}{4(1 + \tan^2\theta) - 3\tan^2\theta} d\theta = 4\int_0^{\pi/4} \frac{\sec^2\theta}{4 + \tan^2\theta} d\theta$$

Let $t = \tan\theta \qquad \Rightarrow dt = \sec^2\theta d\theta$

$$\alpha = 4\int_0^1 \frac{dt}{4 + t^2} = 4\left[\frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right)\right]_0^1 \Rightarrow \quad \alpha = 2\tan^{-1}\left(\frac{1}{2}\right)$$

Let $\beta = \tan^{-1}\left(\frac{1}{2}\right)$. Thus, $\alpha = 2\beta$.

$$\tan\alpha = \tan(2\beta) = \frac{2\tan\beta}{1-\tan^2\beta} = \frac{2\left(\frac{1}{2}\right)}{1-\left(\frac{1}{2}\right)^2} = \frac{1}{1-\frac{1}{4}} = \frac{1}{\frac{3}{4}} = \frac{4}{3}$$

$$\cot\alpha = \frac{1}{\tan\alpha} = \frac{3}{4} = 0.75$$

# CHALLENGER EXERCISES

## Exercise C - I

### SINGLE CORRECT TYPE QUESTIONS

1. The area bounded by regions $y=\sqrt{x}$ and $x=-\sqrt{y}$ and the curve $x^2+y^2=2$ is : (above x-axis)

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}-\frac{1}{3}$ (C) $\frac{\pi}{2}$ (D) None

वक्रों $y=\sqrt{x}$ , $x=-\sqrt{y}$ तथा $x^2+y^2=2$ से घिरा हुआ क्षेत्रफल होगा : (x-अक्ष के ऊपर)

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}-\frac{1}{3}$ (C) $\frac{\pi}{2}$ (D) कोई नहीं

**Ans.** C

**Sol.** $y=\sqrt{x}$ ...(1)

$x=-\sqrt{y}$ ...(2)

$x^2+y^2+2$ ...(3)

POI of (1) & (3)

$x^2+x+-2=0$

$x=-2, 1$

In 1st quadrant

(1,1)

POI of (2) & (3)

$y^2+y-2=0$

$y=-2,1$

In IInd quadrant

$y=1$

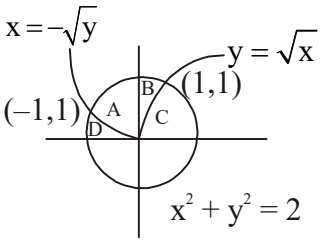
$x=-1$ $(-1,1)$

Let areas are A, B, C, D

from symmetry

A = C & B = D

we need



$= A + B$

$= C + D = \text{area of quadrant} = \dfrac{\pi r^2}{4} = \dfrac{\pi \cdot \left(\sqrt{2}\right)^2}{4} = \dfrac{\pi}{2}$

2. Area bounded by the curve $y = \max\{\sin x, \cos x\}$ and x-axis, between the lines $x = \dfrac{\pi}{4}$ and $x = 2\pi$ is equal to :

(A) $\dfrac{(4\sqrt{2}-1)}{\sqrt{2}}$ sq. units

(B) $(4\sqrt{2}-1)$ sq. units

(C) $\dfrac{(4\sqrt{2}-1)}{2}$ sq. units

(D) None of these

वक्रों $y = \max\{\sin x, \cos x\}$ तथा x-अक्ष, रेखाओं $x = \dfrac{\pi}{4}$ व $x = 2\pi$ के मध्य का क्षेत्रफल होगा :

(A) $\dfrac{(4\sqrt{2}-1)}{\sqrt{2}}$ वर्ग इकाई

(B) $(4\sqrt{2}-1)$ वर्ग इकाई

(C) $\dfrac{(4\sqrt{2}-1)}{2}$ वर्ग इकाई

(D) इनमें से कोई नहीं

**Ans.** A

**Sol.** $y = \max\{\sin x, \cos x\}$, x-axis $x = \dfrac{\pi}{4}$ & $2\pi$

$$A = \int_{\frac{\pi}{4}}^{\pi} \sin x\,dx - \int_{\pi}^{\frac{5\pi}{4}} \sin x\,dx - \int_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} \cos x\,dx + \int_{\frac{3\pi}{2}}^{2\pi} \cos x\,dx$$

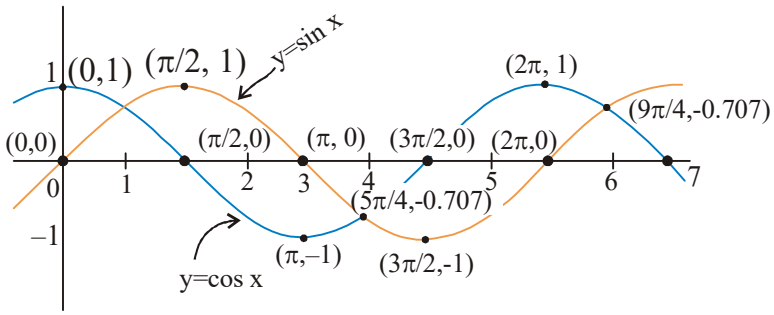
$$= \left(-\cos x\right]_{\frac{\pi}{4}}^{\pi} - \left[-\cos x\right]_{\pi}^{\frac{5\pi}{4}} - \left[\sin x\right]_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} + \left[\sin x\right]_{\frac{3\pi}{2}}^{2\pi}$$

$$= \left[-\cos x\right]_{\frac{\pi}{4}}^{\pi} - \left[-\cos x\right]_{\pi}^{\frac{5\pi}{4}} - \left[\sin x\right]_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} + \left[\sin x\right]_{\frac{3\pi}{2}}^{2\pi}$$

$$= 1 + \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} + 1$$

$$= 4 - \frac{1}{\sqrt{2}}$$

$$= \frac{4\sqrt{2} - 1}{\sqrt{2}}$$



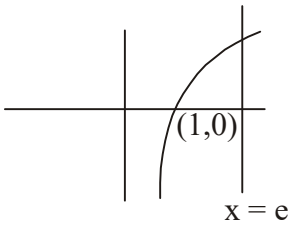

3. If A(n) represents the area bounded by the curve $y = n.\ \ell nx$, where $n \in N$ and $n > 1$, the x-axis and the lines $x = 1$ and $x = e$, then the value of $A(n) + nA(n-1)$ is equal to :

यदि A(n) वक्रों $y = n.\ \ell nx$, (जहाँ $n \in N$ तथा $n > 1$) x-अक्ष तथा रेखाओं $x = 1$ व $x = e$ के मध्य का क्षेत्रफल हो, तो $A(n) + nA(n-1)$ का मान होगा :

(A) $\frac{n^2}{e+1}$ (B) $\frac{n^2}{e-1}$ (C) $n^2$ (D) $en^2$

**Ans.** C

**Sol.** $y = n\ell nx$, $n \in N$, $n > 1$, x-axis




$$A(n) = n\int_1^e \ell n\, x \cdot dx$$

$$= n\left(x\,\ell n\, x - x\right)_1^e = n$$

$$\Rightarrow A(n) = n$$

$$A(n) + nA(n-1) = n + n(n-1)$$

$$= n^2$$

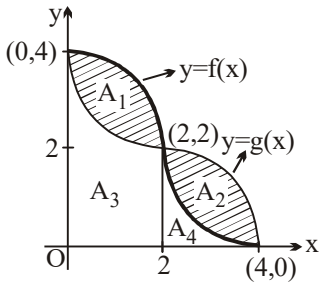
4. Suppose $y = f(x)$ and $y = g(x)$ are two continuous functions whose graphs intersect at the three points (0, 4), (2, 2) and (4, 0) with $f(x) > g(x)$ for $0 < x < 2$ and $f(x) < g(x)$ for $2 < x < 4$. If $\int_0^4 [f(x) - g(x)]dx = 10$ and $\int_2^4 [g(x) - f(x)]dx = 5$, the area between two curves for $0 < x < 2$, is :

माना $y = f(x)$ तथा $y = g(x)$ दो सतत् फलन है जिनके वक्र तीन बिन्दुओं (0, 4),(2, 2) तथा (4, 0) पर प्रतिच्छेद होते हैं साथ ही $f(x) > g(x)$ for $0 < x < 2$ तथा $f(x) < g(x)$ जब $2 < x < 4$. यदि $\int_0^4 [f(x) - g(x)]dx = 10$ तथा $\int_2^4 [g(x) - f(x)]dx = 5$, तो nksuksa oØksa ds e/; {ks=kQy ] v Hkqt jk; $0 < x < 2$ के लिए होगा :

(A) 5 (B) 10 (C) 15 (D) 20

**Ans.** C

**Sol.** Given $\int_0^4 f(x)\,dx - \int_0^4 g(x)\,dx = 10$




$(A_1 + A_3 + A_4) - (A_2 + A_3 + A_4) = 10$

$A_1 - A_2 = 10$ ....(1)

again $\int_2^4 g(x)\,dx - \int_2^4 f(x)\,dx = 5$

$(A_2 + A_4) - A_4 = 5$

$A_2 = 5$ .....(2)

$\therefore$ (1) + (2)

$A_1 = 15$

5. Let 'a' be a positive constant number. Consider two curves $C_1$: $y = e^x$, $C_2$ : $y = e^{a-x}$. Let S be the area of the part surrounding by $C_1$, $C_2$ and the y-axis, then $\lim_{a\to 0} \frac{S}{a^2}$ equals :

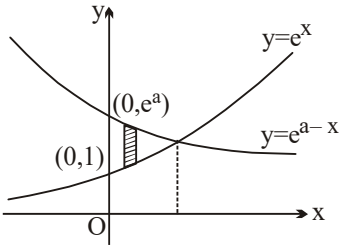
'a' एक धनात्मक अचर है। मान लीजिए दो वक्र है। $C_1$: $y = e^x$, $C_2$ : $y = e^{a-x}$. माना S उस भाग का क्षेत्रफल है जो $C_1$, $C_2$ तथा y-अक्ष }kjk ki fjc) g\$r ks $\lim_{a\to 0} \frac{S}{a^2}$ का मान होगा :

(A) 4 (B) 1/2 (C) 0 (D) 1/4

**Ans.** D

**Sol.** Solving $e^x = e^{a-x}$, we get

$e^{2x} = e^a \Rightarrow \quad x = \frac{a}{2}$




$S = \int_0^{a/2} (e^a \cdot e^{-x} - e^x)\,dx = \left[-(e^a \cdot e^{-x} + e^x)\right]_0^{a/2}$

$= (e^a + 1) - (e^{a/2} + e^{a/2}) = e^a - 2e^{a/2} + 1 = (e^{a/2} - 1)^2$

$\therefore \quad \frac{S}{a^2} = \left(\frac{e^{a/2}-1}{a}\right)^2 = \frac{1}{4}\left(\frac{e^{a/2}-1}{a/2}\right)^2$

$\therefore \quad \lim_{a\to 0} \frac{S}{a^2} = \frac{1}{4}$

6. The area of the region(s) enclosed by the curves $y = x^2$ and $y = \sqrt{|x|}$ is :

वक्रों $y = x^2$ तथा $y = \sqrt{|x|}$ द्वारा परिबद्ध क्षेत्र (क्षेत्रों) का क्षेत्रफल होगा :

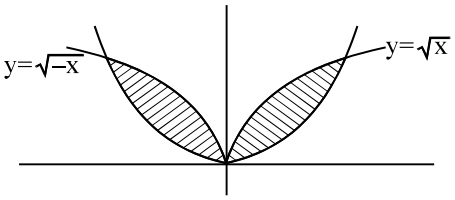
(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) 1

**Ans.** B

**Sol.** $A = \left(\frac{16ab}{3}\right)\cdot 2$

$a = \frac{1}{4};\ b = \frac{1}{4}$

$A = \frac{2}{3}$




7. The slope of the tangent to a curve $y = f(x)$ at $(x, f(x))$ is $2x + 1$. If the curve passes through the point $(1, 2)$ then the area of the region bounded by the curve, the x-axis and the line $x = 1$ is :

वक्र $y = f(x)$ के बिन्दु $(x, f(x))$ पर स्पर्श रेखा की प्रवणता $2x + 1$ है। यदि वक्र बिन्दु $(1, 2)$ से गुजरता है तो वक्र का x-अक्ष तथा रेखा $x = 1$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल होगा :

(A) $\frac{5}{6}$ (B) $\frac{6}{5}$ (C) $\frac{1}{6}$ (D) 1

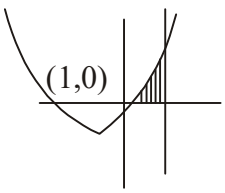
**Ans.** A

**Sol.** $\frac{dy}{dx} = 2x + 1$ (1,2)

$y = x^2 + x + c$

$y = x^2 + x$

$y + \frac{1}{4} = \left(x + \frac{1}{2}\right)^2$



$$A = \int_0^1 \left(x^2 + x\right) dx$$

$$A = \frac{x^3}{3} + \frac{x^2}{2}\Bigg|_0^1$$

$$= \frac{1}{3} + \frac{1}{2}$$

$$= \frac{5}{6}$$

8. The area bounded by the curves $y = -\sqrt{-x}$ and $x = -\sqrt{-y}$ where $x, y \le 0$ :

(A) cannot be determined

(B) is 1/3

(C) is 2/3

(D) is same as that of the figure bounded by the curves $y = \sqrt{-x}$ ; $x \le 0$ and $x = \sqrt{-y}$ ; $y \le 0$

वक्रो $y = -\sqrt{-x}$ तथा $x = -\sqrt{-y}$ जहां $x, y \le 0$ द्वारा परिबद्ध क्षेत्रफल होगा :

(A) ज्ञात नहीं किया जा सकता

(B) 1/3

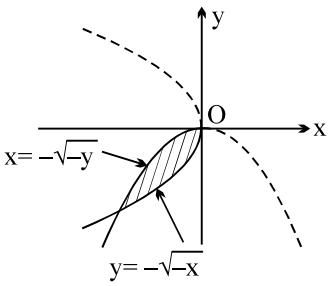
(C) 2/3

(D) वही होगा जो वक्रो $y = \sqrt{-x}$ ; $x \le 0$ तथा $x = \sqrt{-y}$ ; $y \le 0$ द्वारा परिबद्ध है

**Ans.** B

**Sol.** $y = -\sqrt{-x} \Rightarrow y^2 = -x$ where x & y both (–) ve

$x = -\sqrt{-y} \Rightarrow x^2 = -y$ where x & y both (–) ve




Hence $A = \frac{16ab}{3}$

where $a = b = \frac{1}{4}$

$\therefore \quad A = \frac{1}{3} \Rightarrow$ (B)

9. The area bounded by the curve $y = x\,e^{-x}$ ; $xy = 0$ and $x = c$ where c is the x-coordinate of the curve's inflection point, is :

वक्र $y = xe^{-x}$ ; $xy = 0$ तथा $x = c$ द्वारा घिरा क्षेत्रफल होगा, जहाँ वक्र के नति परिवर्तन बिन्दु का x-निर्देशांक c है :

(A) $1 - 3e^{-2}$ (B) $1 - 2e^{-2}$ (C) $1 - e^{-2}$ (D) 1

**Ans.** A

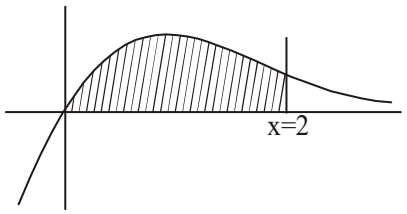
**Sol.** $y = xe^{-x}$

$y' = e^{-x} - x\,e^{-x} = (1 - x)e^{-x} \uparrow$ for $x < 1$

$y'' = -e^{-x} - [e^{-x} - xe^{-x}] = e^{-x}[-1 - 1 + x]$

$= (x - 2)e^{-x}$

for point of inflection $y'' = 0 \quad \Rightarrow \quad x = 2$




$$A = \int_0^2 xe^{-x}\,dx = -\,xe^{-x}\Big|_0^2 + \int_0^2 e^{-x}$$

$= (-2\,e^{-2}) - \left(e^{-x}\right)_0^2$

$= -2\,e^{-2} - (e^{-2} - 1) = 1 - e^{-2} - 2e^{-2} = 1 - 3e^{-2}$

10. Area enclosed by the curve $y = (x^2 + 2x)e^{-x}$ and the positive x-axis is :

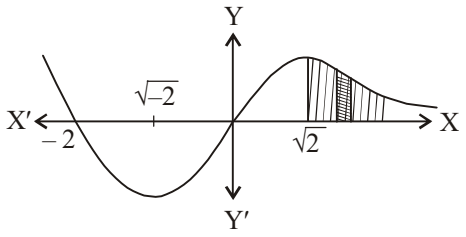
धनात्मक x-अक्ष तथा वक्र $y = (x^2 + 2x)e^{-x}$ द्वारा घिरा क्षेत्रफल होगा :

(A) 1 (B) 2 (C) 4 (D) 6

**Ans.** C

**Sol.** Required area $A = \int_0^{\infty} \underset{\text{(II)}}{e^{-x}}\,\underset{\text{(I)}}{(x^2 + 2x)}\,dx = 4$ Ans.

(I. B. P.)




11. Infinite rectangles each of width 1 unit and height $\left(\frac{1}{n} - \frac{1}{n+1}\right)$ $(n \in N)$ are constructed such that ends of exactly one diagonal of every rectangle lies along the curve $y = \frac{1}{x}$. The sum of areas of all such rectangles, is :

1 इकाई लम्बाई तथा $\left(\frac{1}{n} - \frac{1}{n+1}\right)$ इकाई चौड़ाई (जहाँ $n \in N$) के अनन्त आयत इस प्रकार है कि प्रत्येक आयत का केवल एक fod.kZd s' kh"kZ&Ø $y = \frac{1}{x}$ पर स्थित है, तो इस प्रकार के सभी आयतों के क्षेत्रफल का योग होगा :

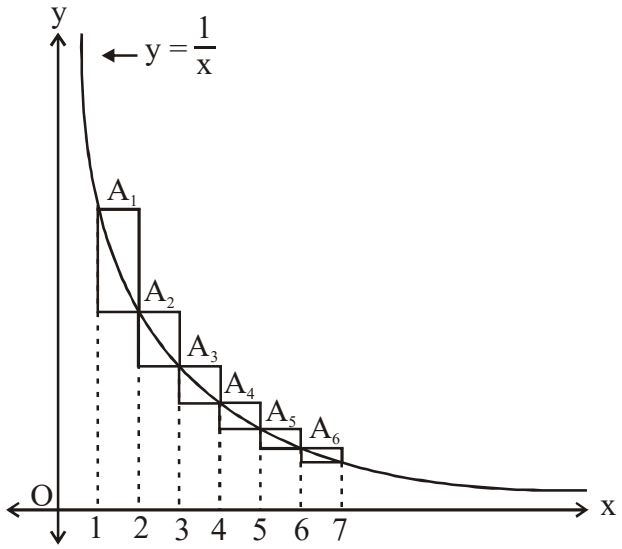
(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) 1

**Ans.** D

**Sol.** Method-I:

$$\text{Required sum} = \lim_{n\to\infty}\left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+.....+\left(\frac{1}{n}-\frac{1}{n+1}\right)\right] = \lim_{n\to\infty}\left(1-\frac{1}{n+1}\right) = 1$$

Method-II:




$$\text{Area of } r^{th} \text{ rectangle } A_r = (\text{height}) \times (\text{width}) = \left(\frac{1}{r}-\frac{1}{r+1}\right)\cdot 1$$

$$\text{Sum of areas} = \lim_{n\to\infty}\sum_{r=1}^{n} A_r = \lim_{n\to\infty}\left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+.....+\left(\frac{1}{n}-\frac{1}{n+1}\right)\right]$$

$$= \lim_{n\to\infty}\left(1-\frac{1}{n+1}\right) = 1$$

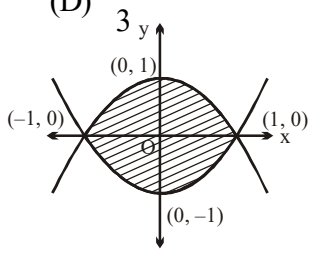
12. Area enclosed by the figure described by the equation $x^4 + 1 = 2x^2 + y^2$, is :

समीकरण $x^4 + 1 = 2x^2 + y^2$ से घिरे क्षेत्र का क्षेत्रफल होगा :

(A) 3 (B) $\frac{8}{3}$ (C) $\frac{7}{3}$ (D) $\frac{10}{3}$

**Ans.** B

**Sol.** We have $y^2 = (x^2 - 1)^2 \Rightarrow y = x^2 - 1$ or $y = 1 - x^2$




$$\text{So, required area} = 4\int_0^1 (1-x^2)dx = 4\left[x-\frac{x^3}{3}\right]_0^1 = \frac{8}{3}$$

13. Let $f:[-1,2]\to[0,\infty)$ be a continuous function such that $f(x)=f(1-x)$ for all $x\in[-1,2]$. Let $R_1=\int_{-1}^{2} x\,f(x)\,dx$, and $R_2$ be the area of the region bounded by $y=f(x)$, $x=-1$, $x=2$, and the x-axis. Then : **[JEE-2011]**

यदि $f:[-1,2]\to[0,\infty)$ एक सतत् फलन इस प्रकार है कि $f(x)=f(1-x)$ जहाँ $x\in[-1,2]$ यदि $R_1=\int_{-1}^{2} x\,f(x)\,dx$ तथा $R_2$ वक्रों $y=f(x)$, $x=-1$, $x=2$, तथा x-अक्ष से घिरा हुआ क्षेत्रफल है, तो : **[JEE-2011]**

(A) $R_1=2R_2$ (B) $R_1=3R_2$ (C) $2R_1=R_2$ (D) $3R_1=R_2$

**Ans.** C

**Sol.** $R_2=\int_{-1}^{2} f(x)dx$, $R_1=\int_{-1}^{2} x f(x)dx$

$=\int_{-1}^{2}(1-x)f(1-x)dx \quad \left(\because \int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right)$

$=\int_{-1}^{2}(1-x)f(x)dx \quad (\text{given } f(x)=f(1-x))$

$=\int_{-1}^{2} f(x)dx-\int_{-1}^{2} x f(x)dx$

or $R_1=R_2-R_1 \quad \Rightarrow 2R_1=R_2$

## Exercise C - II

### SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Find the area bounded by the curve $y = 2x - x^2$ & the straight line $y = -x$.

वक्र $y = 2x - x^2$ तथा सरल रेखा $y = -x$ द्वारा परिबद्ध क्षेत्रफल होगा।

**Ans.** $\frac{9}{2}$

**Sol.** $y = 2x - x^2$

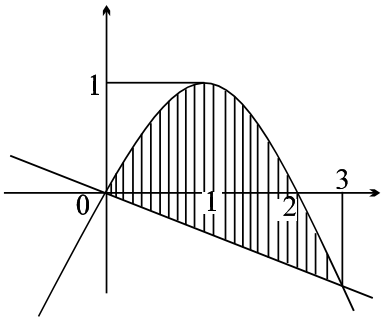
$$\frac{dy}{dx} = 2 - 2x = 0 \Rightarrow x = 1$$

solving $y = 2x - x^2$ and $y = -x$, we have

$$-x = 2x - x^2 \Rightarrow x^2 - 3x = 0 \Rightarrow x = 0 \text{ or } 3$$

$$A = \int_0^3 \left[\left(2x - x^2\right) - (-x)\right] dx = \int_0^3 \left(3x - x^2\right) dx$$

$$= \frac{3x^2}{2} - \frac{x^3}{3}\Big]_0^3 = \frac{27}{2} - 9 = \frac{9}{2} \text{ sq. unit}$$




2. Find the area enclosed by the parabola $y = 1 + x^2$ and a normal drawn to it with gradient $-1$.

परवलय $y = 1 + x^2$ तथा इसके अभिलम्ब जिसका ढाल $-1$ है द्वारा परिबद्ध क्षेत्रफल होगा।

**Ans.** $\frac{4}{3}$

**Sol.** $y = 1 + x^2 \quad \Rightarrow \quad \frac{dy}{dx} = 2x$ [Ans. $A = \frac{4}{3}$ sq. units]

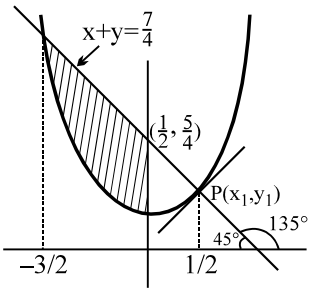
Hence $m = -\frac{1}{2x_1}$ ($m$ = slope of normal at $x_1, y_1$)

$$\therefore \quad x_1 = \frac{-1}{2m} = \frac{1}{2} \quad ; \quad y_1 = 1 + \frac{1}{4m^2} = \frac{5}{4}$$

equation of normal $\quad y - \left(1 + \frac{1}{4m^2}\right) = m\left(x + \frac{1}{2m}\right)$

put $m = -1$, we get $y - \frac{5}{4} = -\left(x - \frac{1}{2}\right)$

$$x + y = \frac{5}{4} + \frac{1}{2} = \frac{7}{4} \qquad \ldots.(1)$$




$$A = \int_{x_1}^{x_2}\left(\frac{7}{4} - x\right) - (1 + x^2)\,dx = \int_{x_1}^{x_2}\left(\frac{3}{4} - x - x^2\right)dx$$

now $1 + x^2 = -x + \frac{7}{4}$

$4x^2 + 4x - 3 = 0 \Rightarrow x = -\frac{3}{2}$ and $x = \frac{1}{2}$

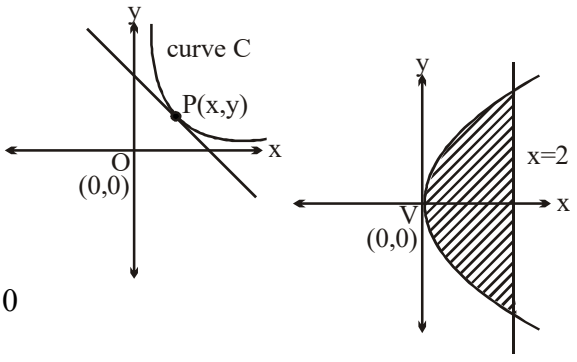
$\therefore \quad A = \int_{-3/2}^{1/2}\left(\frac{3}{4} - x - x^2\right)dx$ simplifying $A = \frac{4}{3}$ sq. units

3. Let C be a curve passing through M (2, 2) such that the slope of the tangent at any point to the curve is reciprocal of the ordinate of the point. If the area bounded by curve C and line $x = 2$ is expressed as a rational $\frac{p}{q}$ (where p and q are in their lowest form), then find $(p + q)$.

वक्र C बिन्दु M (2, 2) से गुजरता है जिसके किसी बिन्दु पर स्पर्श रेखा का ढाल उस बिन्दु की कोटि का व्युत्क्रम है। यदि वक्र C तथा रेखा $x = 2$ द्वारा परिबद्ध क्षेत्रफल $\frac{p}{q}$ है। (जहाँ p तथा q सह अभाज्य है।), तो $(p + q)$ का मान ज्ञात कीजिए।

**Ans.** 19

**Sol.** Let P (x, y) be any point on the curve C.




Now, $\frac{dy}{dx} = \frac{1}{y}$

$\Rightarrow ydy = dx \Rightarrow \frac{y^2}{2} = x + k$

Since the curve passes through M (2,2), so $k = 0$

$\Rightarrow y^2 = 2x$

Hence required area $= 2\int_0^2 \sqrt{2x}\,dx = 2\sqrt{2} \times \frac{2}{3}\left(x^{3/2}\right)_0^2 = \frac{4}{3}\sqrt{2} \times 2\sqrt{2} = \frac{16}{3}$ (square unit)

$\Rightarrow \quad p + q = 19$

4. Let $l$ be the line created by rotating the tangent line to the parabola $y = x^2$ at the point A(1, 1), about A by an angle of $(-\pi/4)$. Let B be the other intersection of the line $l$ with $y = x^2$. If the area enclosed by the line $l$ and the parabola is $\frac{a_1}{a_2}$ where $a_1$ and $a_2$ are coprime, find $(a_1 + a_2)$.

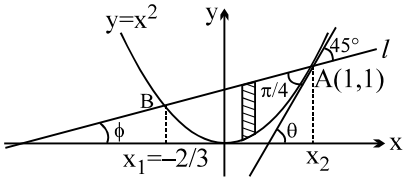
परवलय $y = x^2$ के बिन्दु A(1, 1) पर खीचीं स्पर्श रेखा को बिन्दु A के सापेक्ष $(-\pi/4)$ कोण घुमा देने पर नई रेखा $l$ पुनः परवलय $y = x^2$ को बिन्दु B पर काटती है। यदि रेखा $l$ तथा परवलय द्वारा परिबद्ध क्षेत्रफल $\frac{a_1}{a_2}$ हो ($a_1$ एवं $a_2$ सापेक्षतः अभाज्य है) तो $(a_1 + a_2)$ ज्ञात कीजिए।

**Ans.** 287

**Sol.** $y = x^2$

$$\left.\frac{dy}{dx}\right]_{A(1,1)} = 2x = 2; \quad \text{hence, } \tan\theta = 2 \qquad ....(1)$$

Let the line $l$ makes an angle $\phi$ with the x-axis




then, $\phi = \theta - \frac{\pi}{4}; \quad m_l = \tan\phi = \frac{\tan\theta - 1}{1 + \tan\theta} = \frac{1}{3}$

$\therefore$ equation of the line $l$ is,

$$y - 1 = \frac{1}{3}(x - 1) \quad \Rightarrow \quad 3y - 3 = x - 1 \Rightarrow \quad 3y = x + 2$$

solving line and parabola $y = x^2$,

$$3x^2 = x + 2 \Rightarrow 3x^2 - x - 2 = 0$$

$$x_1x_2 = -\frac{2}{3}; \quad \text{but } x_2 = 1; \quad \therefore \quad x_1 = -\frac{2}{3}$$

$$\therefore \quad A = \int_{-2/3}^{1}\left(\frac{x+2}{3} - x^2\right)dx = \frac{1}{3}\int_{-2/3}^{1}\left(x + 2 - 3x^2\right)dx = \frac{1}{3}\left[\frac{x^2}{2} + 2x - x^3\right]_{-2/3}^{1}$$

$$= \frac{1}{3}\left[\left(\frac{1}{2} + 2 - 1\right) - \left(\frac{2}{9} - \frac{4}{3} + \frac{8}{27}\right)\right] = \frac{1}{3}\left[\frac{3}{2} - \left(\frac{6 - 36 + 8}{27}\right)\right] = \frac{1}{3}\left[\frac{3}{2} + \frac{22}{27}\right]$$

$$= \frac{1}{3}\left[\frac{81 + 44}{54}\right] = \frac{125}{162} = \frac{a_1}{a_2} \quad \Rightarrow \quad a_1 + a_2 = 287$$

5. Let $f(x) = x - x^2$ and $g(x) = ax, a < 0$. If the area bounded by $f(x)$ and $g(x)$ is equal to the area bounded by the curves $x = 3y - y^2$ and $x + y = 3$, then find the value of $|[a]|$

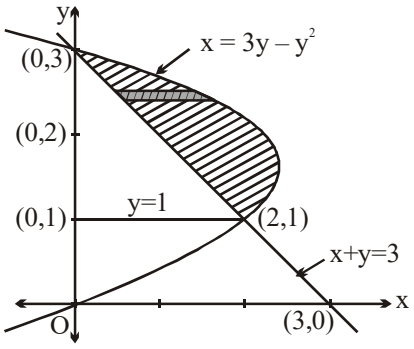
[Note : [ k ] denotes the greatest integer less than or equal to k.]

माना $f(x) = x - x^2$ एवं $g(x) = ax$ है। यदि $f(x)$ व $g(x)$ द्वारा परिबद्ध क्षेत्रफल वक्रों $x = 3y - y^2$ एवं $x + y = 3$ द्वारा परिबद्ध क्षेत्रफल के बराबर है, तो $|[a]|$ का मान ज्ञात कीजिए।

[नोट : [k] महत्तम पूर्णांक को दर्शाता है जो 'k' के बराबर व 'k' से छोटा है]

**Ans.** 1

**Sol.** Area between $x = 3y - y^2$ and $x + y = 3$



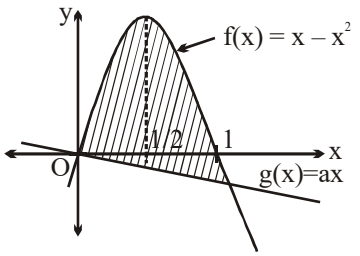

$$A = \int_1^3 \left(3y - y^2\right) - (3 - y)\, dy$$

$$= 4 \cdot \frac{y^2}{2} - \frac{y^3}{3} - 3y \Big]_1^3 = (2 \cdot 9 - 9 - 9) - \left(2 - \frac{1}{3} - 3\right)$$

$$A = \frac{4}{3}$$

Now area bounded by $f(x)$ above the x-axis is $= \int_0^1 \left(x - x^2\right) dx = \frac{1}{6}$, hence $a < 0$ (think)

Given, $\int_0^{1-a} \left((x - x^2) - (a\,x)\right) dx = \frac{4}{3}$




**Note :**
$x - x^2 = ax$
$x^2 + (a-1)x = 0$
$x_1 + x_2 = 1 - a$
but $x_1 = 0$
$\Rightarrow x_2 = 1 - a$

Solving $\frac{1}{6}(1 - a)^3 = \frac{4}{3}$

$(1 - a)^3 = 8$

$1 - a = 2 \Rightarrow a = -1$

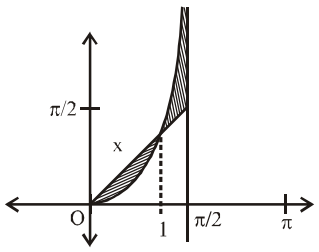
$\Rightarrow \quad |[a]| = 1$

6. If the area bounded by the curves $f(x) = \left(\cos^{-1}|\cos x|\right)^2$, $g(x) = \left(\cos^{-1}|\cos x|\right)$ and $|x| = \frac{\pi}{2}$ is $a\pi^3 + b\pi^2 + c$, then find the minimum value of $(|a| + |b| + |c|)$.

यदि वक्रों $f(x) = \left(\cos^{-1}|\cos x|\right)^2$, $g(x) = \left(\cos^{-1}|\cos x|\right)$ एवं $|x| = \frac{\pi}{2}$ द्वारा परिबद्ध क्षेत्रफल $a\pi^3 + b\pi^2 + c$ है, तो $(|a| + |b| + |c|)$ का न्यूनतम मान ज्ञात कीजिए।

**Ans.** 1

**Sol.** We have $f(x) = \begin{cases} x^2 ; & x \in \left[0, \frac{\pi}{2}\right] \\ (\pi - x)^2 ; & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$; $g(x) = \begin{cases} x ; & x \in \left[0, \frac{\pi}{2}\right] \\ (\pi - x); & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$



Both f and g are periodic with period = π and symmetric about x = π/2

$\therefore$ Required area $= 2\left(\int_0^1 (x - x^2)\,dx + \int_1^{\pi/2} (x^2 - x)\,dx\right)$

$$= 2\left(\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{\pi^3}{24} - \frac{1}{3}\right) - \left(\frac{\pi^2}{8} - \frac{1}{2}\right)\right)$$

$$= 2\left(\frac{\pi^3}{24} - \frac{\pi^2}{8} + \frac{1}{3}\right) = \frac{\pi^3}{12} - \frac{\pi^2}{4} + \frac{2}{3} \equiv a\pi^3 + b\pi^2 + c \text{ (Given)}$$

So, $a = \frac{1}{12}$, $b = \frac{-1}{4}$, $c = \frac{2}{3}$

Hence, $(|a| + |b| + |c|) = \left|\frac{1}{12}\right| + \left|\frac{-1}{4}\right| + \left|\frac{2}{3}\right| = \frac{1}{12} + \frac{1}{4} + \frac{2}{3} = \frac{1+3+8}{12} = \frac{12}{12} = 1$

7. A polynomial function $f(x)$ satisfies the condition $f(x+1) = f(x) + 2x + 1$. Find $f(x)$ if $f(0) = 1$. Find also the equations of the pair of tangents from the origin on the curve $y = f(x)$ and compute the area enclosed by the curve and the pair of tangents.

एक बहुपदीय फलन $f(x)$, $f(x+1) = f(x) + 2x + 1$ का सन्तुष्ट करता है। यदि $f(0) = 1$ तब $f(x)$ ज्ञात कीजिए। साथ ही मूल बिन्दु से वक्र $y = f(x)$ पर स्पर्श रेखा युग्म के समीकरणों का ज्ञात कीजिए तथा वक्र $y = f(x)$ और स्पर्श रेखाओं से परिबद्ध क्षेत्रफल ज्ञात कीजिए।

**Ans.** $f(x) = x^2 + 1$ ; $y = \pm 2x$; $A = \frac{2}{3}$ sq. units

**Sol.** $f(x+1) = f(x) + 2x + 1$

$\Rightarrow f''(x+1) = f''(x) \quad \forall x \in R$

Let $f''(x) = a \quad \Rightarrow \quad f'(x) = ax + b$

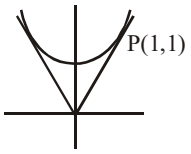
$\Rightarrow f(x) = \frac{ax^2}{2} + bx + c$

$\Rightarrow c = 1 \quad [\because f(0) = 1]$

Now $f(x+1) - f(x) = 2x + 1$

$\Rightarrow \left[\frac{a}{2}(x+1)^2 + b(x+1) + c\right] - \left[\frac{ax^2}{2} + bx + c\right] = 2x + 1$

$\Rightarrow ax + \frac{a}{2} + b = 2x + 1$




on comparing we get $a = 2$,

or $\frac{a}{2} + b = 1 \Rightarrow b = 0$

$\therefore f(x) = x^2 + 1$ ... (i)

Now let equation of tangent be $y = mx$ ... (ii)

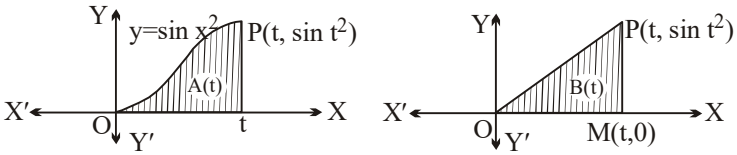
from (i) and (ii)

$x^2 - mx + 1 = 0 \Rightarrow m = \pm 2$

$\therefore$ tangent are $y = 2x$ or $y = -2x$

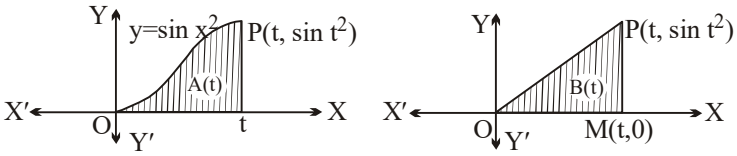
$$A = 2\int_0^1 (x^2 + 1 - 2x)dx = \frac{2}{3}$$

8. The figure shows two regions in the first quadrant.




A(t) is the area under the curve $y = \sin x^2$ from 0 to t and B(t) is the area of the triangle with vertices O, P and M(t, 0). Find $\lim_{t\to 0} \frac{A(t)}{B(t)}$.

चित्र प्रथम चतुर्थांश में दो क्षेत्रो को दर्शाते है।




वक्र $y = \sin x^2$ का $x = 0$ से t में क्षेत्रफल A(t) और B(t) उस त्रिभुज का क्षेत्रफल जिसके शीर्ष O, P और M(t, 0) है तब $\lim_{t\to 0} \frac{A(t)}{B(t)}$ का मान ज्ञात कीजिए।

**Ans.** 2/3

**Sol.** $A(t) = \int_0^t \sin x^2 dx$ & $B(t) = \frac{1}{2} \times t \sin t^2$

$$\lim_{t\to 0}\frac{A(t)}{B(t)}=\lim_{t\to 0}\frac{\int_0^t \sin x^2 dx}{\frac{1}{2}t\cdot\sin t^2}$$

$$=\lim_{t\to 0}\left[\frac{2\int_0^t \sin x^2 dx}{t^3}\right]$$

DLH

$$=\lim_{t\to 0}\frac{2\sin t^2}{3t^2}$$

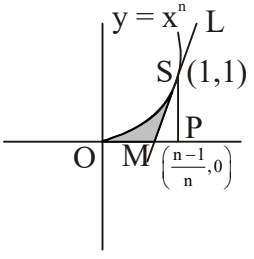
$$=\frac{2}{3}$$

9. Consider the curve $y = x^n$ where $n > 1$ in the 1st quadrant. If the area bounded by the curve, the x-axis and the tangent line to the graph of $y = x^n$ at the point (1, 1) is maximum then find the value of n.

माना प्रथम चतुर्थांशों में वक्र $y = x^n$ जहाँ $n > 1$ है। यदि वक्र $y = x^n$, x-अक्ष और बिन्दु (1, 1) पर स्पर्श रेखा से परिबद्ध क्षेत्र का क्षेत्रफल अधिकतम है तब n का मान ज्ञात कीजिए।

**Ans.** $\sqrt{2}+1$

**Sol.** equation of line L tangent at (1,1)




$y = x^n$

$$\frac{dy}{dx}=nx^{n-1}$$

$$m_T=n$$

$$L: y=nx-n+1 \Rightarrow \text{x-intercept}\left(\frac{n-1}{n},0\right)$$

shaded area

$$A=\int_0^1 x^n dx-\frac{1}{2}\times\left(1-\frac{n-1}{n}\right)\times 1$$

$$A = \frac{1}{n+1} - \frac{1}{2}\left(\frac{1}{n}\right)$$

$$A = \frac{1}{n+1} - \frac{1}{2n}$$

$$A = \frac{n-1}{2n(n+1)}$$

$$\frac{dA}{dn} = \frac{2n(n+1) - 2(n-1)n - 2(n-1)(n+1)}{4n^2(n+1)^2} = 0$$

$= n^2 + n - n^2 + n - n^2 + 1 = 0$

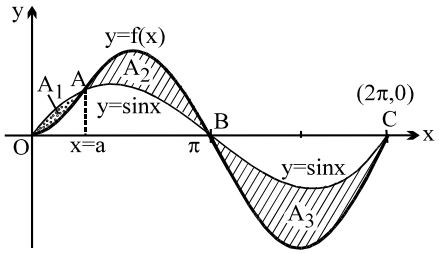
$-n^2 + 2n + 1 = 0$

$n^2 - 2n - 1 = 0$

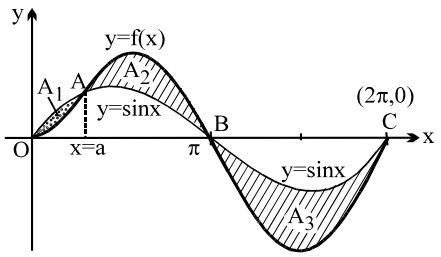
$$n = \frac{2 \pm \sqrt{4+4}}{2}$$

$n = 1 \pm \sqrt{2} \qquad \Rightarrow n \neq 1 - \sqrt{2}\,,\ n = 1 + \sqrt{2}$

10. In the adjacent figure, graphs of two functions y = f(x) and y = sinx are given. y = sinx intersects, y = f(x) at A(a, f(a)); B($\pi$, 0) and C($2\pi$, 0). $A_i$ (i = 1, 2, 3,) is the area bounded by the curves y = f(x) and y = sinx between x=0 and x= a; i = 1, between x = a and x = $\pi$; i = 2, between x = $\pi$ and x = $2\pi$; i = 3. If $A_1 = 1 - \sin a + (a-1)\cos a$, determine the function f(x). Hence determine 'a' and $A_1$. Also calculate $A_2$ and $A_3$.




दिया गया चित्र फलन y = f(x) और y = sin x को दर्शाता है। यदि y = sin x, y = f(x) को A(a, f(a)); B($\pi$, 0) और C($2\pi$, 0) पर प्रतिच्छेदित करता है तथा $A_i$ (i = 1, 2, 3,) वक्रों y = f(x) और y = sin x, x = 0 और x = a; i = 1, x = a और x = $\pi$; i = 2, और x = $\pi$ और x = $2\pi$; i = 3 से क्रमशः परिबद्ध क्षेत्रफल है। यदि $A_1 = 1 - \sin a + (a-1)\cos a$, तथा फलन f(x) ज्ञात कीजिए और a का मान बतलाईये, साथ ही $A_2$ और $A_3$ ज्ञात कीजिए।




**Ans.** f(x) = x sinx, a = 1; $A_1 = 1 - \sin 1$; $A_2 = \pi - 1 - \sin 1$; $A_3 = (3\pi - 2)$ sq. units

**Sol.** $A_1 = \int_0^a (\sin x - f(x)) \cdot dx$ _____(i)

$1 - \sin A + (a-1)\cos a = -(\cos x)_0^a - \int_0^a f(x) \cdot dx$

$\int_0^a f(x) \cdot dx = -\cos a + 1 - 1 + \sin a - (a-1)\cos a$

$\int_0^a f(x) \cdot dx = \sin a - a\cos a$

By Newton Leibnitz Rule

$f(a) = \cos a - [a(-\sin a) + \cos a]$

$f(a) = \cos a + a\sin a - \cos a$

$\Rightarrow f(a) = a\sin a$

$\therefore f(x) = x\sin x$ _____(ii)

from y = sin x and y = x sin x point of intersection is "a"

x sin x = sin x

(x – 1) sin x = 0

$\Rightarrow \sin x = n\pi$ or $x = 1$

Hence a = 1

Now $A_1 = 1 - \sin 1$

$A_2 = \int_1^{\pi} (x\sin x - \sin x)\,dx$

$$A_2 = \int_1^{\pi} (x-1)\sin x \cdot dx$$

$$A_2 = \left(-(x-1)\cos x\right)_1^{\pi} + \int_1^{\pi} \cos x \cdot dx$$

$$A_2 = -\left((x-1)\cos x\right)_1^{\pi} + \sin\pi - \sin 1$$

$$A_2 = -\left[(\pi-1)\cos\pi\right] - \sin 1$$

$$A_2 = \pi - 1 - \sin 1$$

Again $$A_3 = \int_{\pi}^{2\pi} \left|x\sin x - \sin x\right| \cdot dx$$

$$A_3 = 3\pi - 2$$

11. Consider the two curves $y = 1/x^2$ and $y = 1/[4(x-1)]$.

(i) At what value of 'a' ($a > 2$) is the reciprocal of the area of the fig. bounded by the curves, the lines $x = 2$ and $x = a$ equal to 'a' itself?

(ii) At what value of 'b' ($1 < b < 2$) the area of the figure bounded by these curves, the lines $x = b$ and $x = 2$ equal to $1 - 1/b$.

माना दो वक्र $y = 1/x^2$ और $y = 1/[4(x-1)]$ दिये गये हैं।

(i) ५ ६; झ 2द्धके किस मान के लिये दिये गये वक्रों और रेखाओं $x = 2$ व $x = a$ से परिबद्ध क्षेत्र के क्षेत्रफल का प्रतिलोम भी 'a' स्वयं है ?

(ii) ५झ६ ;1 ढ इ ढ 2द्धके किस मान के लिये दिये गये वक्रों तथा रेखाओं $x = b$ और $x = 2$ से परिबद्ध क्षेत्र का क्षेत्रफल $1 - 1/b$ होगा।

**Ans.** $a = 1 + e^2$ , $b = 1 + e^{-2}$

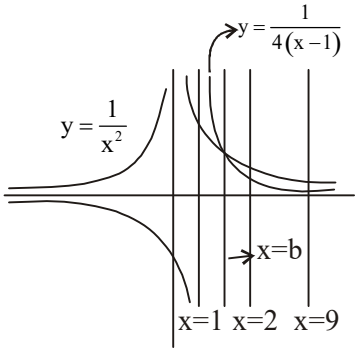
**Sol.** $y = \dfrac{1}{x^2}$ $\qquad$ $y = \dfrac{1}{4(x-1)}$

(i) Point of intersection

$$\frac{1}{x^2} = \frac{1}{4(x-1)}$$

$$x^2 - 4x + 4 = 0$$

$$x = 2$$



$$A = \int_2^a \left( \frac{1}{4(x-1)} - \frac{1}{x^2} \right) dx$$

$$= \frac{1}{4}\log(x-1) + \frac{1}{x}\Big|_2^a$$

$$= \frac{1}{4}\log(a-1) + \frac{1}{a} - \frac{1}{2}$$

Given $A = \frac{1}{a}$

$$\frac{1}{4}\log(a-1) + \frac{1}{a} - \frac{1}{2} = \frac{1}{a}$$

$a - 1 = e^2 \Rightarrow a = 1 + e^2$

(ii) $A = \int_b^2 \frac{1}{4(n-1)} - \left( \frac{1}{x^2} \right) dx$

$$= \frac{1}{4}\log(x-1) + \frac{1}{2}\Big|_b^2$$

$$= +\frac{1}{2} - \frac{1}{b} - \frac{1}{4}\log(b-1)$$

Given $A = 1 - \frac{1}{b}$

$$+\frac{1}{2} - \frac{1}{b} - \frac{1}{4}\log(b-1) = 1 - \frac{1}{b}$$

$$-\frac{1}{4}\log(b-1) = \frac{1}{2}$$

$\log(b-1) = -2 \Rightarrow b = 1 + e^{-2}$

12. Show that the area bounded by the curve $y = \frac{\ell n\, x - c}{x}$, the x-axis and the vertical line through the maximum point of the curve is independent of the constant c.

सिद्ध कीजिये कि वक्र $y = \frac{\ell n\, x - c}{x}$, x-अक्ष और वक्र के उच्चिष्ठ बिन्दु से खींची गयी उर्ध्वाधर रेखा से परिबद्ध क्षेत्र का क्षेत्रफल अचर 'c' से स्वतंत्र है।

**Ans.** 1/2

**Sol.** $y = \frac{\ell nx - C}{x}$

at $y = 0 x = e^c$

$\frac{dy}{dx} = \frac{1-(\ell nx - c)}{x^2} = 0$

$\Rightarrow x = e^{1+c}$

$A = \int_{e^c}^{e^{1+c}} \left(\frac{\ell nx - c}{x}\right) dx$

$A = \frac{(\ell nx - c)^2}{2}\Big|_{e^c}^{e^{1+c}}$

$A = \frac{1}{2}$

13. For what value of 'a' is the area of the figure bounded by the lines, $y = \frac{1}{x}$, $y = \frac{1}{2x-1}$, $x = 2$ and $x = a$, $a > 0$ equal to $\ell n \frac{4}{\sqrt{5}}$?

'a' के किस मान के लिये वक्रों, $y = \frac{1}{x}$, $y = \frac{1}{2x-1}$, $x = 2$ और $x = a$, $a > 0$ से परिबद्ध क्षेत्रफल $\ell n \frac{4}{\sqrt{5}}$ होगा?

**Ans.** $a = 8$ or $\frac{2}{5}\left(6 - \sqrt{21}\right)$

**Sol.** POI $y = \frac{1}{x}$; $y = \frac{1}{2x-1}$; $x = 2$ & $x = a$

$x = 2x - 1$

x = 1

Case-I : a > 2

$$A = \int_2^a \left(\frac{1}{x} - \frac{1}{2x-1}\right) dx$$

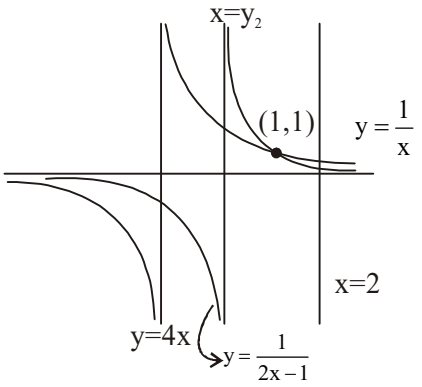
$$\ell n \frac{4}{\sqrt{5}} = \ell nx - \frac{1}{2}\ell n(2x-1)\Big|_2^a$$

$$= \ell na - \frac{1}{2}\ell n(2a-1) - \ell n2 + \frac{1}{2}\ell n3$$

$$\ell n \frac{4}{\sqrt{5}} = \ell n\frac{a}{2} - \frac{1}{2}\ell n\frac{2a-1}{3}$$

$$\frac{1}{2}\ell n\left(\frac{2a-1}{3}\right) = \ell n\frac{a}{2} \times \frac{\sqrt{5}}{4}$$

$$\frac{2a-1}{3} = \frac{a^2 \times 5}{64}$$




$128a - 64 = 15a^2$

$15a^2 - 128a + 64 = 0$

$15a^2 - 120a - 8a + 64 = 0$

$$(15a - 8)(a - 8) = 0 \Rightarrow a = \frac{8}{15}, 8$$

a = 8

Case-II :

1 < a < 2

$$A = \int_a^2 \left(\frac{1}{x} - \frac{1}{2x-1}\right) dx$$

$$\ell n \frac{4}{\sqrt{5}} = \left(\ell nx - \frac{1}{2}\ell n(2x-1)\right)\Big|_a^2$$

$$\ell n \frac{4}{\sqrt{5}} = \frac{1}{2}\ell n\left(\frac{2a-1}{3}\right) - \ell n\frac{a}{2}$$

$$\ell n\frac{2a}{\sqrt{5}} = \frac{1}{2}\ell n\left(\frac{2a-1}{3}\right)$$

$$\frac{4a^2}{5} = \frac{2a-1}{3}$$

$12a^2 = 10a - 5$

$12a^2 - 10a + 5 = 0$

$D = 100 - 240 < 0$

Case-III :

$$\frac{1}{2} < a < 1$$

$$A = \int_a^1\left(\frac{1}{2x-1} - \frac{1}{x}\right)dx + \int_1^2\left(\frac{1}{x} - \frac{1}{2x-1}\right)dx$$

$$\ell n\frac{4}{\sqrt{5}} = \frac{1}{2}\ell n(2x-1) - \ell nx\Big|_a^1 + \left(\ell nx - \frac{1}{2}\ell n(2x-1)\right)\Big|_1^2$$

$$\ell n\frac{4}{\sqrt{5}} = \frac{1}{2}\ell n\frac{a^2}{2a-1} + \frac{1}{2}\ell n\left(\frac{4}{3}\right)$$

$$\frac{16}{5} = \left(\frac{a^2}{2a-1}\right)\frac{4}{3}$$

$12(2a - 1) = 5a^2$

$24a - 12 = 5a^2$

$5a^2 - 24a + 12 = 0$

$$a = \frac{24 \pm \sqrt{24^2 - 4\times 12\times 5}}{10}$$

$$a = \frac{24 \pm \sqrt{12\times 4\times 7}}{10}$$

$$a = \frac{12 \pm 2\sqrt{21}}{5}$$

$a = \frac{2}{5}\left(6+\sqrt{21}\right), \frac{2}{5}\left(6-\sqrt{21}\right)$

$a \neq \frac{2}{5}\left(6+\sqrt{21}\right)$

so, $a = \frac{2}{5}\left(6-\sqrt{21}\right)$

14. For the curve $f(x) = \frac{1}{1+x^2}$, let two points on it are $A\left(\alpha, f(\alpha)\right)$, $B\left(-\frac{1}{\alpha}, f\left(-\frac{1}{\alpha}\right)\right)$ $(\alpha > 0)$. Find the minimum area bounded by the line segments OA, OB and $f(x)$, where 'O' is the origin.

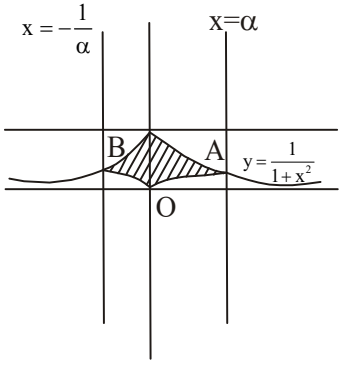
माना $A\left(\alpha, f(\alpha)\right)$, $B\left(-\frac{1}{\alpha}, f\left(-\frac{1}{\alpha}\right)\right)$ $(\alpha > 0)$ वक्र $f(x) = \frac{1}{1+x^2}$ पर दो बिन्दु है तथा रेखाओं OA, OB और वक्र $f(x)$, द्वारा परिबद्ध न्यूनतम क्षेत्रफल ज्ञात कीजिये। (जहाँ 'O' मूलबिन्दु है)

**Ans.** $\frac{(\pi-1)}{2}$

**Sol.** $f(\alpha) = \frac{1}{1+\alpha^2}$

$f\left(-\frac{1}{\alpha}\right) = \frac{\alpha^2}{1+\alpha^2}$

$A = \int_{-\frac{1}{\alpha}}^{\alpha} \frac{1}{1+x^2} - \frac{1}{2}\times\frac{1}{\alpha}\times\frac{\alpha^2}{1+\alpha^2} - \frac{1}{2}\times\alpha\times\frac{1}{1+\alpha^2}$




$A = \tan^{-1}(x)\Big|_{-\frac{1}{\alpha}}^{\alpha} - \frac{1}{2}\frac{\alpha}{1+\alpha^2} - \frac{1}{2}\frac{\alpha}{1+\alpha^2}$

$A = \tan^{-1}\alpha + \tan^{-1}\left(\frac{1}{\alpha}\right) - \frac{\alpha}{1+\alpha^2}$

$A = \tan^{-1}\alpha + \cot^{-1}\alpha - \frac{\alpha}{1+\alpha^2}$

$$A = \frac{\pi}{2} - \frac{\alpha}{1+\alpha^2}$$

$$\frac{dA}{d\alpha} = -\frac{\left(1+\alpha^2\right) - 2\alpha^2}{\left(1+\alpha^2\right)} = 0$$

$1 - \alpha^2 = 0$

$\alpha = 1 \qquad \alpha > 0$

$$A = \frac{\pi}{2} - \frac{1}{2}$$

$$A = \frac{\pi - 1}{2}$$

15. Let 'c' be the constant number such that $c > 1$. If the least area of the figure given by the line passing through the point (1, c) with gradient '*m*' and the parabola $y = x^2$ is 36 sq. units find the value of $(c^2 + m^2)$.

माना 'c' अचर संख्या इस प्रकार है कि $c > 1$. यदि बिन्दु (1, c) से जाने वाली तथा '*m*' प्रवणता की रेखा और परवलय $y = x^2$ से परिबद्ध न्यूनतम क्षेत्रफल 36 वर्ग इकाई है, तो $(c^2 + m^2)$ का मान ज्ञात कीजिये।

**Ans.** 104

**Sol.** Let line $y - c = m(x - 1)$

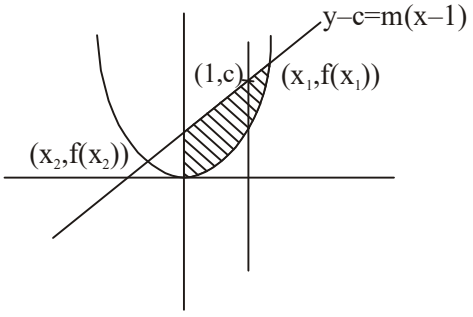
POI

$y = mx + c - m$

& $y = x^2$

$x^2 = mx + c - m$

$$x^2 - mx + m - c = 0 \begin{cases} x_1 \\ x_2 \end{cases}$$




$x_1 + x_2 = m$

$x_1 x_2 = m - c$

$$x_2 - x_1 = \sqrt{m^2 - 4m + 4c}$$

$$x_2^2 - x_1^2 = m\sqrt{m^2 - 4m + 4c}$$

$$x_2^3 - x_1^3 = (x_2 - x_1)\left(x_2^2 + x_1^2 + x_2x_1\right)$$

$$= (x_2 - x_1)\left((x_1 + x_2)^2 - x_1x_2\right)$$

$$= \sqrt{m^2 - 4m + 4c}\left(m^2 - m + c\right)$$

$$A = \int_{x_1}^{x_2}\left(mx + c - m - x^2\right)dx$$

$$A = \frac{m}{2}x^2 + (c - m)x - \frac{x^3}{3}\Bigg|_{x_1}^{x_2}$$

$$A = \frac{m}{2}\left(x_2^2 - x_1^2\right) + (c - m)(x_2 - x_1) - \frac{1}{3}\left(x_2^3 - x_1^3\right)$$

$$A = (x_2 - x_1)\left(\frac{m}{2}(x_2 + x_1) + (c - m) - \frac{1}{3}\left(x_1^2 + x_2^2 + x_1x_2\right)\right)$$

$$A = \sqrt{m^2 - 4m + 4c}\left(\frac{m^2}{2} + c - m - \frac{1}{3}\left(m^2 - m + c\right)\right)$$

$$A = \sqrt{m^2 - 4m + 4c}\left(\frac{m^2}{6} - \frac{2}{3}m + \frac{2}{3}c\right)$$

for minimum area

$$\frac{dA}{dm} = 0$$

$$\sqrt{m^2 - 4m + 4c}\left(\frac{m}{3} - \frac{2}{3}\right) + \left(\frac{m^2 - 4m + 4c}{6}\right)\frac{2m - 4}{\sqrt{m^2 - 4m + 4c}} = 0$$

$$\left(m^2 - 4m + 4c\right)\left(\frac{m - 2}{3}\right) + \frac{\left(m^2 - 4m + 4c\right)}{3}(m - 2) = 0$$

$m = 2$ or $m^2 - 4m + 4c = 0$ (not possible)

so, $m = 2$

$$A_{min.} = \sqrt{4c-4}\left(\frac{4}{6}-\frac{4}{3}+\frac{2}{3}c\right)$$

$$36 = 2\sqrt{c-1}\left(\frac{2}{3}c-\frac{2}{3}\right)$$

$$36 = \frac{4}{3}(c-1)^{\frac{3}{2}}$$

$$(c-1)^{\frac{3}{2}} = 27 \Rightarrow (c-1)^{\frac{1}{2}} = 3$$

$c - 1 = 9 \Rightarrow c = 10$

$c^2 + m^2 = 100 + 4 = 104$

16. Let $A_n$ be the area bounded by the curve $y = (\tan x)^n$ & the lines $x = 0$, $y = 0$ & $x = \pi/4$. Prove that for $n > 2$, $A_n + A_{n-2} = 1/(n-1)$ & deduce that $1/(2n+2) < A_n < 1/(2n-2)$.

यदि $A_n$ वक्र $y = (\tan x)^n$ तथा सरल रेखाओं $x = 0$, $y = 0$ और $x = \pi/4$ से परिबद्ध क्षेत्रफल है, तो सिद्ध कीजिए $n > 2$, $A_n + A_{n-2} = 1/(n-1)$ तथा $1/(2n+2) < A_n < 1/(2n-2)$.

**Ans.**

**Sol.** $A_n = \int_0^{\pi/4} (\tan x)^n \, dx$

$$A_n + A_{n-2} = \int_0^{\pi/4} \left[(\tan x)^n + (\tan x)^{n-2}\right] dx$$

$$= \int_0^{\pi/4} (\tan x)^{n-2} \sec^2 x \, dx = \left[\frac{t^{n-1}}{n-1}\right]_0^1 = \frac{1}{n-1}$$

Also $A_{n+2} < A_n < A_{n-2}$

$$\Rightarrow \frac{1}{n+1} < 2A_n < \frac{1}{n-1}$$

17. Consider the two curves $C_1 : y = 1 + \cos x$ & $C_2 : y = 1 + \cos(x-\alpha)$ for $\alpha \in (0, \pi/2)$; $x \in [0, \pi]$. Find the value of $\alpha$, for which the area of the figure bounded by the curves $C_1$, $C_2$ & $x = 0$ is same as that of the figure bounded by $C_2$, $y = 1$ & $x = \pi$. For this value of $\alpha$, find the ratio in which the line $y = 1$ divides the area of the figure by the curves $C_1$, $C_2$ & $x = \pi$.

दिया है $C_1 : y = 1 + \cos x$ और $C_2 : y = 1 + \cos(x-\alpha)$ for $\alpha \in (0, \pi/2)$ और $x \in [0, \pi]$ दो वक्र हैं। $\alpha$ का मान ज्ञात कीजिये यदि वक्रो $C_1$, $C_2$ और $x = 0$ से परिबद्ध क्षेत्रफल और $C_2$, $y = 1$ और $x = \pi$ से परिबद्ध क्षेत्रफल समान है। इस $\alpha$ के मान के लिये वह अनुपात ज्ञात कीजिये जबकि वक्रों $C_1$, $C_2$ और $x = \pi$ से परिबद्ध क्षेत्रफल को रेखा $y = 1$ विभाजित करती है।

**Ans.** $\alpha = \pi/3$, ratio $= 2 : \sqrt{3}$

**Sol.** $C_1 : y = 1 + \cos x$

$C_2 : y = 1 + \cos(x - \alpha) \qquad \alpha \in \left(0, \frac{\pi}{2}\right),\ x \in [0, \pi]$

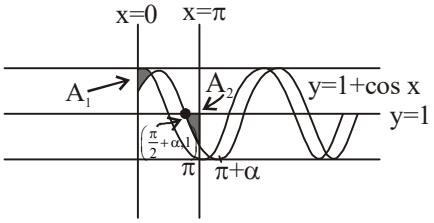
POI of $C_1$ & $C_2$

$\cos x = \cos(x - \alpha)$

$x = |x - \alpha|$

$2\xi = \alpha$

$x = \frac{\alpha}{2}$




Area

$A_1 : C_1, C_2$ & $x = 0$

$A_2 : C_2, y = 1$ & $x = \pi$

$A_1 = A_2$

$$\int_0^{\frac{\alpha}{2}} \left(\cos x - \cos(x-\alpha)\right) dx = \int_{\frac{\pi}{2}+\alpha}^{\pi} -\cos(x-\alpha)\, dx$$

$$\sin x - \sin(x-\alpha)\Big|_0^{\frac{\alpha}{2}} = \sin(x-\alpha)\Big|_{\pi}^{\frac{\pi}{2}+\alpha}$$

$$\sin\frac{\alpha}{2} + \sin\frac{\alpha}{2} - \sin\alpha = 1 - \sin\alpha$$

$$\sin\frac{\alpha}{2} = \frac{1}{2}$$

$$\frac{\alpha}{2} = \frac{\pi}{6}$$

$$\alpha = \frac{\pi}{3}$$

18. For what values of $a \in [0,1]$ does the area of the figure bounded by the graph of the function $y = f(x)$ and the straight lines $x = 0$, $x = 1$ & $y = f(a)$ is at a minimum & for what values it is at a maximum if $f(x) = \sqrt{1-x^2}$. Find also the maximum & the minimum areas.

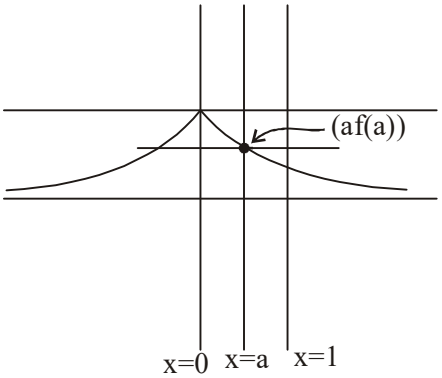
$\in [0,1]$ के किस मान के लिये वक्र $y = f(x)$ सरल रेखाऐं $x = 0$, $x = 1$ और $y = f(a)$ से परिबद्ध क्षेत्रफल न्यूनतम होगा तथा किस मान के लिये यह अधिकतम होगा यदि $f(x) = \sqrt{1-x^2}$ साथ ही अधिकतम और न्यूनतम क्षेत्रफल भी ज्ञात कीजिये।

**Ans.** $a = 1/2$ gives minima, $A\left(\frac{1}{2}\right) = \frac{3\sqrt{3}-\pi}{12}$ ; $a = 0$ gives local maxima $A(0) = 1 - \frac{\pi}{4}$; $a = 1$ gives maximum value, $A(1) = \pi/4$

**Sol.** Given $a \in [0,1]$

$$A = \int_0^a \left(\sqrt{1-x^2} - f(a)\right)dx + \int_a^1 \left(f(a) - \sqrt{1-x^2}\right)dx$$

$$A = \int_0^a \sqrt{1-x^2}\,dx - \int_0^a f(a)\,dx + \int_a^1 f(a)\,dx - \int_0^1 \sqrt{1-x^2}\,dx$$




$$\left[\frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\sin^{-1}(x)\right]_0^a - \left[\frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\sin^{-1}(x)\right]_a^1$$

$-a\,f(0) + (1-a)\,f(a)$

$$= \frac{a}{2}\sqrt{1-a^2} - \frac{1}{2}\sin^{-1}a + \frac{1}{2}\times\frac{\pi}{2} + \frac{a}{2}\sqrt{1\cdot a^2} - \frac{1}{2}\sin^{-1}a$$

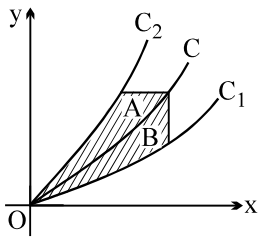
$$-a + (1-a)\sqrt{1-a^2}$$

$$A = \sqrt{1-a^2} - \sin^{-1}a + \frac{\pi}{4}$$

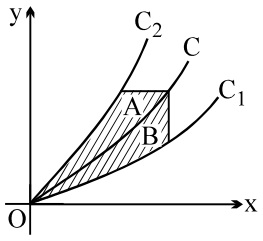
$$\frac{dA}{da} = \frac{-2a}{\sqrt{1-a^2}} - \frac{1}{\sqrt{1-a^2}} = 0$$

$$a = -\frac{1}{2}$$

19. Let $C_1$ and $C_2$ be two curves passing through the origin as shown in the figure. A curve C is said to "bisect the area" the region between $C_1$ and $C_2$, if for each point P of C, the two shaded regions A and B shown in the figure have equal areas. Determine the upper curve $C_2$, given that the bisecting curve C has the equation $y = x^2$ & that the lower curve $C_1$ has the equation $y = x^2/2$.




माना $C_1$ और $C_2$ मूल बिन्दु से जाने वाले वक्र चित्रनुसार है। यदि वक्र C, $C_1$ और $C_2$ के क्षेत्र को C पर प्रत्येक बिन्दु P के लिये समद्विभाजित करता है, तथा चित्र में दो परिबद्ध क्षेत्रफल A और B समान क्षेत्रफल वाले दिये गये हैं। तब ऊपरी वक्र $C_2$ ज्ञात कीजिये। जबकि वक्र C और $C_1$ के समीकरण क्रमशः $y = x^2$ और $y = x^2/2$ दिये गये हैं।




**Ans.** $(16/9)\,x^2$

**Sol.** $C : y = x^2$

$C_1 : y = \dfrac{x^2}{2}$

Let $C_2 : y = f(x)$

Let point of curve C is $(\alpha, \alpha^2)$

$A_1 = B$

$$\int_0^{\alpha^2} (C - C_2)\,dy = \int_0^{\alpha} (C - C_1)\,dx$$

$$\int_0^{\alpha^2} \left(\sqrt{y} - f^{-1}(y)\right)dy = \int_0^{\alpha} \left(x^2 - \frac{x^2}{2}\right)dx$$

$$\left(\alpha - f^{-1}\left(\alpha^2\right)\right)2\alpha = \frac{\alpha^2}{2}$$

$$\alpha - f^{-1}\left(\alpha^2\right) = \frac{\alpha}{4}$$

$$f^{-1}\left(\alpha^2\right) = \frac{3}{4}\alpha$$

$$f\left(\frac{3}{4}\alpha\right)=\alpha^2$$

$$f(x)=\frac{16}{9}x^2$$

$$\because \frac{3}{4}\alpha=x$$

$$\alpha=\frac{4}{3}x$$

$$\therefore\ y=\alpha^2=\frac{16}{9}x^2$$

20. Given f (x) = $\int_0^x e^t(\ln\sec t-\sec^2 t)\,dt$ ; g (x) = – $2e^x$ tan x. Find the area bounded by the curves $y=f(x)$ and $y=g(x)$ between the ordinates $x=0$ and $x=\frac{\pi}{3}$.

माना $f(x)=\int_0^x e^t(\ln\sec t-\sec^2 t)\,dt$ ; $g(x)=-2e^x\tan x$ तब $y=f(x)$, $y=g(x)$ और $x=0$ तथा $x=\frac{\pi}{3}$ से परिबद्ध क्षेत्रफल ज्ञात कीजिए।

**Ans.** $e^{\pi/3}\log 2$ sq. units

**Sol.** $f(x)=\int_0^x e^t\left(\ell n\sec t-\sec^2 t\right)$

$$=\int_0^x e^t\left(\ell n\sec t+\tan t\right)-e^t\left(\tan t+\sec^2 t\right)dt$$

$$=e^t\left(\ell n\sec t-\tan t\right)\Big|_0^x=e^x\left(\ell n\sec x-\tan x\right)$$

Area between f(x) & g(x)

$$A=\int_0^{\frac{\pi}{3}}\left(f(x)-g(x)\right)dx$$

$$= \int_0^{\frac{\pi}{3}} e^x \left(\ell n \sec x + \tan x\right) dx$$

$$= e^x \ell n \sec x \Big|_0^{\frac{\pi}{3}}$$

$$= e^{\frac{\pi}{3}} \ell n 2$$

21. Let $f: [0, \infty) \to R$ be a continuous and strictly increasing function such that $f^3(x) = \int_0^x t f^2(t)\, dt$, $\forall\ x > 0$. Find the area enclosed by $y = f(x)$, the x-axis and the ordinate at $x = 3$.

माना $f : [0, \infty) \to R$ सतत् तथा निरतंर वर्धमान फलन इस प्रकार है कि $f^3(x) = \int_0^x t f^2(t)\, dt$, $\forall\ x > 0$, तब $y = f(x)$, x-अक्ष और $x = 3$ से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

**Ans.** 3/2

**Sol.** $f^3(x) = \int_0^x t f^2(t)\, dt$

$f(0) = 0$

$f \Rightarrow$ strictly increasing

$3f^2(x)\, f'(x) = x f^2(x)$

$$f'(x) = \frac{x}{3}$$

$$f(x) = \frac{x^2}{6} + C$$

$f(0) = C = 0$

$$f(x) = \frac{x^2}{6}$$

$$A = \int_0^3 \frac{x^2}{6} dx$$

$$= \frac{x^3}{18}\Big|_0^3 = \frac{3}{2}$$

## COMPREHENSION TYPE QUESTIONS

**Paragraph for Question nos. 22 to 24**

If f is smooth on [a, b], the length of the curve y = f(x) from a to b is $L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = \int_a^b \sqrt{1+(f'(x))^2}\,dx$ .....(i)

At point on a curve where dy/dx fails to exist, dx/dy may exist and we may be able to find the curve 's length by expressing x as a function of y and applying the following analogue of Eq.(i) Formula for the length of a smooth

curve $x = g(y),\ c \le y \le d$ $L = \int_c^b \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy = \int_c^d \sqrt{1+(g'(y))^2}\,dy$ .....(ii)

यदि f अन्तराल [a, b] में अवकलनीय है तथा a से b में वक्र y = f(x) की लम्बाई निम्न प्रकार से परिभाषित की जाती है

$$L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = \int_a^b \sqrt{1+(f'(x))^2}\,dx \qquad \text{.....(i)}$$

यदि वक्र के किसी बिन्दु पर dy/dx विद्यमान नहीं है तथा dx/dy विद्यमान हो सकता है तो हम वक्र की लम्बाई x को y के फलन के रूप esa O;Dr djds fuEu iz dkj ls Kkr dj l drs gSA Eq.(i) की मदद से वक्र की लम्बाई $x = g(y),\ c \le y \le d$

$$L = \int_c^b \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy = \int_c^d \sqrt{1+(g'(y))^2}\,dy \qquad \text{.....(ii)}$$

22. The length of the curve $y = \frac{4\sqrt{2}}{3}x^{3/2} - 1,\ 0 \le x \le 1$ is :

(A) $\frac{13}{6}$ (B) 2 (C) $\frac{1}{12}$ (D) none of these

वक्र $y = \frac{4\sqrt{2}}{3}x^{3/2} - 1,\ 0 \le x \le 1$ की लम्बाई होगी :

(A) $\frac{13}{6}$ (B) 2 (C) $\frac{1}{12}$ (D) कोई नहीं

**Ans.** A

**Sol.** $y = \frac{4\sqrt{2}}{3}x^{\frac{3}{2}} - 1 \qquad 0 \le x \le 1$

$$\frac{dy}{dx} = 2\sqrt{2}x^{\frac{1}{2}}$$

$$L = \int_0^1 \sqrt{1+8x}\,dx$$

$$= \frac{2}{3}\frac{(1+8x)^{\frac{3}{2}}}{8}\Bigg|_0^1 = \frac{2}{3}\times\frac{27}{8} - \frac{1}{12}$$

$$= \frac{27}{12} - \frac{1}{12}$$

$$= \frac{13}{6}$$

23. The length of the curve $y = (x/2)^{2/3}$ from $x = 0$ to $x = 2$ is :
(A) 3.37 (B) 2.27 (C) 2.0 (D) none of these

वक्र $y = (x/2)^{2/3}$ की $x = 0$ तथा $x = 2$ के मध्य की लम्बाई होगी :
(A) 3.37 (B) 2.27 (C) 2.0 (D) कोई नहीं

**Ans.** B

**Sol.** $y = \left(\frac{x}{2}\right)^{\frac{2}{3}} \Rightarrow x = 0 \text{ to } y = 2$

$$\frac{dy}{dx} = \frac{2}{3\left(\frac{x}{2}\right)^{\frac{1}{3}}} \qquad x = 0 \text{ not defined}$$

$x = 2y^{\frac{3}{2}} \Rightarrow y = 0 \text{ to } y = 1$

$$\frac{dx}{dy} = 3y^{\frac{1}{2}}$$

$$L = \int_0^1 \sqrt{1+9y}\,dy$$

$$= \frac{2}{3}\frac{(1+9y)^{\frac{3}{2}}}{9}\Bigg|_0^1$$

$$= \frac{2}{3}\times\frac{10^{\frac{3}{2}}}{9} - \frac{2}{3\times 9}$$

$$= \frac{2\times 31{\cdot}62}{27} - \frac{2}{27}$$

$= 2.266$

24. The length of the curve $y = (1/3)\,(x^2+2)^{3/2}$ from $x = 0$ to $x = 3$ is :

(A) 10 (B) 12 (C) $\frac{1}{12}$ (D) none of these

वक्र $y = (1/3)\,(x^2+2)^{3/2}$ की $x = 0$ से $x = 3$ के मध्य की लम्बाई होगी :

(A) 10 (B) 12 (C) $\frac{1}{12}$ (D) कोई नही

**Ans.** B

**Sol.** $y = \frac{1}{3}\left(x^2+2\right)^{\frac{3}{2}}$ $x = 0$ to $x = 3$

$$\frac{dy}{dx} = \frac{1}{2}\left(x^2+2\right)^{\frac{1}{2}}\cdot 2x$$

$$L = \int_0^3 \sqrt{1+\frac{1}{4}\left(x^2+2\right)\cdot 4x^2}\,dx$$

$$L = \int_0^3 \sqrt{1+x^4+2x^2}\,dx$$

$$= \int_0^3 \left(x^2+1\right)dx$$

$$= \frac{x^3}{3} + x \Big|_0^3$$

$= 9 + 3 = 12$

**Paragraph for Question nos. 25 to 28**

Let $f(x)$ be a differntiable function such that $f(x+y) = e^x f(y) + e^y f(x)$ all x,y and $f'(0) = 1$

माना $f(x)$ अवकलनीय फलन इस प्रकार है कि $f(x+y) = e^x f(y) + e^x f(x)$ सभी x,y के लिये तथा $f'(0) = 1$

25. f(x) has :

(A) maximum (B) minimum

(C) both maximum and minimum (D) neither maximum nor minimum

f(x) रखता है :

(A) उच्चिष्ठ (B) निम्निष्ठ

(C) उच्चिष्ठ तथा निम्निष्ठ (D) ना तो उच्चिष्ठ न ही निम्निष्ठ

**Ans.** B

**Sol.** $f(x + y) = e^x f(y) + e^y f(x)$

$f'(0) = 1 \qquad f(0) = 0$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{e^x f(h) + e^h f(x) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{e^x f'(h) + e^h f(x)}{h}$$

$$= e^x f'(0) + e^0 f(x)$$

$$f'(x) = e^x + f(x)$$

$$\frac{dy}{dx} - y = e^x$$

I.F. $= e^{-x}$

$$y \cdot e^{-x} = \int dx$$

$y = (x + c)e^x$

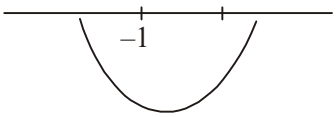
$x = 0, y = 0$

$0 = c \Rightarrow c = 0$

$y = x e^x$

$$\frac{dy}{dx} = e^x (x+1) = 0$$

$x = -1$




minima

26. The range of f(x) is :

f(x) का परिसर होगा :

(A) R (B) $[0, \infty]$ (C) $\left(-\frac{1}{e}, 1\right)$ (D) $\left[-\frac{1}{e}, \infty\right)$

**Ans.** D

**Sol.** $f_{min}(-1) = -\frac{1}{e}$

Range $\left[-\frac{1}{e}, \infty\right)$

27. $\lim\limits_{x \to -\infty} f(x)$ is :

(A) 0 (B) 1 (C) –1 (D) non existent

$\lim\limits_{x \to -\infty} f(x)$ होगा :

(A) 0 (B) 1 (C) –1 (D) विद्यमान नहीं है

**Ans.** A

**Sol.** $\lim\limits_{x \to -\infty} f(x) = \lim\limits_{x \to -\infty} xe^x$

$= \lim\limits_{x \to -\infty} \frac{x}{e^{-x}}$

$= \lim_{x \to -\infty} \frac{1}{-e^{-x}}$

$= 0$

28. The area bounded by the curve $y = f(x)$ and the x–axis is :

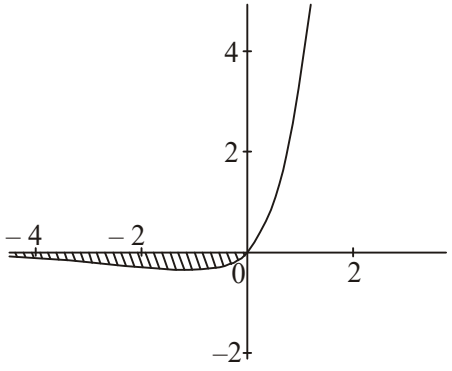
वक्र $y = f(x)$ तथा x–अक्ष द्वारा परिबद्ध क्षेत्रफल होगा :

(A) 1 (B) $\frac{1}{2}$ (C) 2 (D) e

**Ans.** A

**Sol.** Required area $= \int_{-\infty}^{0} \left| x \cdot e^x \right| dx = -\left( e^x (x-1) \right)_{-\infty}^{0} = -\left( e^0 (0-1) - 0 \right)$

= 1 sq. units




## MATCH THE COLUMN TYPE QUESTION(S)

29. Match the following column :

| **Column I** | **Column II** |
|---|---|
| (A) The area between the parabolas $y^2 = 4ax$ and $x^2 = 8ay$ is | (P) 27 |
| (B) Area bounded by the curve $y = (x-1)(x-2)(x-3)$ and x-axis lying between the ordinates $x = 0$ and $x = 3$ is equal to | (Q) $2\pi$ |
| (C) The area cut off the parabola $4y = 3x^2$ by the straight line $2y = 3x + 12$ in sq. units is | (R) $\frac{32}{3}a^2$ |
| (D) The area of the region satisfying $\lvert x \rvert + \lvert y \rvert \le \sqrt{\pi}$ is | (S) $\frac{11}{4}$ |

निम्न कॉलम को सुमेलित कीजिए :

| **Column I** | **Column II** |
|---|---|
| (A) परवलयों $y^2 = 4ax$ तथा $x^2 = 8ay$ के मध्य का क्षेत्रफल होगा | (P) 27 |
| (B) वक्र $y = (x - 1)(x - 2)(x - 3)$, x-अक्ष तथा रेखाओं $x = 0$ और $x = 3$ के मध्य क्षेत्रफल होगा | (Q) $2\pi$ |
| (C) परवलय $4y = 3x^2$ तथा सरल रेखा $2y = 3x + 12$ के मध्य का क्षेत्रफल होगा | (R) $\frac{32}{3}a^2$ |
| (D) वक्र $\lvert x\rvert + \lvert y\rvert \le \sqrt{\pi}$ से घिरा हुआ क्षेत्रफल होगा | (S) $\frac{11}{4}$ |

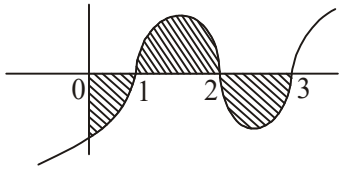
**Ans.** A - R, B - S, C - P, D - Q

**Sol.** (A) $y^2 = 4ax$

$x^2 = 8ay$

$$A = \frac{16}{3} \times a \times b$$

$$A = \frac{32}{3}a^2$$

(B) 


$$A = -\int_0^1 (x-1)(x-2)(x-3) + \int_1^2 (x-1)(x-2)(x-3) - \int_2^3 (x-1)(x-2)(x-3)\,dx$$

$$\int (x-1)(x-2)(x-3)\,dx = \int \left(x^3 - 6x^2 + 11x - 6\right)dx$$

$$= \frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x$$

$$A = -\left[\frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x\right]_0^1 + \left[\frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x\right]_1^2 - \left[\frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x\right]_2^3$$

$$A = \frac{9}{4} + \frac{1}{4} + \frac{1}{4} = \frac{11}{4}$$

(C) Point of Intersection

$4y = 3x^2$ & $2y = 3x + 12$

$3x^2 = 6x + 24$

$x^2 - 2x - 8 = 0$

$x = 4$ & $x = -2$

$y = 12$ $y = 3$

$$A = \int_{-2}^{4}\left(\frac{3x+12}{2} - \frac{3x^2}{4}\right)dx$$

$$= \left(\frac{3x^2}{4} + 6x - \frac{x^3}{4}\right)_{-2}^{4}$$

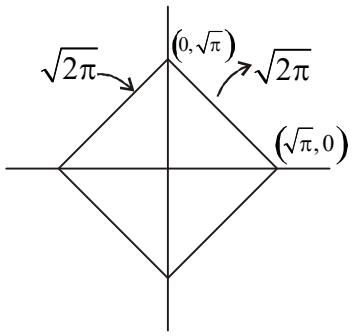
$= 20 - (-7)$

$= 27$

(D) $|x| + |y| \le \sqrt{\pi}$

$A = \sqrt{2\pi} \times \sqrt{2\pi}$

$= 2\pi$




30. Match the following column :

| **Column I** | **Column II** |
|---|---|
| (A) The cosine of the angle between the curves $y = 3^{x-1} \ln x$ and $y = x^x - 1$ at their point of intersection on the line $y = 0$, is | (P) 0 |
| (B) The area bounded by the curves $x = -4y^2$ and $(x - 1) = -5y^2$ is | (Q) 1 |
| (C) The value of the integral $\int_0^{\pi/2} (\sin x)^{\cos x}(\cos x \cot x - \ln(\sin x)^{\sin x})\,dx$, is | (R) $\frac{4}{3}$ |

(D) A continuous function $f: [1, 6] \to [0, \infty)$ is such that $f'(x) = \frac{2}{x + f(x)}$ (S) $2\ ln\ 6$

and $f(1) = 0$, then the maximum value of f cannot exceed

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| | **Column I** | **Column II** |
|---|---|---|
| (A) | वक्रों $y = 3^{x-1}\ ln\ x$ और $y = x^x - 1$ के मध्य रेखा $y = 0$ से प्रतिच्छेद बिन्दु पर कोज्या कोण का मान है | (P) 0 |
| (B) | वक्रों $x = -4y^2$ और $(x - 1) = -5y^2$ से परिबद्ध क्षेत्रफल है | (Q) 1 |
| (C) | समाकलन $\int_0^{\pi/2} (\sin x)^{\cos x} (\cos x \cot x - \ln(\sin x)^{\sin x})\, dx$ का मान है | (R) $\frac{4}{3}$ |
| (D) | एक सतत् फलन $f: [1, 6] \to [0, \infty)$ इस प्रकार है कि $f'(x) = \frac{2}{x + f(x)}$ और $f(1) = 0$, तब f का मान अधिकतम मान से अधिक नहीं हो सकता | (S) $2\ ln\ 6$ |

**Ans.** (A) Q, (B) R, (C) Q, (D) S

**Sol.** (A) $y_1 = 3^{x-1} \ell nx$ $\qquad y_2 = x^x - 1$

Point of intersection $y = 0$

$x = 1$

$$\frac{dy_1}{dx} = \frac{3^{x-1}}{x} + 3x^{x-1} \ell n3 \ell nx$$

$(1, 0)$

$m_1 = 1$

$$\frac{dy_2}{dx} = x^x (1 + \ell nx)$$

(1,0)

$m_2 = 1$

Angle $\theta = 0$

$\cos \theta = 1$

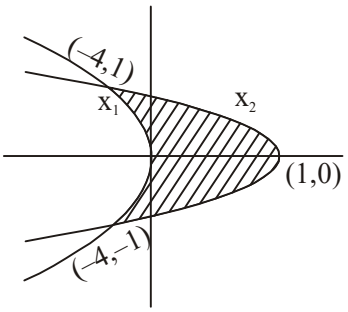
(B) $x = -4y^2$ $\qquad x - 1 = -5y^2$

Point of intersection $\qquad -4y^2 - 1 = -5y^2$

$y^2 = 1$

$y = \pm 1$

$$A = \int_{-1}^{1} (x_2 - x_1)\,dy$$




$$= \int_{-1}^{1} \left[1 - 5y^2 + 4y^2\right] dy$$

$$= \int_{-1}^{1} \left(1 - y^2\right) dy$$

$$= 2\int_{0}^{1} \left(1 - y^2\right) dy$$

$$= 2\left(y - \frac{y^3}{3}\right)_0^1 = \frac{4}{3}$$

(C) $$\int_{0}^{\frac{\pi}{2}} \sin x^{\cos x} \left(\cos x \cdot \cot x - \ell n\left(\sin x\right)^{\sin x}\right) dx$$

$$= \int_{0}^{\frac{\pi}{2}} \frac{d}{dx}\left(\sin x^{\cos x}\right) dx$$

$$= \left(\sin x\right)^{\cos x}\Big|_0^{\frac{\pi}{2}}$$

$= 1$

(D) $f'(x) > 0$ so $f_{max}$ at $x = 6$

$$f'(x) = \frac{2}{x + f(x)}$$

$$\frac{dy}{dx} = \frac{2}{x + y}$$

$x + y = t$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx}-1=\frac{2}{t}$$

$$\frac{tdt}{2+t}=dx$$

$$\int\left(1-\frac{2}{2+t}\right)dt=\int dx$$

$$t-2\log(2+t)=x+C$$

$$x+y-2\log\left(2+x+y\right)=x+C$$

$$y-2\log\left(2+x+y\right)=C$$

$x=1 \qquad y=0$

$$-2\log 3=C$$

$$y-2\log\left(2+x+y\right)=-2\log 3$$

$$y=2\log\left(\frac{2+x+y}{3}\right)$$

at $x=6$

$$y=2\log\left(\frac{8+y}{3}\right)$$

$$\frac{8+y}{3}=e^{\frac{y}{2}}$$

$$8+y=3e^{\frac{y}{2}}$$

$$y<e^{\frac{y}{2}}$$

$$8+e^{\frac{y}{2}}\geq 3e^{\frac{y}{2}}$$

$$8\geq 2e^{\frac{y}{2}}$$

$$e^{\frac{y}{2}}\leq 4$$

$$\frac{y}{2}\leq \ell n4$$

$y \le 2\ell n4$

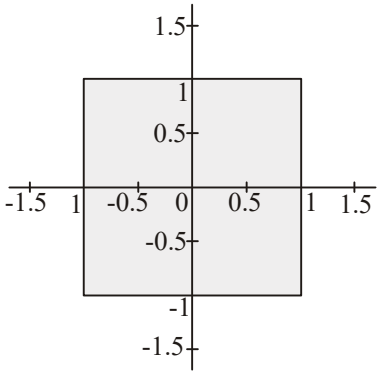
31. Match the following column :

| **Column I** | **Column II** |
|---|---|
| (A) Area of region formed by points (x, y) satisfying $[x] = [y]$, $0 \le x, y \le 10$, [.] denote greatest integer function is | (P) 2 |
| (B) Area of region formed by points (x, y) satisfying $\max(|x|, |y|) \le 1$ is | (Q) 4 |
| (C) If area of region formed by points (x, y) satisfying $|y + x| \le 1$, $|y - x| \le 1$ and $2x^2 + 2y^2 \ge 1$ is $k - \frac{\pi}{2}$, then k = | (R) 6 |
| (D) If area of region formed by points (x, y) satisfying $4 \le x^2 + y^2 \le 2(|x| + |y|)$ is 2k, then k = | (S) 10 |
| | (T) 12 |

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| **Column I** | **Column II** |
|---|---|
| (A) खा, ञ्ज ख्ला ए$0 \le x, y \le 10$ को सन्तुष्ट करने वाले बिन्दुओं से निर्मित क्षेत्र का क्षेत्रफल होगा | (P) 2 |
| (B) उं ग ; ब्बाद्यद्यद्य

(B) Given condition




$\max(|x|, |y|) \le 1$

There will be a square formed by lines

$x = \pm 1$ and $y = \pm 1$

So side of square = 2 units

required area = 4 sq. units

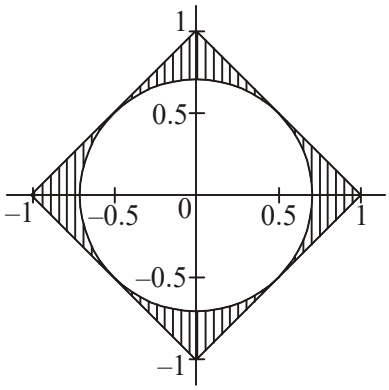
(C) Given constraints

$|y + x| \le 1$ _____(1)

$|y - x| \le 1$ _____(2)

fromo (1) and (2) a square will be formed having vertices $(0, \pm 1)$ and $(\pm 1, 0)$

So area of square $A_1 = 2$ sq. units




Also $x^2 + y^2 \ge \frac{1}{2}$

area of circle $A_2 = \pi \times \frac{1}{2}$

required area $= A_1 - A_2$

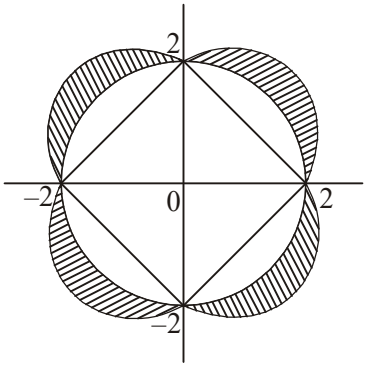
$= 2 - \frac{\pi}{2}$

$\therefore k = 2$

(D) Given constraints

$4 \le x^2 + y^2 \le 2(|x| + |y|)$

$A_1$ = area of square formed by $|x| + |y| \ge 2$




$A_1 = \left(2\sqrt{2}\right)^2 = 8$ square units

$A_2$ = area of circle formed by $x^2 + y^2 \ge 4$

$A_2 = \pi(4)$ square units

$A_3$ = area of shape for med by $x^2 + y^2 \le 2(|x| + |y|)$

$A_3 = 4 \times$ semi circles $= 4 \times \frac{\pi}{2} \times \left(\sqrt{2}\right)^2$

$= 4\pi$ square units

Required area $= A_3 - (A_2 - A_1)$

$= 4\pi - (4\pi - 8)$

$= 8$ square units

ATQ $2k = 8$

$k = 4$